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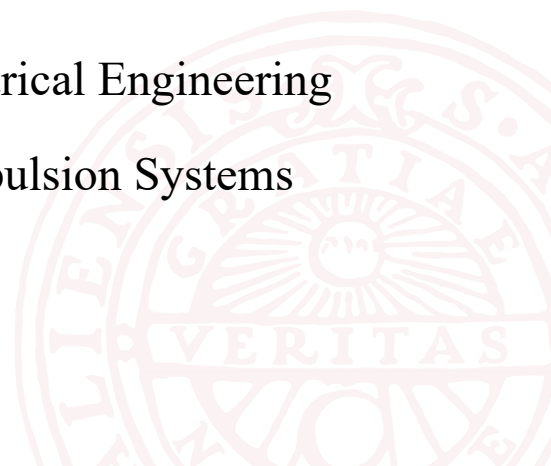
TORQUE-VECTURING CONTROL FOR A SERIES-HYBRID ARTICULATED MINE TRUCK WITH FOUR IN-WHEEL MOTORS

Master's Degree Thesis

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Master's Programme in All-Electric Propulsion Systems





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Abstract

Underground mine trucks are essential for safe and continuous ore haulage, they operate in confined tunnels, steep gradients and low ventilation, so reliability, maneuverability and energy use matter. For decades these vehicles have used internal combustion engine (ICE) propulsion and hydraulic articulation. In recent years these have started to transition from ICE propelled to hybrid and fully electric drivetrains. They have proven to improve the cycle time and reduce energy use and losses, yet they require new control strategies. Virtual simulation is therefore needed to develop and test operational control and performance before hardware. They expose the limitations at the early stage and allow system level validation under repeatable scenarios.

This thesis presents a model-based control framework that integrates steering to articulation conversion, load transfer aware torque vectoring control and direct yaw moment control for a series hybrid, centrally articulated mining truck with four in-wheel motors. The framework is implemented in MATLAB/Simulink as modular subsystems. The layout follows planar articulated vehicle dynamics, and the control supervisory layers within the simulation control loop reflect torque vectoring methods for electric all-wheel drive vehicles. Motor torque and speed limits from data sheets are set for the saturation and power limits used by the torque distributor (allocator).

The study is carried out on an existing Epiroc's underground truck with a series hybrid configuration and four in-wheel motors. Direct access to or adjustment of the inverter and motor electrical parameters is unavailable because the traction units are supplied by external manufacturers. The control model therefore commands individual torques by estimating normal forces and by requesting a stabilizing yaw moment rather than by modifying electrical control loops. Longitudinal and lateral accelerations are computed, normal forces per wheel are derived from geometry and load transfer, and the total torque is distributed according to the load with an adjustable weighting parameter. A small yaw moment term proportional to the articulation angle shifts the left and the right torques while preserving the total torque. An advanced variant estimates the yaw rate and the sideslip to form a feasible yaw moment request to allocate torques within the traction, power, and rate limits.

The simulation across representative maneuvers (0-25 km/h, payloads up to 66 tones and grade) shows feasible yaw tracking and enrichment of outer wheel torques in turns and load dependent torque distribution between each motor. The model supports energy use distribution and provides a practical tool for the electrification of Epiroc's underground vehicles.

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Sammanfattning

Underjordiska gruvtruckar är avgörande för säker och kontinuerlig malmtransport; de verkar i trånga tunnlar, på branta lutningar och under begränsad ventilation, vilket gör att tillförlitlighet, manövrerbarhet och energianvändning är avgörande. I årtionden har dessa fordon använt förbränningsmotorbaserad (ICE) framdrivning och hydraulisk ledstyrning. Under senare år har de börjat övergå från ICE-drift till hybrida och helt elektriska drivlinor. Dessa har visat sig förkorta cykeltiden och minska energianvändning och förluster, men de kräver nya styrstrategier. Virtuellt simulering behövs därför för att utveckla och testa operativ styrning och prestanda innan hårdvara tas i bruk. Den blottlägger begränsningarna i ett tidigt skede och möjliggör validering på systemnivå under reproducerbara scenarier.

Detta examensarbete presenterar ett modellbaserat styrningsramverk som integrerar omvandling från styrning till artikulation, lastöverföringsmedveten vridmomentsfördelning (torque vectoring) och direkt girmomentreglering för en seriehybrid, centralt ledat gruvtruck med fyra hjulnavsmotorer. Ramverket implementeras i MATLAB/Simulink som modulära delsystem. Upplägget följer plan dynamik för ledat fordon, och de överordnade lagren speglar metoder för vridmomentsfördelning i elektriska fyrehjulsdrivna fordon. Begränsningar för motormoment och varvtal enligt leverantörsdata fastställs för de mättnader och effekttak som används av allokeraren.

Studien genomförs på en befintlig underjordstruck från Epiroc med seriehybridkonfiguration och fyra hjulnavsmotorer. Direkt styrning av växelriktaren och motorns elektriska parametrar är inte tillgänglig eftersom dragenheterna tillhandahålls av externa tillverkare. Styrmodellen kommenderar därför individuella moment genom att uppskatta normalkrafter och genom att begära ett stabiliserande girmoment i stället för att modifiera elektriska reglerkretsar. Longitudinella och laterala accelerationer beräknas, hjulvisa normalkrafter härleds utifrån geometri och lastöverföring, och det totala momentet fördelas i enlighet med lasten med en justerbar fördelningsviktning. En liten girmomentterm proportionell mot ledvinkeln förskjuter vänster- och högermomenten samtidigt som det totala momentet bibehålls. En avancerad variant skattar girhastigheten och slirvinkeln (sideslip) för att bilda en genomförbar girmomentsbegäran som allokerar moment inom dragkrafts-, effekt- och ändringshastighetsbegränsningar.

Den simuleringsbaserade valideringen indikerar förbättrad hjulvis momentfördelning och momentvektorerad artikulation samt bättre manövrerbarhet över representativa körprofiler. Modellen stödjer fördelningen av energianvändning och utgör ett praktiskt verktyg för elektrifieringen av Epirocs underjordiska fordon.

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Acronyms and Abbreviations

DYC	Direct Yaw Moment Control
EKF	Extended Kalman Filter
IMM	Interacting Multiple Model Filter
LQR	Linear Quadratic Regulator
LPV	Linear Parameter Varying
MPC	Model Predictive Control
MPCA	Model Predictive Control Allocation
NMPC	Nonlinear Model Predictive Control
PID	Proportional Integral Derivative
QP	Quadratic Program

Symbols, Notions and Units

Φ - <i>phi</i>	articulation angle at the hinge in radians (rad)
$\dot{\phi}$ - <i>phi-dot</i>	Articulation rate in radians per second (rad)
FL, FR, RL, RR	Front Left, Front Right, Rear Left, Rear Right
r	Vehicle yaw rate in radians per sec (rad/s)
β - <i>beta</i>	Vehicle sideslip angle in radians (rad)
a, b	Distance from vehicle to center of gravity to from front and rear axles reference lines in meters (m)
u	Longitudinal speed in meters per second (m/s)
v_y	Lateral speed in meters per second (m/s)
R_w	Effective wheel radius in meters (m)
w_f, w_r	Track width of front and rear modules in meters (m)
M_z	Direct yaw-moment request in newton meters (Nm)
T_i	Wheel torques at wheel ‘ <i>i</i> ’ in newton meter (Nm)
T_{tot}	Total wheel torque demand in newton meters, equal to sum of torque of all wheels torques (Nm)
$F_{x,i}$	longitudinal force at wheel <i>iii</i> in newtons, taken as $F_{x,i} = T_i/R_w$ (N)
$F_{y,i}$	Lateral forces at wheel ‘ <i>i</i> ’ in newtons (N)
$F_{z,i}$	Normal load at each wheel ‘ <i>i</i> ’ in newtons. $F_{z,f}$ and $F_{z,r}$ are the axle normal loads (N)
m	Mass in kilograms (kg)
I_z	Yaw moment of inertia in kilograms meter squared
C_f, C_r	Front and rear cornering stiffness in newtons per radians
K_R, C_R	Hinge stiffness and damping in newton meter per radian and newton meter second per radian

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Chapter 1

1. Introduction

The extraction and the transport of underground minerals depend mostly on mine-specific vehicles, which are designed for safety, high payload, and reliable operation in confined drifts. In their operating environment, they must turn tightly, climb steep ramps, and should cope with variable grip and poor visibility. Historically, control of these vehicles has been difficult because diesel powertrains and hydraulically articulated frames respond slowly and couple strongly to terrain and load. In recent years, electronically managed hybrid and fully electric platforms have improved torque responses and maneuverability, which reduces the operator's workload and provides more software-defined control of the whole vehicle.

This thesis develops and evaluates a simulation-based torque control architecture for a centrally articulated underground mine truck with four in-wheel traction motors in a series-hybrid configuration. The work was carried out at Epiroc Rock Drills AB in Örebro, Sweden, using an industrial plant model representative of the Mine Truck MT66 eDrive class in MATLAB/Simulink. The architecture follows a clear structure that can be implemented within existing interfaces.



Figure 1: Epiroc MT66 eDrive underground haul truck [1]

The simulation modeling is carried out as three parts. First part, a steering-to-articulation module maps the operators steering input to a desired articulation angle and rate using speed-dependent limits. Second, a torque-vectoring module, that will allocate the commanded total drive torque to the four-wheel motors using real-time estimates of the longitudinal and the lateral load transfer to determine each wheel's normal force. A small yaw moment term, which is proportional to articulation, shifts the left and right torques while preserving the total torque. The third part is a yaw controller, which forms a feasible yaw-moment request from the parameters like articulation, speed, and estimated yaw states, then uses a constrained allocation to satisfy the total torque and yaw targets within the motor saturation and power caps.

The objective of this thesis is to improve the yaw stability, traction use, and maneuverability for underground haulage, using only the torque command interfaces, since low-level inverters and motors direct control of parameters are unavailable in production platforms. Validation is performed in MATLAB and Simulink using representative speed and steering profiles.

1.1. Background

The underground haulage demands vehicles that can move heavy payloads safely through narrow drifts and steep ramps while keeping the predictable handling for the operator. Centrally articulated trucks meet these requirements by splitting the chassis into front and rear frames joined by a hinge. This geometry enables tight turns, yet this also couples steering, lateral load transfer, and yaw dynamics in ways that challenge the stability and traction control. Traditional diesel powertrains with hydraulic articulation only offer limited torque bandwidth and no wheel authority, which forces conservative operation and increases reliance on braking to correct path errors. The shift towards the hybrid and electric platforms have introduced faster torque production at each wheel and improved sensing, which makes software supervisory control both feasible and necessary.

This thesis is built on an industrial plant model representative of Epiroc's Mine truck class vehicle in MATLAB/Simulink, with a series hybrid power generation system and four in wheel traction motors for the vehicle propulsion. The control problem is shaped by the practical constraints. The controller must act through total torque and per wheel torque requests only, because the access to low level inverter and motor parameters are not accessible. Actuator limits on torque and power must be always observed. The steering interface is hydraulic and is reaction based, so the driver's input must be converted to a desired articulation angle and rate, then needs to be tracked under loop in the presence of sensor delay and noise. Robust allocation of the torque requires estimation of the normal force at each tire under changing payload and curvature, since the available longitudinal force scales with the vertical load. Yaw allocation must be added through a small left right torque difference that preserves the total drive request.

1.2. Objective

The goal is to design and validate a supervisory control architecture in MATLAB/Simulink that can command the individual wheel torques to improve the stability and maneuverability of the centrally articulated mine truck, while operating strictly through the torque request interfaces. The architecture must run at a fixed sample time and remain compatible with the existing inverter controllers and plant models used at Epiroc.

The first objective is to formalize a steering input to the articulation angle and rate mapping with speed scheduling, signal conditioning, and to have a minor closed loop tracking of the articulation sensor. The second objective is to estimate the per wheel normal forces from the longitudinal and lateral accelerations, then allocate the total drive torque in proportion to the load while preserving the commanded sum. The third objective is to generate a bounded yaw moment request from the articulation and vehicle speeds and to embed this request in the torque allocation, so that the left and right torques create the desired yaw effect without violating any limits. The fourth objective is to enforce the motor torque and power constraints through saturation and power capping, and to ensure that the allocator feasibility under the varying payload and friction changes. The fifth objective is to implement the complete system in MATLAB and Simulink with masked parameters and clear interfaces, then to evaluate the performance on representative underground driving tasks, using yaw rate tracking, torque summation error, wheel torque saturation time, and the smoothness of commanded torques.

1.3. Research Question

The research seeks to optimize the integration of the torque vectoring control strategy in individual motors, also aid the hydraulic actuation system while ensuring better maneuverability and efficient energy management. The following research questions guide the study:

- How can the torque-vectoring control strategy be integrated with the existing model of the MT66 eDrive plant model?
- How do torque-vectoring strategies dynamically manage articulation and variable load distributions across the independent wheel motors to ensure optimal traction, efficiency and performance?

1.4. Delimitation

This study is confined to the development and simulation of a software-based torque vectoring system module integrated into the existing model of the MT66 eDrive plant model in

MATLAB/Simulink. The approach focuses solely on modifications within the control of the simulation plant model and does not involve any direct changes in the electrical control parameters of inverter and the motor control system and other physical components of the powertrain. The simulation is conducted in MATLAB/Simulink, and validation is carried out on available data from Epiroc.

- **Operating Conditions:** The study primarily focuses on steady-state and low-speed scenarios encountered in underground mining operations, such as cornering, variable load conditions, and uneven terrains. Extreme operating conditions, including high-speed dynamics or highly irregular surfaces are excluded to maintain the feasibility of the project within the given timeframe.
- **Physical Testing Scope:** While the control strategies developed in this thesis will be validated through solely using simulations in MATLAB/Simulink.
- **Terrain and Environmental Factors:** Simulations will employ predefined datasets and mining scenarios that will not fully encompass the diversity of real-world mining environments.
- **System-Level Focus:** The emphasis of this thesis is on control strategies for optimizing torque power distribution among wheel motors and their integration with the hydraulic articulation mechanism. But the control model or information regarding hydraulic articulation mechanism in the results will not be discussed.
- Some parameters used for the modeling part are assumptions and calculated round figure values, due to the lack of information regarding vehicle physical and geometric information. Data that are publicly available were only used for the entire simulation.

Chapter 2

2. Literature Study

This chapter reviews a comprehensive study of the fundamentals which are necessary to understand how the actual vehicle is, simulation-based modeling, reference generation, yaw-moment control, torque allocation, state estimation, energy-aware supervision and articulation-specific issues for the articulated vehicles with in-wheel motors.

2.1. Introduction

The Mine truck MT66 eDrive is a large underground haul truck that combines a high-power diesel prime mover with a full electric driveline making it a series hybrid configuration. The machine can carry up to a 66-ton payload, uses steer by wire articulation, and employs wet disc service and park brakes at the wheel ends. Propulsion is electric by wheel motors. Two high power generators feed a 750 V direct current link that supplies the four traction inverters and the four-wheel end machines. A dedicated electric motor powers the hydraulic auxiliaries. The platform is monitored and coordinated by an onboard control system with automation ready features [1].

This truck is an articulated heavy vehicle with a centrally mounted hinge and four independent hub motors. The hinge replaces the conventional front-wheel steering and couples the motion of the front and the rear modules. The powertrain provides fast longitudinal force control at each wheel, but each of the hub motors are bound by the parameters like the current, voltage, thermal, and rate limits that are frequently active during the long duty cycles [2][3]. A clean separation of tasks across the layers simplifies design and supports the formal constraint handling.

The architecture adopted in this review has three layers. A reference generator that converts the driver input and speed into a yaw-related target such as a yaw-rate reference, and optionally a sideslip target or an articulation target. A yaw-moment controller computes the direct yaw-moment request M_z to track the reference while regulating the side-slip when required. A constrained allocator maps the pair $[T_{tot}, M_z]$ into wheel torques 'T' within per-motor and tire limits. This decomposition aligns with a comprehensive torque-vectoring review and with the control allocation literature [4], [5]. In settings with the tight friction margins and actuator limits, the convergence and safety depend on a shared understanding of constraints between the yaw layer and the allocator, with predictive layers used when limits evolve over a horizon [3], [6].

The same decomposition appears abstractly in the control allocation process, where a high-level controller outputs a virtual effort and a dedicated allocator maps that effort to the actuators through a mixing model that manages the constraints and reconfiguration [4]. The layered architecture is also consistent with the supervisory views of the torque vectoring in the electric all-wheel drive vehicles, where the dynamics and powertrain layers have coordination via well-defined interfaces [7]. These sources provide stable coordination that the remainder of the review uses to organize the models and methods [4], [5], [7].

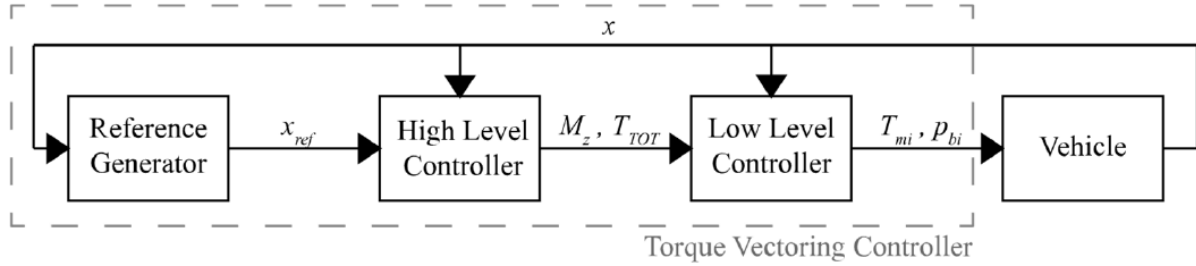
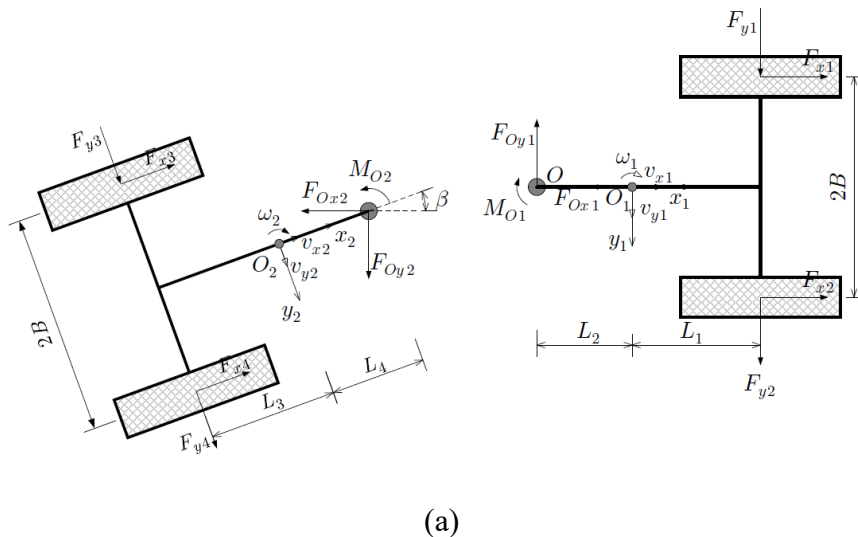


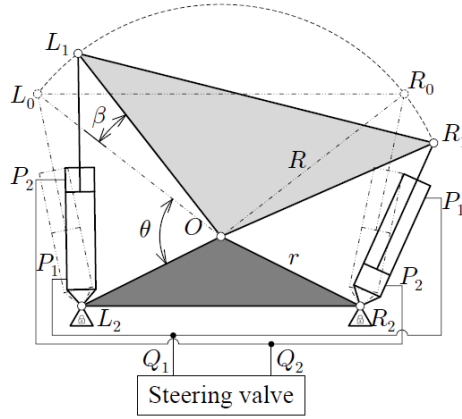
Figure 2: Three-layer control flow: reference generator, yaw controller and torque allocator [5]

2.2. Modeling: Articulated Vehicles

2.2.1 Planar geometry, hinge kinematics, and hydraulics

In the case of an articulated vehicle two independent sources provide compatible planar models, which anchor the state selection and linearization points for the controller design [8], [9]. The geometry places a hinge between the two rigid modules with distances from the hinge to the modules reference axles and to the center of gravity of the vehicle. Articulation angle Φ defines the relative yaw between the modules. The hydraulic articulation actuator is modeled as a cylinder pair acting through a lever arm at the hinge, driven by a pressure source through a servo valve. These features justify a first or second order actuation block with the saturation and rate limits determined by the hydraulic schematic and measured pressure dynamics [8], [9].





(b)

Figure 3: Articulation hinge geometry and hydraulic actuation: (a) Planar layout and axle locations, (b) Opposed-cylinder actuator [8].

Articulation constraint and module yaw rate: The articulation constraint relates to the front and rear module yaw angles and therefore their yaw rates. In the tractor–trailer formulation, the articulation rate equals the difference between the front and rear yaw rates [10]:

$$\dot{\Phi} = r_{front} - r_{rear} \quad (2.1)$$

In an articulated-steer formulation, the articulation angle is the difference of the module yaw angles [9]

$$\Phi = \psi_1 - \psi_2 \quad (2.2)$$

with $r_{front} = \dot{\psi}_1$ and $r_{rear} = \dot{\psi}_2$. The kinematic continuity at the hinge links module translation velocities and yaw rates. In compact vector form as [9]

$$v_h = v_f + r_{front} \hat{k} * r_{h|f} = v_r + \hat{k} * r_{rear} \hat{k} * r_{h|r} \quad (2.3)$$

where v_h is the hinge-point velocity, v_f and v_r are the front and rear modules center-of-gravity velocities, $\hat{k}$ is the unit vector normal to the plane, $r_{h|f}$ and $r_{h|r}$ are the position vectors from each module center to the hinge. Taking the lateral components gives the scalar relations that tie $v_{y,f}$, $v_{y,r}$, r_{front} and r_{rear} through the known geometric arms. These identities fix the variable that matter for the directional control and allow discrete-time updates that remain consistent with the hinge kinematics [9].

Hydraulic actuation at block level: The articulation hydraulics is treated at a block level with a pressure source, a servo valve and a cylinder pair acting through a lever arm at the hinge [8], [9]. The path introduces finite bandwidth, saturation and rate limits. For control design in the linear region, it is practical to represent the valve to cylinder dynamics as a first or second order actuator:

$$G_{act}(s) = \frac{K_{act}}{1 + r_{act}s} \text{ or } G_{act}(s) = \frac{K_{act}}{1 + \zeta\omega_n s + \omega_n^2 s^2} \quad (2.4)$$

with K_{act} , r_{act} , ζ and ω_n identified from the pressure and hinge-angle responses. This representation captures the bandwidth limitations without requiring detailed valve-flow identification (out of scope from this thesis literature review stage), yet it preserves the saturation and the rate limits that must be respected by the reference generator and by the yaw-moment controller [8], [9]. This hydraulic block level model is used only to bound reference rates for steering to articulation, no hydraulic controller is designed in this thesis.

2.2.2 Lateral tire force in small-slip region

The yaw stability controllers and the sideslip observers typically operate in a region where the lateral tire forces are locally linear in slip angle. A Dugoff-type or an effective cornering-stiffness law in the small-slip region is used to construct the linear relations between the lateral forces and slip angles. The articulated-vehicle stability study states this compactly as a linear tire law that applies around the small slip angles and moderate lateral acceleration [13]. In aligned notation [13]:

$$F_{y,i} = -C_i \alpha_i, \quad \text{valid in small - slip region} \quad (2.5)$$

Here $F_{y,i}$ is the lateral force at the axle or the wheel group i in newtons, C_i is the effective cornering stiffness in newtons per radian, and α_i is the slip angle in radians. Within the linear region C_i will be treated as a constant for control design. Its slow dependence on the normal load can be handled by the process of gain scheduling when needed.

For the independently driven articulated buses the linear tire forces across the axles can be collected in a compact vector form that supports the controller synthesis and allocation. The independently driven bus model writes the axle forces as [11]:

$$\begin{bmatrix} F_{y,f} \\ F_{y,m} \\ F_{y,r} \end{bmatrix} = \begin{bmatrix} C_f & 0 & 0 \\ 0 & C_m & 0 \\ 0 & 0 & C_r \end{bmatrix} \begin{bmatrix} \alpha_f \\ \alpha_m \\ \alpha_r \end{bmatrix} \quad (2.6)$$

In this expression f , m and r denote the front, middle and rear axles of an articulated platform that has three axle drive. The slip angle α_f , α_m and α_r are the linear functions of the local lateral velocities and the yaw rates defined by the planar geometry and the hinge kinematics in Section 2.1 discussed before, with the steering angles entering on steered axles as in the cited model [11]. The diagonal structure reflects the assumption that each axle's lateral force depends on its own slip angle in the small-slip region, which is the standard basis for the linear controllers and observers on an articulated vehicle [11][8].

The linear-region assumption will hold for the moderate maneuvers where tire utilization stays well below the friction limit. As utilization increases, the lateral force relation starts to become nonlinear. Controllers then either schedule gains using the estimated stiffness and load or switch

to predictive optimization that enforces the slip and friction constraints explicitly. Both approaches have been documented for direct yaw moment control and integrated stability control in electric vehicles [5][6].

2.3. Unifying Linear Dynamics Model

The upper layer requires a compact state space representation that connects the steering and direct yaw moment inputs to the yaw rate, lateral speed, and articulation. These two sources meet that need. The tractor–trailer study derives a fourth order single track linear model under the small angle and linear tire assumptions and provides an explicit matrix form [10]. The independently driven bus study provides a three degree of freedom compact form $M\dot{x} = D\dot{x} + Fu$ and defines inputs that include the steering and front and rear direct yaw moments [11].

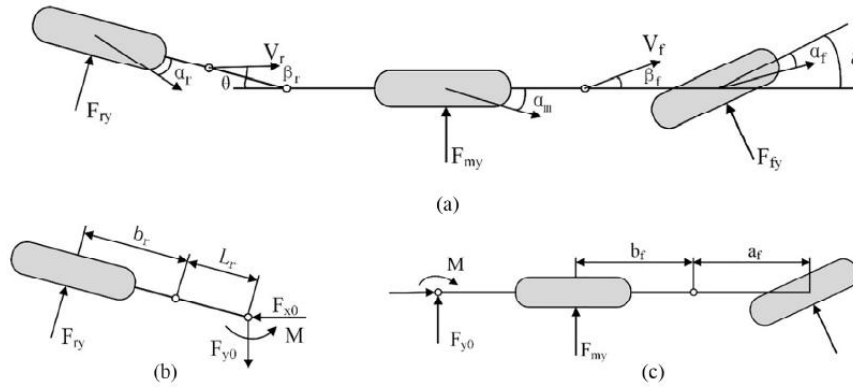


Figure 4: Linear tire model in the small-slip region and compact mass-state from $\dot{x} = D\dot{x} + Fu$ [11].

We adopt the following compact model for the dynamics layer, written in our notation and consistent with small angle linear tire region, adapted from the compact mass form and the input structure in the two sources [11][10]:

$$M(\Phi)\dot{x} = D(\Phi, u)\dot{x} + Fu_c, \quad x = \begin{bmatrix} v_y \\ r \\ \Phi \end{bmatrix}, \quad u_c = \begin{bmatrix} d \\ M_{z,f} \\ M_{z,r} \end{bmatrix} \quad (2.7)$$

Here ‘ d ’ is the steering signal in the passenger car models, used as an assumption. On an articulated platform, this d will be replaced by the commanded articulation angle or its rate, with the articulation actuator dynamics modeled as described above. The matrices M , D , and F depend on the geometry, mass properties, cornering stiffness, and the speed of the vehicle. The dependence on Φ appears through the geometry and dynamic load transfer. In steady conditions with small articulation, M and D can be treated as a constant for the design and is scheduled

and allows slip and actuator constraints to be handled in the optimizer [6]. The torque vectoring review summarizes the same pattern within a broader aspect of the yaw controllers [5].

Starting from the compact linear dynamics in the mass form $M\dot{x} = D\dot{x} + Fu_c$ with inputs $u_c = [d, M_{z,f}, M_{z,r}]^T$ and states $x = [v_y, r, \Phi]^T$ [11], and from the equivalent single track linearization in the tractor-trailer derivation [10], the steady response follows by setting $\dot{x} = 0$. With the direct yaw moment setpoints disabled during the reference generation, the steady state satisfies:

$$0 = Dx_{ss} + Fu_c, \quad u_c = \begin{bmatrix} d \\ 0 \\ 0 \end{bmatrix} \quad (2.9)$$

Solving gives the steady state as, $x_{ss} = -D^{-1}Fu_c$. Selecting the yaw rate components with the row selector $e_r = [0 \ 1 \ 0]$ yields a scalar gain from the steering to yaw rate:

$$r_r = e_r x_{ss} = -e_r D^{-1}F \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} d \triangleq G_r(u, params)d \quad (2.10)$$

where $G_r(u, params)$ is the steady gain that depends on the forward speed u , geometry, mass properties, and the cornering stiffnesses as they appear in D and F [10][11]. On an articulated platform the input d is the commanded articulation angle or a filtered and rate limited map of the driver command to the articulation actuator, as motivated by the hydraulic bandwidth discussion and the small angle operating region.

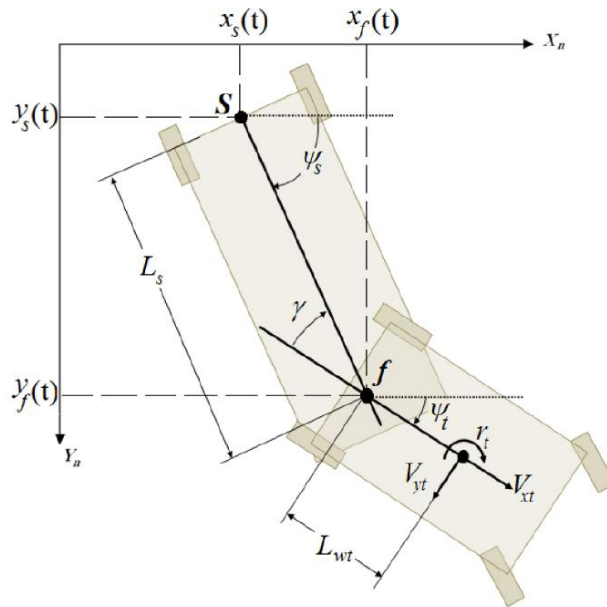


Figure 6: Planar motion of an articulated vehicle with hinge constraint [10]

Equation (2.10) is the steady yaw rate reference used by the yaw moment controller in the linear region. The gain G_r is naturally scheduled with the speed and with the parameter estimates. It is extracted from the same matrices used for the control design, so it remains consistent with the model basis. When a geometric articulation reference is preferred, the tractor-trailer work provides a construction based on the fifth wheel kinematics that produces desired Φ_r , together with the steady state yaw rate from the linear model, which together define the reference triple $[r_r, v_{y,r} = 0, \Phi_r]$ [10].

2.4.2 Geometric articulation reference for articulated combination

For a combination vehicle, the desired articulation angle Φ_r can be generated so that the rear end of the semitrailer follows the fifth-wheel path after a finite delay. The construction in [10] defines that delay from the geometry and speed. With L_s the distance from the fifth wheel to the trailer rear axle and V_x the longitudinal speed, the delay and its discrete equivalent are given as:

$$t_d = \frac{L_s}{V_x}, \quad j = \frac{t_d}{\Delta T} \quad (2.11)$$

Which are built from the fifth-wheel kinematics and the sampling time ΔT used in the controller implementation [10]. Using the same kinematic frame, the desired articulation is produced by a geometric predictor $\Phi_{geom}(\cdot)$ that advances the fifth-wheel motion by t_d and enforces coincidence of the trailer rear point with that delayed fifth-wheel path, given by:

$$\Phi_r(t) = \Phi_{geom}(t, u, params) \quad (2.12)$$

where Φ_{geom} has been constructed from the longitudinal and lateral fifth-wheel velocity relations together with the delay mapping [10]. Under the same construction, the desired lateral velocity is set to zero, and the desired yaw rate is the steady value given by the linear model used in the upper layer. Collecting the three references in our model notation gives:

$$\begin{bmatrix} r_r \\ v_{y,r} \\ \Phi_r \end{bmatrix} = \begin{bmatrix} r_{ss}(u, params) \\ 0 \\ \Phi_{geom}(t, u, params) \end{bmatrix} \quad (2.13)$$

This restates the content in the reference vector [10]. Equation 2.13 is used when the combination needs to be represented explicitly in the reference generator. In case of a single articulated unit such as the mine trucks, a smoothed mapping from the driver's command through the steering to articulation Φ is sufficient if it completely take into consideration of the actuator rate and position limits and if it is filtered to avoid exciting the hydraulic dynamics in the hinge articulation path [8][6].

2.4.3 Sideslip targets and stability factor

A stability factor modifies yaw-rate reference so that the lateral stability margins are preserved as the speed of the vehicle increases. Designs that coordinate with the active steering and direct yaw-moment control based on a single-track model include this factor explicitly in the reference generator, where it scales the steady yaw-rate gain as a function of speed [12]. In our case, with ‘ u ’ the longitudinal speed of the vehicle and ‘ d ’ the driver articulation demand command, when it enters the reference block, we will take the yaw-rate reference as:

$$r_r = \frac{G_0(u, params)}{1 + K_s u^2} d \quad (2.14)$$

where $G_0(u, params)$ is the steady gain from the steering input to yaw rate in the underlying linear model, and $K_s \geq 0$ is the stability factor that decreases the yaw-rate sensitivity as the speed rises [12]. The function $G_0(\cdot)$ carries the model dependence on the geometry, mass properties and cornering stiffness, and r_r is given by radians per second and the d is given in radians.

A sideslip reference is often set to zero in order avoiding the cause of large sideslips in normal operating conditions, i.e., $\beta_r = 0$, but a nonzero β_r can be scheduled in path tracking or crosswind rejection is required (out of scope from this thesis). In case when we use an integrated model predictive controller, then the reference generator may output both the r_r and β_r , while the optimizer will enforce sideslip and friction constraints directly within the prediction horizon [6].

2.5. Yaw-Moment Control Strategies

2.5.1 Baseline proportional-integral-derivative control with feedforward

A classical direct yaw moment controller combines the steady-feedforward with feedback of the yaw-rate errors, and they can also include the sideslip regulation when a reliable estimate is available. The torque vectoring reviews these structures and shows the observer aided variants and the gain scheduling that keeps the loop effective across the different speeds of the vehicle [5]. A practical baseline tracks the reference yaw rate r_r and produces a direct yaw-moment request M_z that a separate allocator will turn into the wheel torques. The law is given by:

$$M_z = K_p(r_r - r) + K_i \int_0^t (r_r(r) - r(r)) dr + K_d(\dot{r}_r - \dot{r}) + M_{z,ff} \quad (2.15)$$

where r is the measured yaw rate, K_p , K_i and K_d are the proportional, integral and derivative gain scheduled with the speed and load of the vehicle. $M_{z,ff}$ is a steady feedforward computed from the linear model and the current operating point of the vehicle [5]. The steady feedforward term can be obtained by solving the linear small slip model at equilibrium for the yaw moment that produces $r = r_r$ with $\dot{r} = 0$ and $v_y = 0$, which places most of the burden on the

feedforward and leaves the feedback terms to reject disturbances and modeling errors[10]. Sideslip β can be neglected with a secondary adjustment that reduces the M_z when $|\beta|$ exceeds a certain threshold, provided that the β is available from a kinematics or Kalman filter estimator [5][13].

With two implementations we can improve robustness without changing the basic form from equation (2.15). First is to compute the derivative term on a filtered yaw rate to avoid the noise amplification. A common choice is a first order low pass on r ; given by:

$$\dot{r}_f = \frac{1}{\tau_d}(r - r_f), \text{ use } \dot{r} \approx \dot{r}_f \quad (2.16)$$

with $\tau_d > 0$ tuned so that the effective derivative bandwidth is below the sensor noise corner. Second, protect the integral term with back calculation anti windup so that the integrator state does not drift when the allocator or plant saturates. With $\text{sat}(\cdot)$ the saturation operator reflecting allocator limits $M_z^{\min}$ and $M_z^{\max}$ the integrator updates become:

$$\dot{I} = (r_f - r) - k_{aw}(M_z - \text{sat}(M_z, M_z^{\min}, M_z^{\max})) \quad (2.17)$$

And $K_I I$ replaces the integral term in equation (2.15). The limits $M_z^{\min}$ and $M_z^{\max}$ are supplied by the allocation layer and reflect the motor current, power and tire friction margins so that the yaw loop respects the feasibility at the actuator level [3], [4], [14][2].

The baseline controller in Equation (2.15) is simple to implement and works well when the actuators have headroom and when the tires operate in the linear region. It is also a good inner loop inside a coordinator that blends the active steering and direct yaw moment. Such coordinators shift the authority to steering at higher speeds or when the allocator approaches saturation, which improves tracking while keeping the sideslip within safe bounds [12].

2.5.2 Control: Nonlinear and integrated MPC

When the tire and actuator constraints are active or near active, predictive control is more effective because it enforces constraints explicitly. A nonlinear model predictive controller can use a single-track or a double-track vehicle model with nonlinear tires and formulate a constrained optimal control problem that is solved in real time. The resulting control action tracks the r_r and respects bounds on states such as the sideslip and on inputs such as the M_z . Integration of real time solvers reduces the decision time to fit the automotive sampling intervals, and the computed generalized efforts can be distributed to the wheel torques by a constrained allocator [15].

In the case of an integrated stability and traction controller, it uses MPC to co-manage traction and yaw stability. In that structure, the reference generator supplies r_r . The optimizer outputs the longitudinal and differential efforts which meet the reference while respecting the slip and actuator limits. This architecture will match the layered view adopted here and demonstrates that the predictive optimization can sit in the yaw layer while a lower layer transforms the

generalized moments into wheel actions [6]. The torque-vectoring reviews the linear quadratic, LPV, and MPC approaches and helps bound the design space [5].

2.5.3 Variable-structure control for snaking suppression

In articulated vehicles a phenomenon known as the snaking mode can occur at higher speeds when the hinge damping and stiffness are small. A variable-structure controller can stabilize that issue by using rear axle differential torque ΔT as the input (out of scope from this thesis, for general information). A planar linearized model includes the states $[v_y, r_f, \Phi, \dot{\Phi}, \Delta\omega_f, \Delta\omega_r]^T$ and an input ΔT that directly affects the articulation dynamics [16]. Although the structure is different from the three-layer design, the insight is valuable. The rear torque bias adds effective yaw stiffness and damping and suppresses the snaking oscillation. The mechanism supports the use of direct yaw moment as an assist channel for stability when the hinge actuation is bandwidth limited [16].

2.5.4 Coordination of direct yaw moment and active steering

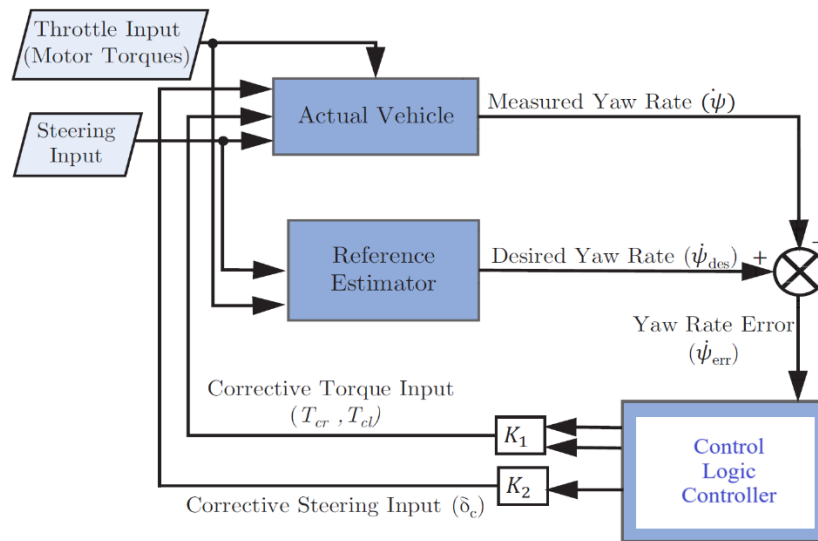


Figure 7: Supervisor coordinating articulation and direct yaw moment through a rule base [12].

A supervisory layer can coordinate the active steering and direct yaw moment in a way that improves tracking and preserves feasibility. The supervisor evaluates a normalized yaw-rate error and then schedules the authority between the articulation actuator and torque vectoring. The design follows the integrated scheme reported for the single-track vehicles with direct yaw moment and active steering, where a fuzzy rule base shapes the split of authority and enforces the explicit bounds on the corrective yaw moment [12]. The same structure applies when steering is realized by a hydraulic articulation actuator rather than a steered axle. Authorities

should shift towards the articulation channel as the speed increases and as the allocator approaches saturation, because steering modifies the lateral tire forces directly, whereas the torque vectoring has limited effect when lateral forces are near their limit [12].

Defining the normalized yaw rate error in our notation as:

$$e_r^n = \frac{r_r - r}{\max(|r_r|, \varepsilon_r)} \quad (2.18)$$

where r_r is the reference yaw rate, r is the measured yaw rate and $\varepsilon_r > 0$ is a small constant that void the division by zero. This follows the normalized error used to drive the supervisor in the integrated design with the active steering and torque vectoring [12].

The supervisor maps the e_r^n , the vehicle speed u , and allocator state indicator into a commanded direct yaw moment. A convenient expression that captures the reported behavior is:

$$M_{z,cmd} = \text{sat}(F(e_r^n, u, \text{sat}_{alloc}), -M_{z,max}, M_{z,max}) \quad (2.19)$$

where $F(\cdot)$ denotes the rule-based mapping, sat_{alloc} is a saturation indicator from the allocator, and $M_{z,max}$ is the maximum admissible corrective yaw moment (computed from μ and N_i based traction limits and electric constraints discussed in sections 2.6 and 2.8.2). The saturation in Equation (2.19) implements the explicit bound on the corrective torque reported in the integrated controller, which protects feasibility when the tire or motor limits are tight [12]. The bound $M_{z,max}$ is computed from the friction and actuator constraints provided by the allocator and can be reduced in real time when the sideslip estimates predict that the friction margin is small [12][4].

Once $M_{z,cmd}$ is set, the allocator computes the left and right wheel torques on each axle. For a single axle with full track width w and effective wheel radius R_w , the left-right split that produces the commanded yaw moment while meeting a given total axle drive torque $T_{d,axle}$ is given as:

$$T_L = \frac{T_{d,axle}}{2} - \frac{M_{z,cmd}R_w}{w}, \quad T_R = \frac{T_{d,axle}}{2} + \frac{M_{z,cmd}R_w}{w} \quad (2.20)$$

which restates the relation reported for the integrated active-steering and the torque-vectoring controller, adapted to our notations [12]. For two axles, the same relation is applied per axle with their respective track widths, and the front-rear share of $M_{z,cmd}$ follows the three-degree-of-freedom formulation where the separate front and rear direct yaw-moment inputs are available to the allocator [11].

The controller or the so-called supervisor can also schedule authority across the two channels. At higher speeds and near the allocator saturation it increases the steering contribution and reduces the torque-vectoring contribution, because the steering has stronger leverage on the lateral tire forces in that operating region. At lower speeds and away from the saturation it allows a more direct yaw moment, which reduces the yaw-rate error without overtaxing the hydraulic system [12]. This trade has been supported with field evidence. In a study where

direct yaw moment assists steering, the addition of direct yaw moment reduced the hydraulic steering pressure by about 41.2 percent for the tested accumulator setting, while preserving the original steering process. This shows that a coordinated design can substitute part of the hydraulic effort with the wheel-torque-generated yaw moment during transients [17].

2.6. Constrained Allocation of Wheel Torques

2.6.1 Control allocation and static formulation

The allocator converts $[T_{tot}, M_z]$ into wheel torques while considering the constraints on torque limit, power limit and rate, and by considering the tire friction limits. The control allocation presents a canonical structure in which the plant is linear in virtual inputs, and the allocator maps the virtual efforts to actuators through a mixing model, with constraints enforced in the allocator [4]. A ground-vehicle study formulates this mapping as a quadratic program that balances the tracking objectives and effort while enforcing bounds and handling reconfiguration under the saturation or failure [14].

In our notation the static mapping at a single instant can be given as:

$$\min_T \frac{1}{2} (GT - v)^T W (GT - v), \quad \text{subject to } T_{min} \leq T \leq T_{max}, \Delta T_{min} \leq \Delta T \leq \Delta T_{max}, \quad (2.21)$$

where $T = [T_{FL}, T_{FR}, T_{RL}, T_{RR}]^T$ collects the left and right wheel torques on the front and rear axles. The vector $v = [T_{tot}, M_z]^T$ holds the requested total drive and the requested direct yaw moment. The vehicle in study here does not have a regeneration system, hence the negative longitudinal effort is provided by Jake Braking and mechanical brakes, so the lower torque bound is $T_{min} = 0$ for all wheels. The matrix G is the mixing matrix that maps wheel torque to the two virtual efforts. The matrix W is positive definite and sets the relative priority between the matching T_{tot} and matching M_z . The elementwise bounds T_{min} and T_{max} encodes the per wheel torque limits from the inverter and machines. The rate bound use $\Delta T = T - T^-$, where T^- is the previous allocation and they enforce actuator slew limits via ΔT_{min} and ΔT_{max} [4][14].

The mixing G follows from geometry and from the relation between the longitudinal force and torque. Using $F_{x,i} = T_i/R_w$ with R_w the wheel radius, the generalized efforts are:

$$T_{tot} = T_{FL} + T_{FR} + T_{RL} + T_{RR} \quad (2.22)$$

$$M_z = \frac{w_f}{R_w} (T_{FR} - T_{FL}) + \frac{w_r}{R_w} (T_{RR} - T_{RL}) \quad (2.23)$$

where w_f and w_r are the front and the rear track widths. Equations (2.22) and (2.23) assume the sign convention that a positive M_z results from a positive right minus left torque difference. If moments are referenced about the articulation hinge rather than about the vehicle center, the

half arms are measured from the hinge for each module. These relations are consistent with the actuation topologies used in the model predictive control allocation for electric vehicles [3].

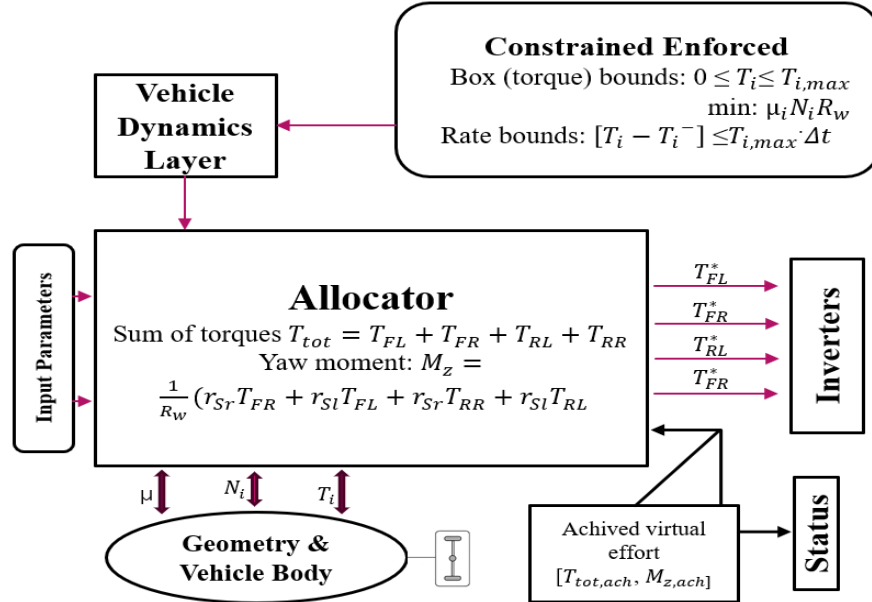


Figure 8: Effectiveness matrix mapping $[T_{tot}, M_z]$ mapped to per-wheel torques [4][14][18].

Here in yaw-moment row, it uses the hinge-aware lever arms, so in the small-angle limit these should be reduced to axle half-tracks about the hinge. Whereas in general they follow the validated pseudo lever-arm expressions $r_{sl} = \frac{s}{2} + l \tan(\Phi/2)$ and $r_{sr} = \frac{s}{2} - l \tan(\Phi/2)$, for the inner and outer sides, here track width w_f and w_r are equal, so half of track width gives s and l is the longitudinal distance from hinge to axle centerline. With the equation (2.22) and (2.23), a convenient form for the mixing matrix can be given as:

$$G = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -r_{sl} & r_{sr} & r_{sl} & -r_{sr} \\ \frac{1}{R_w} & \frac{1}{R_w} & \frac{1}{R_w} & \frac{1}{R_w} \end{bmatrix}$$

So that $GT = [T_{tot}, M_z]^T$. The quadratic program in the equation (2.21) is a direct specialization of the formulations surveyed in the control allocations for over actuated systems and applied to the ground vehicles with constrained actuators [4][14].

When it comes to practical aspects the constraints extend Equation (2.21) without changing its convexity. The tire friction can be expressed as $|T_i| \leq \mu N_i R_w$, where μ is the estimated friction coefficient and N_i is the normal load at wheel i . The power and thermal limits are captured by $-P_{max,i}^{regen} \leq \omega_i T_i \leq P_{max,i}^{mot}$, with the motor speed ω_i . These inequalities restrict the feasible torque under high speed or during the regeneration and are important in duty cycles which leads to heating of the machines and inverters [4][3]. Reconfiguration under a fault is achieved by tightening the bounds for a failed actuator to zero torque or by removing the corresponding

column of G . The allocator then redistributes the effort among the remaining wheels while continuing to track v according to the weights in W [14].

The static program in Equation (2.21) is efficient and is well suited for high-rate implementation. It places all inequality handling in the allocator, which keeps the upper layer simple, and model based, and it admits degradation under saturation or faults through the bounds and weights as reported in the control allocation literature [4][14].

2.6.2 Baseline coaxial torque and reconciliation with energy-optimal split

In case of an articulated vehicle with four hub motors, the nominal strategy is to command equal torque on the coaxial wheels during the steering process. The powertrain study defines the steering-judgment parameter as $Q(t)$, which is determined by measuring the wheel speeds and the effective rolling radii. Another parameter that is used is slip-judgment parameter $S(t)$ which is determined from the wheel torques. The supervisory layer uses these indicators to detect the steering anomalies and to find if there is any potential slip, and moreover to protect the DC link during the regeneration transients. When a prototype testing was carried out [2], the same source confirms that the speed-ratio mapping is accurate. Also, the closed-loop torque control converges to equal coaxial torques under normal conditions [2].

An energy-efficient torque vectoring can favor the skewed torques even in steady cornering conditions. An energy model which is built from the measured branch loss maps shows that the minimum-loss operating point at a small yaw-moment demand corresponds approximately to zero torque on the inner side. A simple relation links the desired direct yaw moment, the total drive torque, the track width, and the wheel radius. In our notation this is given as [19]:

$$M_z \approx \frac{1}{2} T_{tot} \frac{w}{R_w} \quad (2.24)$$

Equation (2.24) serves as a guideline for determining the appropriate amount of differential torque that needs to be applied before engaging auxiliary drive units. This threshold helps to optimize torque distribution without the premature activation of any additional components. The same study presents the mathematical expressions for calculating the lateral torque (also referred to as side torque) and for allocating the torque to individual wheels. These formulas apply to states with either the two-motor or the four-motors configurations, depending on the vehicle's operating mode.

To determine when the supervisor should remain in the equal-coaxial mode and when it needs to be intervened, we will have to restate the steering and slip guards from the point of an articulated electric-drive strategy. The steering-judgment parameter must lie inside a calibrated band for treating the event as a normal steering. In our notation [2]:

$$Q_{min} \leq Q(t) \leq Q_{max} \quad (\text{normal steering region}) \quad (2.25a)$$

The slip-judging parameter compares a monitored wheel torque $T_k(t)$ to the average torque on the two wheels which share the same axle side. It can be given as [2]:

$$S(t) \triangleq \frac{T_k(t)}{T_{coax}(t)} \quad (2.26)$$

Also, a guard band defines the acceptable slip margin for the nominal mode, given by [2]:

$$S_{min} \leq S(t) \leq S_{max} \quad (2.27)$$

In case, if either of the inequalities in equations (2.25) or (2.27) is not satisfied, the controller derates and redistributes the torque to the driveline to protect the DC link and to prevent the chance of excessive slip. When both the indicators remain within their prescribed limits and when the thermal conditions are also stable, then the torque allocator may proceed with the energy-efficient torque-split distribution pattern[19]. This method will resolve the requested yaw moment and the total drive torque into right side torques (T_R) and left side torques (T_L). Using the standard geometry with the track width w and wheel radius R_w , the relation can be given as [19]:

$$T_R = \frac{1}{2}T_{tot} + \frac{1}{2}\frac{R_w}{w}M_{z,ref}, \quad T_L = \frac{1}{2}T_{tot} - \frac{1}{2}\frac{R_w}{w}M_{z,ref} \quad (2.28)$$

When all the four motors are active, each side torque is split between the front and rear wheel by a calibrated front ratio γ_f lies between $[0, 1]$. In our notation [19]:

$$T_{FR} = \gamma_f T_R, \quad T_{FL} = \gamma_f T_L \quad (2.28)$$

$$T_{RR} = (1 - \gamma_f)T_R, \quad T_{RL} = (1 - \gamma_f)T_L$$

Also, when one axle is energized in a two-motor operation state, the entire side torque will be placed on that axle. For front only activation, it can be given as [19]:

$$T_{FR} = \frac{1}{2}T_{tot} + \frac{1}{2}\frac{R_w}{w}M_{z,ref}, \quad T_{FL} = \frac{1}{2}T_{tot} - \frac{1}{2}\frac{R_w}{w}M_{z,ref} \quad (2.29)$$

$$T_{RR} = 0, \quad T_{RL} = 0$$

The activation thresholds select the two-motor versus four active motors as a function of the total torque demand and the stability request. These thresholds are defined in the energy-efficient method and are referenced in the supervisory logic [19]. The authority split and the activation decision are placed inside the three-layer torque-vectoring architecture where the dynamics layer produces a yaw-moment request and the powertrain layer enforces the traction and the thermal limits through the allocation and supervision [7].

2.7. State Estimation for Yaw Rate, Sideslip and Parameters

Having a reliable estimation of yaw rate and sideslip is a preliminary request for both the baseline and advanced operation. This estimation also informs constraint sharing between the yaw-moment controller and the allocator. A kinematic method reconstructs sideslips from the measured accelerations and yaw rate using the kinematic relations. It requires inexpensive sensors and is valid away from the friction limit, which makes it a practical baseline aligned with a PID yaw loop [13]. Dynamic observers reduce bias from parameter drift. An augmented EKF estimates states and uncertain parameters online, such as the front and rear cornering stiffness. It provides the process and measurement models and the adaptation rules and reports improved accuracy under the changing loads and temperatures [20].

An adaptive EKF further improves the reliability across a wide range of sideslips and friction by adjusting covariances using a nonlinearity measure and by switching between a fixed linear tire model and an adaptive one [21]. Near the limits, interacting multiple model filters run multiple models in parallel with Markov switching. A state-dependent transition matrix, that improves mode switching under the sudden friction changes, and tables of results will document performance across operating conditions [22].

Having a reliable estimation of the yaw rate and the sideslip is a prerequisite for both the baseline and advanced operation. A kinematic method can be considered suitable for baseline operation but keeping away from the friction limit [13]. For having the robustness to the tire parameter drift in our small-angle linear region, we can adopt an augmented EKF that estimates the states and cornering stiffness online [20]. Longitudinal speed u is treated as measured or is slowly varying. The augmented state gathers the lateral velocity v_y , yaw rate r , and tire parameters, here we will be only considering the cornering stiffnesses C_f and C_r . The control input will be the steering or articulation command, denoted by ϕ depending on the plant model.

Augmented state:

Let $x \triangleq [v_y, r]^T$ and $\theta \triangleq [C_f, C_r]^T$. Define the augmented vector $z \triangleq [x^T, \theta^T]^T$.

Minimal measurement models: These capture the bicycle-type lateral and yaw dynamics with the parameter augmentation and standard internal measurements[20], in the notation of our model:

$$z_{k+1} = f^{aug}(z_k, u_k) + w_k \quad (2.30)$$

$$y_k = h^{aug}(z_k, u_k) + v_k \quad (2.31)$$

EKF predict and correction in compact form: We will be using the standard EKF recursion with Jacobians, which are computed from f^{aug} and h^{aug} as in the source [20]:

$$\hat{z}_{k|k-1} = f^{aug}(\hat{z}_{k-1|k-1}, u_{k-1}), \quad \hat{z}_{k|k} = \hat{z}_{k|k-1} + K_k[y_k - h^{aug}(\hat{z}_{k|k-1}, u_k)] \quad (2.32)$$

Explicit parameter update block: We take only the adaptation line that is needed by our allocator and yaw cap.

$$\hat{\theta}_{k|k} = \hat{\theta}_{k|k-1} + K_{k,\theta} [y_k - h^{aug}(\hat{z}_{k|k-1}, u_k)] \quad (2.33)$$

The estimator outputs $\hat{r}$, $\hat{v}_y$, and $\hat{C}_f$, $\hat{C}_r$. Sideslip is reconstructed as $\hat{\beta} \approx \arctan(\hat{v}_y/u)$ in a small angle region, then used with the measured from the EKF to cap M_z requests and to tighten the allocator feasibility bounds when friction margin shrinks [6], [20].

2.8. Torque Vectoring Estimation: Control model

In this section we will discuss the torque-vectoring functions and formulations for vehicles that use four independent wheel-hub motors, and it specializes in the powertrain layer to operate under the hard electric drive limits and the actuator faults. Here we will assume a layered structure. First a driver layer which issues the command for the operations. Then we have a vehicle-dynamics layer that delivers the desired direct yaw moment M_z together with the total drive torque effort T_{tot} . And finally in the loop a powertrain layer converts $\{T_{tot}, M_z\}$ into the per-wheel torques while enforcing the parameters machine current, torque-rate, voltage, temperature, and DC-link constraints. This separation matches the multi-layer torque-vectoring architecture reported for electric all-wheel-drive vehicles [7]. It is also consistent with distributed-drive articulated platforms that emphasize DC-link protection and supervisory slip monitoring during regeneration (out of scope from thesis) [2][18].

2.8.1 Problem data and mapping for hub motors

At any given sampling instant, the dynamic layer should supply the virtual effort vector:

$$v \triangleq \begin{bmatrix} T_{tot} \\ M_z \end{bmatrix},$$

And the allocator computes the wheel-torque vector as:

$$T \triangleq [T_{FL}, T_{FR}, T_{RL}, T_{RR}]^T$$

The kinematic and the geometric mapping from wheel torques to $\{T_{tot}, M_z\}$ can be written compactly as:

$$GT = v, \quad \text{mapping is given by Equations (2.22) \& (2.23)}$$

where G is the mixing matrix that collects the total-effort on a row and the yaw-moment on another row, which is defined about the chosen reference. In the case of hub-motor articulated

platforms, the yaw-row will use the hinge-aware arms, and the total-effort row sums all the four torques, which will match the powertrain layer interfaces [7].

In a practical aspect, the machine limits, DC-link power limits, and possible actuator faults make some requests infeasible. Hence the allocator should compute a feasible T , which satisfies equality as closely as possible without violating the hard bounds.

2.8.2 Road friction (μ) and normal-load observer

This section explains the practical aspects of estimating the road friction and the per-wheel normal loads for a hub-motor articulated platform. This computation stays within the small-angle operating region and the small-slip operating condition used for the linear upper layer, and it explains how the resulting estimates are shared with the yaw moment controller and the allocator to enforce the required feasibility. This theory is built from the observer and regime-switching results that we have from the sideslip and EKF literature and on the planar articulated-vehicle models for the load transfer and geometry [21][20].

Friction μ estimator based on C_f and C_r adaptations

Residual-based friction inference on linear model: Will be using the same bicycle model that underlies the augmented EKF for the lateral velocity and yaw rate. We will compare the predicted lateral acceleration with the measured value. A persistent signed error means the linear tire law is nearing saturation or that the effective cornering stiffness has shifted. The augmented EKF in [20] already adapts C_f and C_r in the small slip region. Convert these adapted stiffnesses and their covariances into a conservative friction margin by insisting that the lateral force demand stays below μN . In practice this becomes a per axle friction cap with a safety factor, which remains consistent with the process and measurement models in [20].

Reliability under changing μ with an adaptive EKF: The adaptive EKF [21] tunes its covariances with a nonlinearity metric and switches between a fixed linear tire law and an adaptive tire law. This helps in improving the robustness of the sideslip and yaw estimates during the changes in friction. Use the nonlinearity metric and the innovation statistics as confidence flags for the residual-based friction cap. When the metric indicates stronger nonlinearity, then lower will be the shared friction cap even if the fitted cornering stiffness appears unchanged. This helps when friction drops quickly [21].

Regime detection with an IMM: The interaction of multiple model filters runs several models in parallel with the Markov switching and use a state dependent transition matrix to follow the abrupt friction changes. Treat the IMM mode probabilities as a qualitative friction indicator. When the probability of a low friction mode rises, it tightens the traction limits in the allocator and reduces the yaw moment request. This gives a second trigger that does not depend on a single tire fit [22].

Mild excitation checks: Small steering or torque steps can create informative transients without leaving the validity of the linear model. The torque vectoring survey notes that reference blocks

and the observers often rely on such gentle excitations to detect the margin loss while staying in the linear region [5]. These inputs should be only used when conditions are safe. They are not needed if the residual and IMM cues already show a clear trend.

Normal Load Observers for payload and ramp effects

Parametric load transfer from planar articulated models: Estimate the per wheel normal loads from a static distribution corrected by the longitudinal and the lateral transfer and by the articulation geometry. The planar articulated sources gives the compatible kinematics and equations of motion with hinge location, axle distances to the hinge, and the track widths [8][9]. Using these parameters with measured longitudinal and lateral accelerations, grade, and articulation angle we can propagate normal load changes. The result is a set of per axle or per wheel profiles N_i that match the same geometry used in the dynamics layer.

Scheduling with operating conditions: Include speed, payload, and articulation in the transfer map. Payload will alter the static term and the yaw inertia. Variation in the ramps and bank angles shift the gravity components that the drive transfer. When suspension sensing is not available, low pass filtered accelerations, and the hinge angle are adequate to form a conservative N_i estimate that follows the slow payload changes and the terrain effects [8], [9].

Alignment with allocation needs: Model predictive allocation studies for electric vehicles treat normal force as a primary input for the traction limits and for energy related choices [3]. Even if a predictive allocator is not used, feed the same N_i estimates to the static allocator so that its traction caps reflect the articulation and grade. This keeps the lower layer consistent with the actuation topology and for the constraint handling [3].

Sharing estimates with yaw loop and the allocator

Signals and rates: Give the per wheel estimates and the normal load estimates as $\hat{\mu}_i$ and $\hat{N}_i$ at the estimator rate. Also publish the confidence indicators derived from the EKF or the IMM mode probabilities.

Allocator interface: Use $\hat{\mu}_i$ and $\hat{N}_i$ as the traction limit in the allocator constraints. This is the same form used in the static constrained allocation and agrees with vehicle level constraint handling in the control allocation survey and in EV allocation work [4][14]. When estimator confidence is low, reduce the traction caps with a safety factor so that the allocator will automatically derate the affected side.

Yaw moment capping and authority scheduling: We need to translate a drop in the friction margin into a cap on the requested direct yaw moment before mapping it to the corresponding wheel torques. Combine three cues, first the friction cap implied by the EKF parameter block and its innovations [20]. Second, the adaptive EKF nonlinearity metric [21]. And third the IMM probability of the low friction mode [22]. When any cue crosses a threshold, scale the yaw moment request and shift the authority towards the steering channel, and also alert the supervisor that the allocation is near saturation. This mirrors the integrated stability designs that prefer steering when the torque vectoring has little authority near the tire limit [12], and it follows the layered coordination principle in the torque vectoring architectures [5][7].

Response prioritization: During the operation if the allocator activates the traction constraints, then the slack in the virtual effort equality should need to quantify those dynamically to reflect the priorities. The supervisor or the controller should then soften the reference, and it needs to give more weight to the articulation channel, which matches the recommended behavior in the all-wheel electric architectures under tight margins [18]. The same sequence should occur when the payload rises or when a ramp varies the front axle load. The N_i observer reflects these changes, traction caps tighten, and the allocator returns a smaller achievable yaw moment that remains physically consistent.

2.8.3 Virtual estimation to Individual Wheel Torques

The data from each of the control steps, the nominal mapping that delivers the virtual effort and the constrained correction that guarantees the feasibility under traction and actuator limits are all computed dynamically to determine the final torque to the wheels.

Assemble bounds and mapping constraints: During the execution at each sample, set the per-wheel upper bound as the minimum of the machine limit and a traction-based limit. The articulated-vehicle allocator must prioritize the per-drive torque limits, and these limits are communicated to the allocation block, and they need to be enforced as a point of saturation. A standard way to compute the traction bounds is to map the tire friction margin μN into an available longitudinal force and then into a torque bound using the wheel radius, exactly as done in EV control with combined tire-capacity constraints [18][6]. Also, include the rate limits by constraining the change in torque over the sample, which is a normal extension used in the control-allocation formulation process [4].

Two equalities will complete the mapping the wheel torques to virtual effort. First, the sum of the four wheel torques equals the requested total longitudinal effort T_{tot} . And second, the differences between the right and left torques on each of the axles, scaled by hinge-aware moment arms and divided by R_w , realize the requested direct yaw moment M_z . In allocation terms these two relations are the rows of the effectiveness or mixing matrix [18].

Nominal Split: When none of the bounds are active, then compute a nominal wheel-torque vector which satisfies both the equalities exactly while minimizing a weighted quadratic measure of effort on the individual torques. This is the weighted least-squares inner mapping that is used in the allocation layers [18][7]. Implement the nominal mapping in two stages consistent with the effectiveness matrix. First, choose the left and the right side torques whose sum equals T_{tot} . Set their difference so that, after distributing to front and rear with the same hinge-based arms, the resulting moments generate the commanded M_z . In the second stage distribute each of the side torques to the front and the rear wheels based on the current articulation geometry. The result will be a smooth nominal vector which will coordinate with the hinge layout. In case of steady state steering, a consistent coaxial-torque pattern will provide a benign nominal split, and it preserves the drivability on the articulated platforms with hub motors[2], [18].

Feasibility through a small constrained weighted correction: Testing the nominal vector against all the limits are required, if any of the limits are violated, then it requires to solve a small constrained weighted problem program that changes the torques as little as possible while enforcing the two mapping equalities, the box limits, the rate limits, and the per-wheel traction caps. Here box limits bound each of the wheel torques between its lower and upper admissible values and the rate limits bound the change of each wheel torque across a sample. Additionally, addition of a two-element slack to the equalities with strong penalties will be effective, because then the allocator will remain solvable under the positive-torque and finite-friction constraints [18]. In case we are getting a nonzero slack then it means the exact total effort and the yaw moment cannot be met. And finally, it will return the feasible torques to the individual wheels T_i and at the same time it will report the total effort and yaw moment generated by corresponding T_i , so that the upper layer can scale the next request [7], [18].

2.9. Summary

The literature review has adopted a three-layer architecture for a centrally articulated vehicle with four in-wheel motors, for determining the torques to the individual wheels. The reference generator produces a yaw-rate target from the articulation demand and speed. A yaw-moment controller supplies a direct yaw-moment request M_z that can track this target. Then a constrained allocator will convert $\{T_{tot}, M_z\}$ into the individual wheel torques while honoring the motor and tire limits.

The models that are used for these studies uses a planar hinge kinematics and the linear tire forces in the small-slip region. A compact dynamic form $M\dot{x} = Dx + Fu$ supports the reference design and feedback. The coordination policies within the model will shift the priorities towards the articulation at high speed or when the allocator approaches saturation, because sharp changes in steering will directly affect lateral forces. State estimation will supply the yaw rate, sideslip, and cornering-stiffness updates, together with the road-friction μ and the normal-load N_i estimates used as a major feasibility caps. The descriptions of hydraulic articulation discussed in the studies are provided only for the context on bandwidth and limits, but these are not included in the Simulink scope or in the control blocks.

The allocator enforces per-wheel torque bounds from machine capability and traction capacity, with traction capacity $\mu_i N_i R_w$ for wheel radius R_w . The rate bounds in the constraints will limit the torque change per sample. Two equalities form the 4→2 effectiveness map. It first sums the four torques to the requested total effort T_{tot} . The second generates the requested M_z from the left–right torque differences through hinge-aware half-track arms r_{sl} and r_{sr} that depend on the articulation angle Φ . When all the limits are inactive, then a weighted least-squares map will map a smooth nominal split. If any of the limit is active, a short constrained correction adjusts torques minimally, uses a small slack on the equalities for feasibility, and reports the achieved $\{T_{tot,ach}, M_{z,ach}\}$ upstream. The actual vehicle model in this thesis excludes regeneration while braking, so the lower torque bounds are taken as zero and the negative longitudinal effort is provided by the mechanical brakes.

The individual wheel torques will follow the dynamic axle total loads and the yaw-moment shares shaped by the geometry and traction. Each axle will receive a portion of T_{tot} . On that axle the left and right torques equal half the axle total, minus or plus the axle's share of M_z scaled by R_w and the hinge-aware lever arm set by Φ . Values are clipped by the caps $\mu_i N_i R_w$ and by motor limits, which ties commanded torques directly to the normal forces and the articulation angle.

Chapter 3

3. The Existing Model

3.1. Physical, Electrical and Driveline Specification

The machine is an underground haul truck with a nominal payload of 66,000 kg. This vehicle platform uses a series-hybrid configuration, hence a compression-ignition engine will deliver the mechanical power to the two AC generator units. The generators whose three-phase output is actively rectified to a shared direct-current link with a nominal operating level of 750 V (DC bus hub). Four independent traction converters draw from this link and energize the four wheel-end electric motors. Each of the traction motors are rated near 420 kW in continuous power operation. A separate electric machine supplies all the hydraulic auxiliaries. The chassis uses spring-applied, hydraulically released wet disc brakes on each of the hubs. The front suspension travel is about 140 mm. Load-sensing pumps supply steering and other services. The vehicle platform comes with automation-ready supervision, these features define the operating envelope used in the model and match the published platform description [1][23].

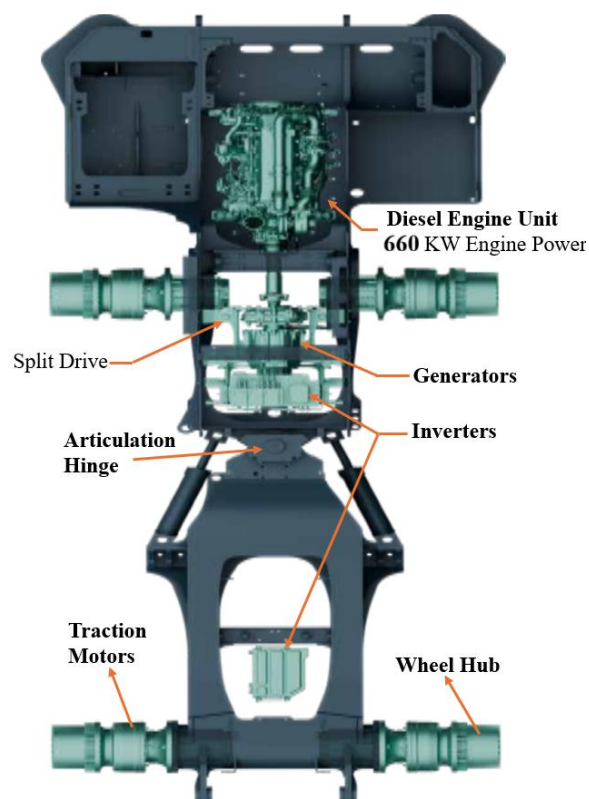


Figure 9: MT66 eDrive driveline and chassis basic overview [1].

Alternator excitation is ramped from zero by the controller and the isolation contactors sequence the safe connection of the generator bridge rectifies to the 750 V DC link, and subsequently to the traction inverters, preventing sudden inrush and ensuring proper synchronization. The converters are responsible for regulating the phase currents in the four-wheel motors. Mechanical power from the motors passes through the hub reductions (gear ratio 53:1) to the wheels and then to the ground. The hydraulic system operates near 23 MPa, that's the maximum operational range when the vehicle articulates and this can be compared to the Yaw Moment required values when introducing the torque vectoring[1]. For the operations of the accessory electrical systems, they run at 24 V_{DC} and it uses isolation and monitoring that are specifically suitable for mine operations [23].

The thermal management systems maintain component integrity during duty operations. The engine cooling circuit rejects engine and generator losses. Power-electronic converters are liquid-cooled. The wheel-end machines use liquid and conduction paths that are integrated with the hub structure. The mechanical wet disc brakes are oil-cooled, and they permit repeated service braking during long declines [1]. The heat-rejection capabilities that are defined in the control system unit for both the ICE and inverters will bound the steady-state tractive effort at lower speeds. The provision of supervisory protection will reduce the allowed torque limit when the thermal indicators approach calibrated thresholds [23]. This behavior is reflected and is explained in the control layer described in 3.2.

The body consists of a forward module with engine, generators, and the front axle, and a rear module with the dump box and the front and rear axle. A central hinge connects the two modules. Steering is command-by-wire (SBW) at the operator interface with hydraulic actuation at the hinge. The cabin meets ROPS and FOPS requirements. The telemetry and logging reports the production metrics and internal states for the diagnostics and for planning. The platform's operational controls are defined within the manufacturer's smart series control system and can be integrated with the underground automation toolchains.

Hydraulic articulation actuation: The steering command from the human-machine interface displaces a metering stage that ports the flow from a load-sensing supply to the two opposed double-acting cylinders that are mounted across the hinge. The load-sensing loop always maintains an approximately constant metering differential. The commanded spool displacement will set the orifice area and therefore the flow. Let x_v be the spool displacement and Δp_{LS} be the maintained differential. Then a chamber flow can be represented as:

$$Q = C_d A_o(x_v) \sqrt{\frac{2}{\rho}} \Delta p_{LS} \quad (3.1)$$

where C_d is the discharge coefficient, A_o is the effective area and ρ is the fluid density. Opposed cylinders will convert the flow into stroke rate $\dot{s} = (Q_A - Q_B)/(A_A + A_B)$, A_A and A_B are the chamber areas. Chamber pressures p_A and p_B produces the cylinder force, $F = p_A A_A - p_B A_B - F_f$ where F_f lumps the seal friction and snubbing. Hitch geometry gives $s = f(\Phi)$ with articulation angle Φ and $\dot{\Phi} = \dot{s}/f'(\Phi)$. The hydraulic yaw moments about the hinge is:

$$M_{hyd}(\Phi, \dot{\Phi}) = \sum_{i=1}^2 \sigma_i l_i(\Phi) (\rho_{Ai} A_{Ai} - \rho_{Bi} A_{Bi}) \quad (3.2)$$

with $l_i(\Phi)$ the lever arm and the installation sign σ_i . The coupled bodies obey:

$$I_z \ddot{\Phi} + c_h \dot{\Phi} + k_h \Phi = M_{hyd} - M_{road} \quad (3.3)$$

with equivalent hinge inertia I_z , damping c_h , any centering stiffness k_h , and resisting moment M_{road} . The end-stop cushions, cross-port relief and the anti-cavitation features bind pressures during reversals and during full lock. These arrangements will match the steer-by-wire metering with load-sensing supply and opposed cylinders reported for equipment in this vehicle class [23], [24].

The geometric relations for the cylinder lengths, chamber volume, and the flow as functions of articulation angle Φ follow from the hinge layout using the law of cosines, and these are used to compute $\dot{\Phi}$ from the measured cylinder flows. The left and right-side cylinder flows are not equal in general, even with symmetric mount positions, that is because the lever arms can vary with Φ (just reaction force). The hinge geometry is parameterized by the power-frame joint vector r_{pf} , the load-frame joint vector r_{lf} , the zero-angle offset α , and the cylinder lengths r_{cr} and r_{cl} . The chamber volumes $V_r(\Phi)$ and $V_l(\Phi)$ are defined from these cylinder lengths together with the bore and rod areas, and differentiating the volumes yields the per-cylinder flow. The left side is obtained from the right by the substitutions $\Phi' = -\Phi$ and $\frac{d}{dt} \Phi' = -\frac{d}{dt} \Phi$ [25]. These definitions support the mappings $s=f(\Phi)$ and $\dot{\Phi}$ from the measured flow, and they also justify the statement that left and right flows generally differ with angle because the lever arms $l_i(\Phi)$ are angle dependent [23], [25].

Numerical details of the platform:

The internal-combustion engine is the primary source of energy for all the operations on the on-board. It is a six-cylinder unit capable of producing around 660 kW at 1700 rpm and it can deliver a peak torque of roughly 4070 Nm at 1400 rpm. These envelopes the total available power, thus it brackets the operating region of the two traction generators, nominally each individual generators can generate up to 420 kW and this sets the expectations for the fuel-rate mapping and the thermal loading for the purpose duty-cycle studies [23].

The four traction converters (inverters) on board manages the individual wheel motors control and operation. Each converter runs from the 750 V direct-current link and is liquid cooled. The converters are capable of 430 kW continuous operation and has peak operational capability of 540 kW for a short rated period, and can operate over a bus range from 500 V to 750 V. The wheel-end motors used on the vehicle are in the class of Dana TM4 Sumo HP. The available continuous torque is about 2050 Nm with peaks near 2500 Nm, and maximum electrical speed is roughly 4500 rpm [23]. These ratings defines the feasible electrical-speed window in the controller and the torque-rate limits enforced by the control layer.

The overall length of the vehicle is approximately 11,447 mm with the standard configuration. The total width of the vehicle is 3500 mm and the berm-crossing height is about 422 mm. The dump box achieves a nominal 62° tip angle. The distance from front axle to the hinge is near 3,850 mm and the distance from the hinge to the rear axle is near 1976 mm. These values support the kinematic checks on axle loads during dump and articulation events. The net unladen mass of the vehicle is approximately 43,000 kg. The front axle share in the unladen condition is about 30,300 kg and the rear axle share is about 12,700 kg[1]. These values, together with the 66,000 kg payload capacity, define the suspension and brake sizing envelopes and inform the initial tire load set-points of the model.

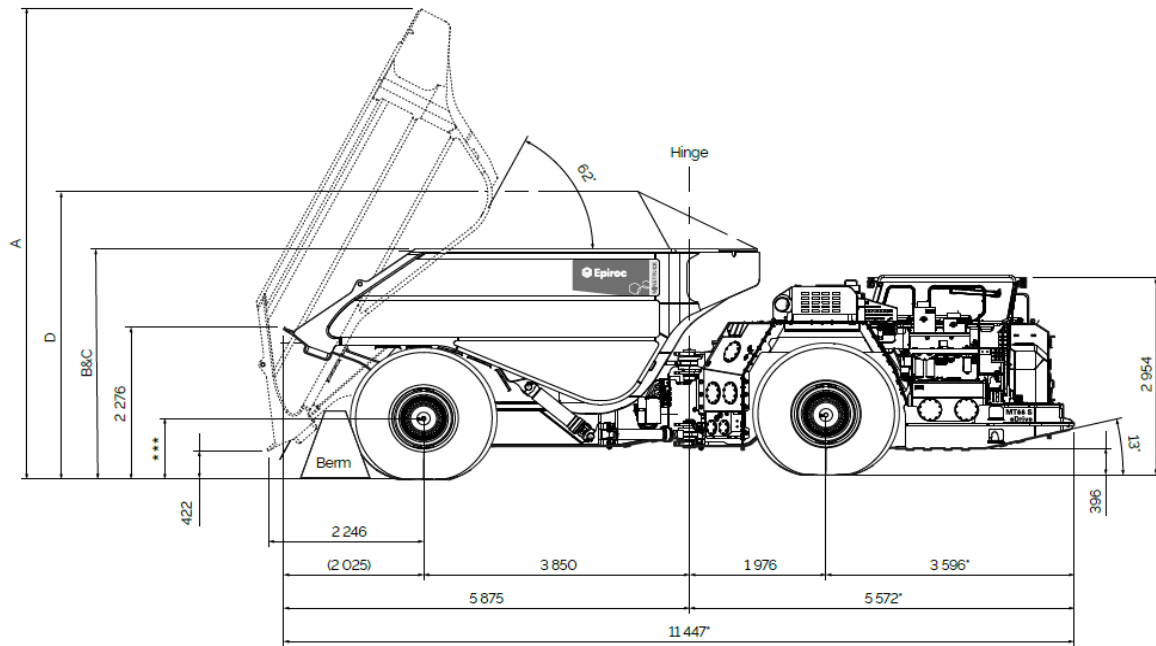


Figure 10: Vehicle Geometric and key dimensions of the MT66 eDrive [1]

Standard tires that are used on all four wheels 35/65R33. The nominal section width is about 923 mm. Overall diameter is about 2056 mm. The loaded tire radius lies in between the 920 to 960 mm range. The rolling circumference of the tires is nearly 6135 mm [23]. These parameters are used to convert the wheel torque to tractive effort and to estimate the vehicle speed. Wheel hubs are heavy-duty planetary units sized for the wheel-motor torque envelope with a gear ratio of 53:1. A representative family is the Kessler R111, selected to transmit the continuous and the peak torques listed above while preserving bearing life under heavy mining duty [1].

3.2. Simulink-Simscape model of the Vehicle

The Top-Level model comprises the subsystems for the operator's intent, the digital controller system, propulsion controller, traction conversion and the mechanical environment. Internal

library names and private blocks labels will not be disclosed. Subsystems are described by their functions.

3.2.1 Human interface and supervision

The primary interface module that aggregates the major operator's (Driver) inputs into a typed signal sets all the other controllers. This subsystem includes a throttle command (0 to 1), a service brake command (0 or 1), a gear selection or direction selection (Forward (+1) – Neutral (0) - Reverse (-1)), a speed limit selection, and a direct articulation command (between +45 and -45 degrees). A supervisory module will evaluate these signals against the defined interlocks and the environmental constraints defined within the system. The supervisor exposes an enable for the propulsion control. These also impose the machine-level limits, such as a maximum vehicle speed to which the vehicle is limited to, and the retarder coordination for safety functions. Propulsion commands are ignored until this enable is asserted.

Sampling is uniform within the control layer. The supervisor runs at a slower rate than the propulsion controller. Typical practice places the supervision at tens of milliseconds and the propulsion at a few milliseconds. All the measurements from the mechanical environment are buffered and time-stamped before entry into the control computations.

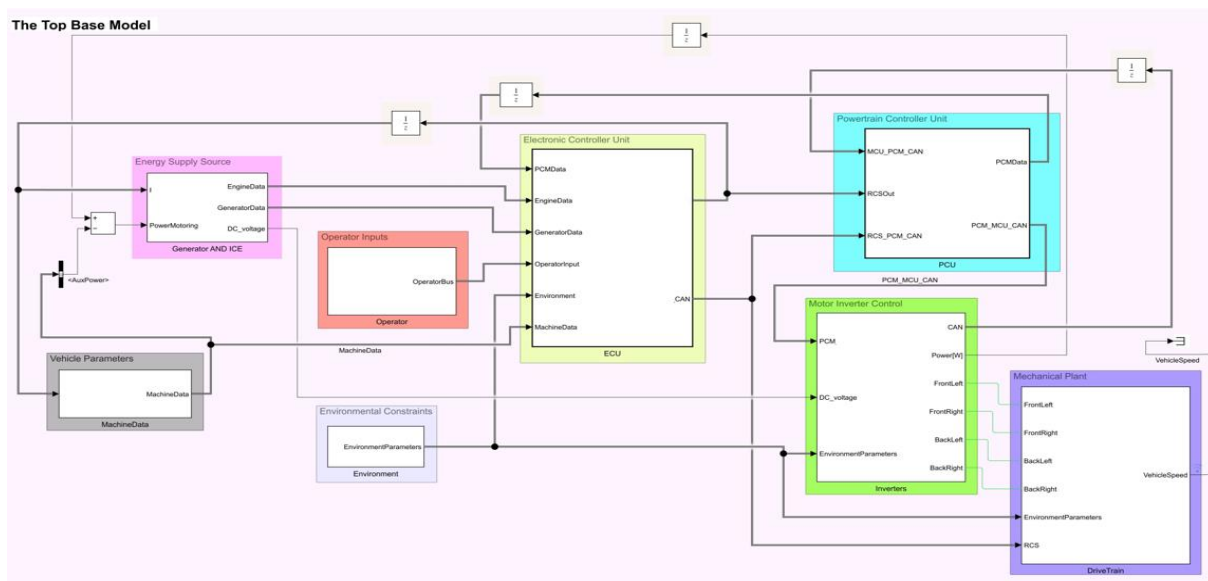


Figure 11: The Top-Level MATLAB/Simulink model of the MT66 eDrive

3.2.2 Signals presented to propulsion control and parameterization

An input assembly collects all the signals that are required for the propulsion computation. The per-wheel electrical speeds ω_i are read from the motor controller feedback pathway. The direct

current link voltage V_{DC} is computed from the high voltage sources and the bus sensors. The propulsion enables flag from supervision is passed through. A parameter defined within a bus will store the supply constants such as the pedal filter time constants, torque ceilings, thermal thresholds, and admissible speed bounds.

All assumptions follow the production practice. Electrical speed is represented in rad/s. All voltages are represented in volts. Torque is represented in newton meters. Limits are applied in the electrical torque domain upstream of the motor controller inner loops. Anti windup is implemented at the torque-request level to avoid integrator growth in lower-level loops in the plant model.

3.2.3 Propulsion computation from throttle to four torque references

The propulsion computation has three stages. The state-machine logic will request and grants the converters operation based on the supervision status and the measured link conditions. A torque chain will map the scalar throttle command to all four per-wheel torque references while enforcing all the other constraints. An output packager will emit network messages to all four motor controllers individually and publish diagnostic flags.

From Pedal to Normalized Demand: Let $u_{thr} \in [0, 1]$ denote the normalized throttle command. Let $s_{dri} \in \{-1, +1\}$ denote the selected direction. The average electrical speed can be given as:

$$\bar{\omega} = \frac{1}{4} \sum_{i \in \{FL, FR, RL, RR\}} \omega_i \quad (3.4)$$

The speed scheduled gain $g(\bar{\omega})$ (a lookup function of $\bar{\omega}$) scales the response near standstill and when it's near high speed. The dimensionless normalized request is given as:

$$r_n = s_{dri} g(\bar{\omega}) u_{thr} \quad (3.5)$$

This variable expresses the demanded propulsive effort independent of the absolute hardware limit.

Pedal conditioning and mapping provide the smoothing of the accelerator signal input. First, the debouncing imposes a short time requirement before a sudden change is accepted and a small deadband around zero will remove the sensor jitter and prevent any unintended creep. Next, a low pass filter will remove the high frequency content that the inverters and motors cannot follow. The cutoff frequency is chosen to keep the response quick while avoiding the formations of torque chatter at very low speed.

Constraint enforcement: A feasibility filter is a deterministic limiting block that maps the normalized request r_n to a feasible r_n^* by enforcing speed, temperature, voltage, power-margin and rate constraints consistent with the 750 V DC link and machine limits. The normalized request is bounded $r_n \in [-1, 1]$, near standstill q dead zone is applied so that $|r_n| < \epsilon$ is floored

to 0 to avoid chatter, otherwise $|r_n|$ is capped at 1 (and -1 for reverse). It provides overspeed ceilings based on admissible electrical speed of the machines as a function of V_{DC} and temperature. It reduces the demand when the DC link bus voltage V_{DC} falls below a threshold to preserve the bus stability and when the total generator power margin is low. If a machine-specific speed limit is active, the filter will apply an additional ceiling. The output r_n^* lies inside a speed and power dependent envelope that is consistent with the 750 V link and the four 420 kW traction machines, rate limiting is enforced on the requested torque slope [23].

Brake blending and Thermal derating: When the service brake is applied, that is when the inputs from the operator u_{brk} is above zero, then controller immediately downscales the propulsion with a smooth factor $f_{brk}(u_{brk}) \in [0,1]$. During operation when the operator applies smaller brake inputs, then it only trims the total drive torque to the wheels. Whereas when a full brake is applied, it drives the propulsion toward zero so that the braking authority will become absolute. At the same time, each wheel will evaluate continuously its thermal margin and maps it to $\alpha_i \in [0,1]$, where lower values will reflect hotter hardware or tighter limits. The per-wheel propulsion request is then given as:

$$\widetilde{r}_{n,i} = \alpha_i f_{brk}(u_{brk}) r_n^* \quad (3.6)$$

This formulation guarantees that the electric drive will never function over the brake signal, prioritizing the stopping performance, and protects the components under high temperature. The rate limits and filtering will smooth the transition into and out of braking, so that the vehicle will remain stable during creep, hill hold, and at brake release.

Per-wheel shaping for Drivability: Each wheel will shape its own torque request with a first-order filter and demanding that needs to the inverter controller. Let ω_i be the electrical speed measure at wheel ‘ i ’. The filter time constant depends on the sign of power flow and on ω_i . Near zero speed, the generating time constant is chosen larger than the motoring time constant. This will prevent the abrupt reversals of torque as the request will cross through zero. The output of the filter will be a continuous and bounded reference T_i^{ref} . These references will coordinate cleanly with the spring-applied, hydraulically released brakes during crawl and at stop, which helps in avoiding lurch and chatter at very low speeds.

Traction constraints acting as Gates: The auxiliary logic will build the traction constraints that will gate both the feasibility filter and the per-wheel shaping. Ratios of the measured electrical speeds across the left–right and the front–rear are filtered to create an allowable speed window for each of the wheel. Each window will define the minimum and the maximum electrical speeds that are consistent with the surface conditions and the current geometry. A cross-side coupling emulation will then prevent one side from free-spinning while the opposite side will be torque-starved. These constraints are not capable of synthesizing a yaw moment neither is able distribute the torque for handling. They bound and smooth a single propulsion request when grip varies, which stabilizes the driveline under split- μ conditions.

Output packaging for the Motor Controllers: The controller will collect the four references T_i^{ref} together with the operation status and any active derating flags. It will then serialize these quantities into network messages that are addressed to each of the four motor controllers. Each

motor controllers are designed to continuously perform a local checks on the electrical speed, direct-current link voltage, and the device temperature, and then tracks its torque references with an inner control loop. The energy to the system is drawn from the shared 750 V direct-current link supplied by the two generator units and delivered to the four wheel-end motors [1].

3.2.4 Driveline, ground contact and feedback closure

The mechanical plant in the top model contains the chassis, hub reductions, shafts, wheel end assemblies, and a tire–ground element. The longitudinal force at each of the wheel is reported as a function of slip and normal load. The normal load varies with the static weight distribution and with the load transfer during acceleration and braking. The model reads the individual wheel speeds and the speed about the center of the vehicle and returns them through the motor controller feedback pathway. The propulsion loop closes at the propulsion sampling rate.

The service brake system used in the model is brake by wire. The wet disc brakes are spring applied type and are hydraulically released. The brake pressure commands pass through a supervisory coordinator that ensures a consistent relation with the electric torque. When the brake tests or safety holds are active, the coordinator will set an inhibit which reduces or zeros the electric torque reference. The propulsion computation honors this inhibit.

3.2.5 Interactions among subsystems and emergent behavior

The human interface defines high level intent and is directly fed to the Electronic Controller Unit. The Electronic Controller Unit (ECU) checks these commands against all the pre-defined interlocks and the environmental conditions, then issues a propulsion enable command and defines any applicable machine limits dynamically. Only when the ECU is permissive, that means when all the required safety and operating checks are satisfied, then only the Powertrain Controller Unit (PCU) can execute the propulsion computation. The PCU maps the throttle input u_{thr} and the direction s_{dri} to the four-wheel torque references while enforcing the protections for the direct current link voltage V_{DC} , admissible electrical speed, torque-rate limits, thermal state, and the traction windows.

The motor controllers will translate those references into their corresponding phase currents using the inner torque loops. The mechanical plant then converts the electromagnetic torque to the tractive force through the gear reductions and finally the tires. The sensors provide measurements of the wheel speed, DC link voltage, temperature and articulation angle. These signals are sampled at the control rate to close the feedback loop. In this current plant configuration, the system only operates as a single input propulsion system without the intentional yaw moment generation. Vehicle acceleration and deceleration will follow the total demanded torque, bounded by V_{DC} , generator margin, thermal limits, and traction constraints.

Chapter 4

4. Methodology and Simulation

The project is done as a three-part control architecture for the torque vectoring of the centrally articulated in-wheel motor vehicle. Part one is designed to simulate the driver's steering intent into its corresponding physically bounded articulation target and closes an articulation-angle loop around the measured hinge state. Part two couples the articulation and the yaw dynamics to produce a feasible direct yaw-moment command that respects the operating envelope of the vehicle plant model. Part three distributes the total drive torque to the four in-wheel machines using the per-wheel normal-force cues together with the commanded yaw moment. The entire architecture is designed and simulated in the MATLAB/Simulink, and Simscape software with explicit units, explicit sampling, and a single source of truth for the tunable parameters. Block diagrams and equations are stated within the Methodology for traceability, whereas the code listings will appear in the Appendix and are cross-referenced in the procedure steps.

4.1. Software Environment and Model Configuration

Table 1: Configuration of Software Environment and Simulation Settings

Configuration Item	Specification (symbol/Value)	Technical note
Software	MATLAB/Simulink R2023b	
Physics toolboxes	Simscape Multibody; Simscape Electrical	Plant modeling (multibody + powertrain)
Model scope	Plant (multibody + powertrain)	Controller executed in discrete domain
Data management	Simulink data dictionary linked via Model Explorer	Single authoritative source for defined input parameters
Start time	0 s	
Stop time	inf (Infinity)	Enables long scenarios without reconfiguration
Solver	Fixed-step	
Solver algorithm	Auto (Discrete)	
Sample time	$T_s = 0.01$ s (10 ms)	Base rate for control tasks
Timing Specifications	Explicit per-block discrete sample times in control subsystems	
Logging configuration	Simulink data logs, SI units	Consistent units in all exported signals

4.2. Modeling and Simulation Workflow

The High-level diagram of the Torque Vectoring model is shown in Figure 12. The simulation procedure is organized in three parts:

- Steering-to-articulation.** Convert steering command to articulation angle/rate via the steering-hinge map and actuator model.
- Reference torques from normal forces.** Estimate per-wheel normal forces and distribute the driver-requested total torque proportionally, preserving the total.
- Torque vectoring.** Compute a feasible yaw-moment request and superimpose a left-right correction to obtain the final torque commands within traction/power/rate envelopes.

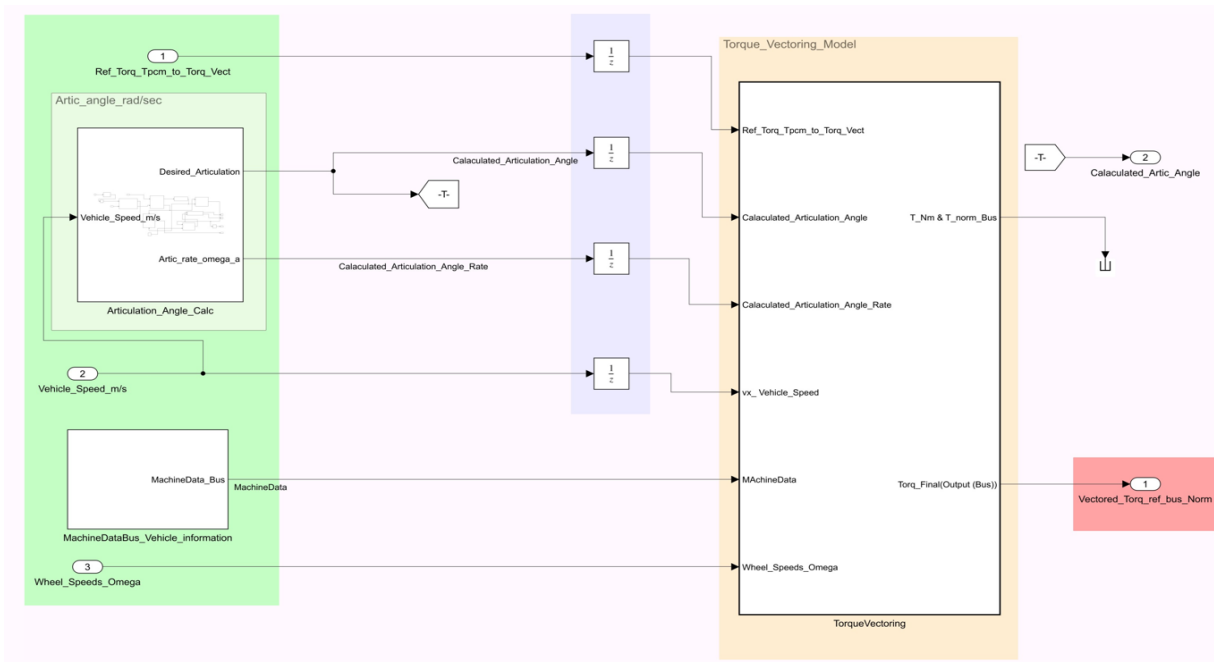


Figure 12: Torque-vectoring control architecture Simulink layout

4.2.1 Steering-to-articulation mapping and closed loop control

Figure 13 shows the simulated block diagram of the articulation mapping, the block runs at $T_s = 0.01$ s and converts the operator's steering input to an articulation angle command. Inputs are the steering wheel angle and rate at the CAN boundary, the vehicle speed at the vehicle CoG, and the measured articulation. The mapping computes a speed-dependent half-lock, a desired articulation angle and rate, and issues a feedback unitless controller output $u(t)$ to the plant. All detailed signal definitions, filtering, static/dynamic maps, limits, and PID tuning are specified in the subsequent subsections.

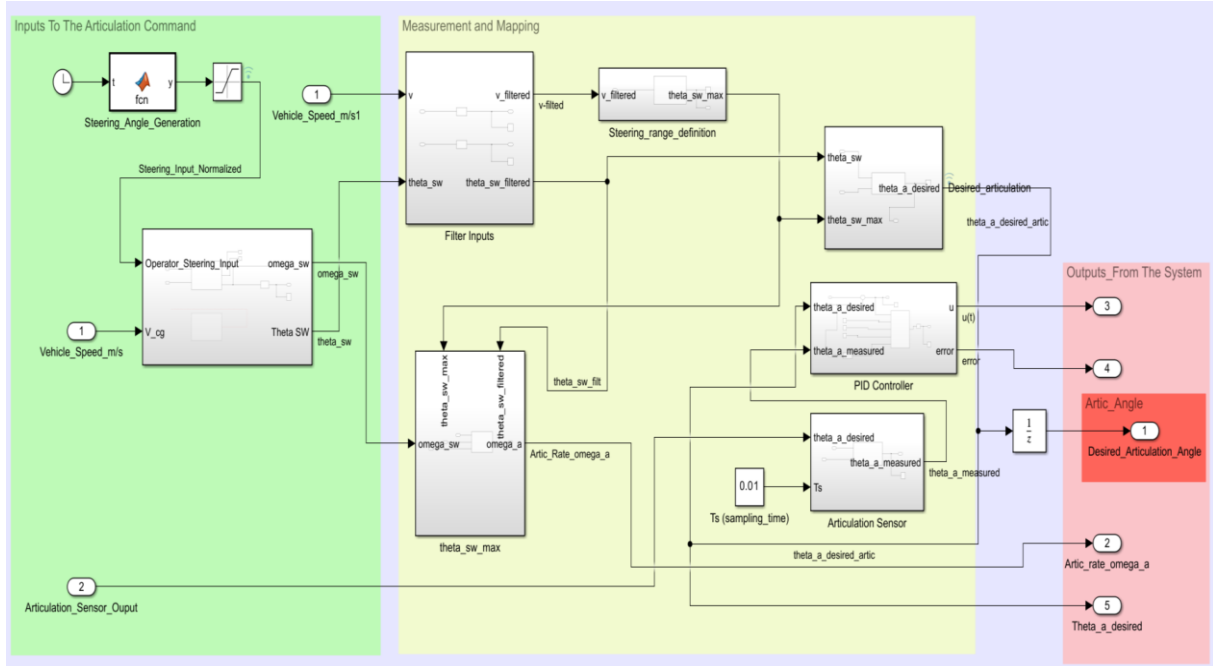


Figure 13: Speed-scheduled mapping from steering command to articulation reference

- I. **Inputs and input filtering:** As mentioned above the vehicle speed signal $v(t)$ will be provided from the top model. The upstream signal source is defined in meters per second, it will be converted to kilometer per hour before the steering computation. The speed signal and the steering angle are each passed through a discrete second-order Butterworth low-pass filter with a 45Hz cutoff at a 100Hz sampling rate. The filter coefficients are precomputed offline with the standard design calls and are then applied in two discrete filter blocks with the sample time T_s . The outputs from this are the v_{filt} and $\theta_{sw,filt}$. Filtering process will reduce the noise before normalization and differentiation, which could improve numerical behavior.
- II. **Steering half-lock as a function of speed:** The steering range uses the calibrated piecewise gain in relation to the speed of the vehicle, where the gain decreases linearly with speed up to 8 km/h and then holds constant. The normalized gain and the lock-to-lock range are computed exactly as described in Equations (4.1) to (4.3). The half-lock in radians is then obtained via Equation (4.4). This logic is implemented in a single MATLAB Function block that accepts the filtered speed in kilometers per hour and returns $\theta_{sw,max}(v)$ in radians. The function block will implement the calibrated gain law and scales it to its normalized form (0-1 value), linearly interpolates that to the lock-to-lock steering range, and converts the result to radians. The Appendix contains the function listing and a screenshot of the block mask.

$$Gain(v) = \begin{cases} 5.2 - 0.05v, & v \leq 8 \\ 4.8, & v > 8 \end{cases} \quad (4.1)$$

The constants 5.2 and 4.8 are calibration anchors chosen for an affine, speed-dependent steering gain $Gain(v)$ that maps via a normalized weight to prescribed lock-to-lock steering-wheel

ranges. On the Epiroc platform, the steering controller's stationary calibration uses $Gain = 5.2$ to reproduce the validated six-turn lock-to-lock steering-wheel geometry. Enforcing $Gain(0\text{km/h}) = 5.2$ and $Gain(8\text{km/h}) = 4.8$ yields $Gain(v) = 5.2 - 0.05v$ for $v \leq 8$, with a clamp at 4.8 for $v > 8$. This preserves the low-speed steering-wheel range while attenuating steering sensitivity across the low-to-moderate speed regime.

The normalized gain is: (4.2)

$$NG(v) = \frac{Gain(v) - 4.8}{0.4}$$

The lock-to-lock steering range in degrees is: (4.3)

$$T(v) = NG(v) * 2160 + (1 - NG(v)) * 2360.02$$

This means, the steering range from lock-to-lock at 0 km/h is 2160 degrees (six full rotations), but from speeds above 0 km/h till 8 km/h it linearly increases from 2160 to 2360 degrees, and beyond 8 km/h it remains constant at 2360 degrees.

The speed-dependent steering range in degrees is: (4.4)

$$\theta_{sw,max}(v) = \frac{T(v)}{2} * \frac{\pi}{180}$$

- III. **Static steering-to-articulation map:** The static mapping will convert the steering command to a desired articulation using a bounded sine law. First a normalized steering ratio $x(t)$ is generated, that is by dividing the filtered steering angle in degrees by the current half-lock expressed in degrees. This will provide us with $x = \pm 1$ at full steering in either direction (left or right), as in Equation (4.5). The desired articulation $\theta_{a,desired}(t)$ can be given as A_{max} multiplied by the sine of $\frac{\pi}{2}x(t)$, see Equation (4.6). A MATLAB Function block is used to implement this mapping. It takes $\theta_{sw,flt}$ in degrees, $\theta_{sw,max}$ in radians, and A_{max} in degrees for convenience at the mask as the inputs to the block. Inside the function block, A_{max} will be converted to radians, the half-lock is converted to degrees for a consistent ratio, the bounded sine is applied, and the output is returned in radians. The saturation inherent to the sine function will prevent any nonphysical articulation commands.

$$x(t) = \frac{\theta_{sw}(t)}{180/\pi(\theta_{sw,max}(v(t)))} \quad (4.5)$$

$$\theta_{a,desired}(t) = A_{max} \sin\left(\frac{\pi}{2}x(t)\right) \quad (4.6)$$

- IV. **Dynamic map for the articulation rate:** The analytic rate map relates steering-wheel rate to articulation rate. The expression is Equation (4.7). It multiplies the steering-wheel rate in radians per second by a cosine term that depends on the normalized steering and scales by the inverse of $\theta_{sw,max}(v)$. The implementation is carried in a function block which accepts the filtered steering angle in degrees, the steering-wheel rate in degrees per second, and the half-lock in radians. Inside the block the steering-wheel rate is first converted to radians per

second. The normalized steering $x(t)$ is formed and then the cosine term is evaluated, and finally the scale factor is applied to the resultant. The output $\omega_a(t)$ is used for diagnostics and can support future feed-forward.

$$\omega_a(t) = \frac{\pi^2}{8\theta_{sw,max}(v(t))} \cos\left(\frac{\pi}{2}x(t)\right) \frac{\pi}{180}(\omega_{sw}(t)) \quad (4.7)$$

- V. **Sensing, control and integration for articulation loop:** The measured articulation $\theta_{a,measured}(t)$ is generated by a sensor model aligned with the controller sampling with the same sampling period $T_s = 0.01$. A five-sample circular buffer gives a fixed delay of 0.05 s. Zero-mean noise with a standard deviation of 0.2° , converted to radians, is added to the output. Persistent internal state enforces one-sample-in and one-sample-out behavior at T_s . If the sensor output remains identically zero after the start-up, then the typical cause would be vectorized inputs, a sample-time mismatch, or having the reset line being held high.

$$e(t) = \theta_{a,desired}(t) - \theta_{a,measured}(t) \quad (4.8)$$

$$u(t) = K_p e(t) + K_I \int e(t) + K_D \frac{de(t)}{dt} \quad (4.9)$$

The controller regulates articulation with a discrete PID in parallel form with control error is $e(t)$ at T_s . The output $u(t)$ follows the standard PID law with a first-order derivative filter. Anti-windup is enabled whenever saturation follows the controller. During the initial bring-up, the controller output will be symmetrically limited and later can be aligned with the actuator's physical bounds based on the on-bound vehicle parameters. A practical initial tuning is $K_p = 3$, $K_I = 5$, and $K_D = 0.45$. Increase K_p to quicken response, raise K_I to remove bias, and use K_D to reduce overshoot. If the derivative path amplifies the noise, then increase the derivative filter's time constant at the same time reduce its filter coefficient. A reset clears the integrator when $|\theta_{a,measured}|$ falls below a small threshold ε . This reset is issued as a short positive-edge pulse, and the same Boolean can momentarily zero the steering command to re-center the request.

The resultant $\theta_{a,desired}$ per sample is taken as the final output from the system as the Articulation Command in radians, that is $\theta_{a,desired}(t) = \text{Articulation Angle } \phi'$.

4.2.2 Torque vectoring from normal forces

The preceding design part established a speed dependent steering map, a bounded articulation target. The present section will explain the base for one to four torque splitters with a torque-vectoring subsystem that respects the Normal Forces based on load transfer and, when enabled, adds a basic corrective yaw moment derived from the articulation state from the previous part. The simulated block diagram of this is shown in Figure 14. All symbols are defined on first use, and the units follow SI.

- I. **Role and interfaces of the torque-vectoring subsystem:** The Torque Vectoring subsystem receives the commanded total drive torque reference T_{PCU} in newton meters from the Power Train Unit of the top-level plant model, the articulation angle Φ in radians, the optional articulation rate $\dot{\Phi}$, and the vehicle speed about the center of gravity V_{CG} in meters per second as the inputs to the system. A structured input bus Machine Data will supply all the parameters that are required by the subsystem and the load-transfer model. The field parameters used are the mass ' m ', gravity ' g ', center-of-gravity height ' h_{CG} ', the CG-to-axle distances for front and rear ' a ' and ' b ', the axle to hinge distance at the front and rear as a_h and b_h , the front and rear track widths w_f and w_r , and optionally the front and rear roll stiffness values $K\phi_f$ and $K\phi_r$.

The outputs are the four wheel torques $\{T_{FL}, T_{FR}, T_{RL}, T_{RR}\}$, ordered front-left, front-right, rear-left and rear-right. The subsystem is defined as an atomic unit at the controller rate T_s , because in the on-board vehicle the rate should be compatible with the State-Machine's rate. The sign conventions are standard, the x-axis points towards forward, the y-axis points to the vehicle's left, and a positive articulation Φ corresponds to a left turn. A positive lateral acceleration a_y therefore shifts vertical load to the right side.

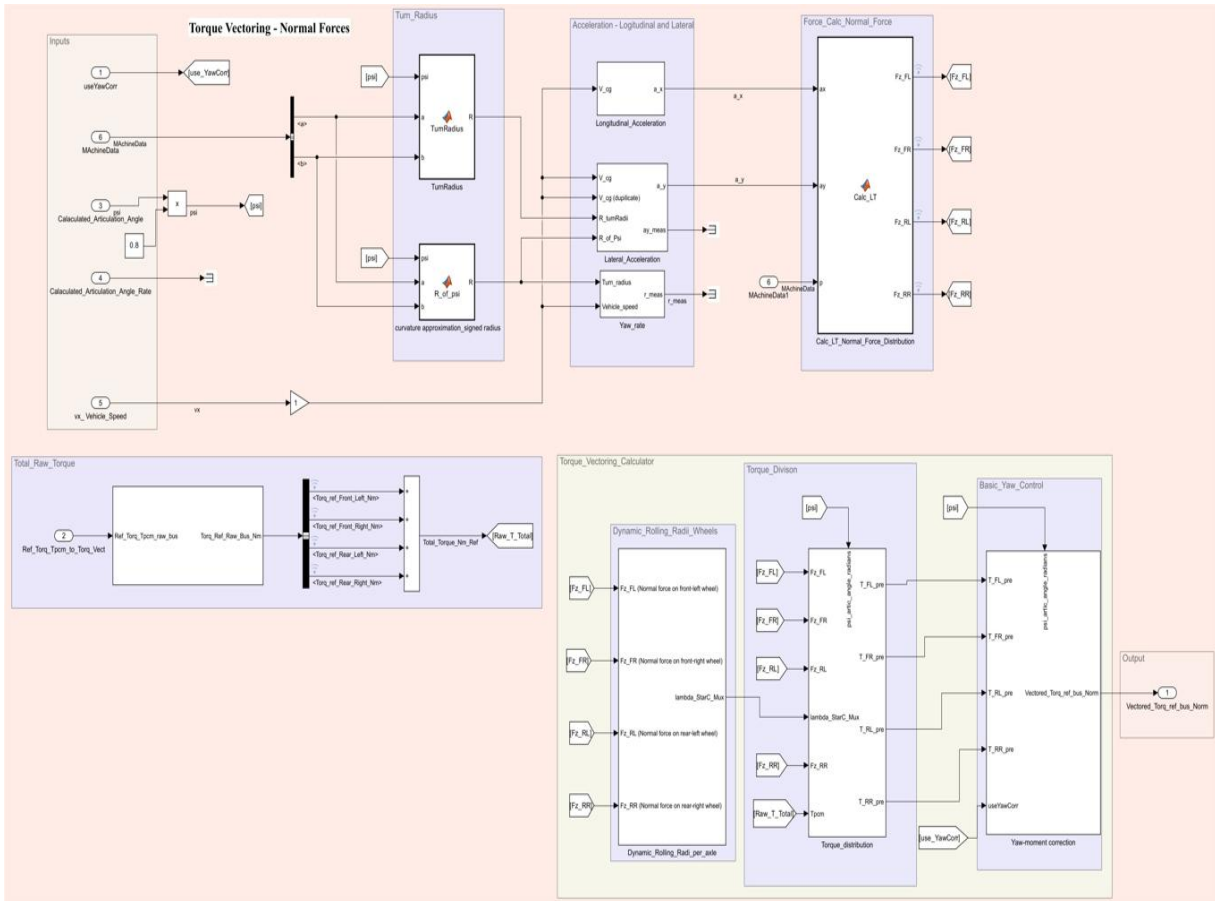


Figure 14: Torque distribution by normal-load weighting simulation layout

- II. **Kinematic quantities required by the model:** Two accelerations drive the load-transfer model, the longitudinal acceleration and the lateral acceleration. The longitudinal acceleration is given as $a_x = \frac{dV_{cg}}{dt}$. This is computed with a discrete derivative at T_s and is filtered consistently with the rest of the control layer, as shown in the Simulink diagram Figure 15. The lateral acceleration is calculated from an articulation-based turning radius, as this is a centrally articulated vehicle. For a centrally articulated vehicle, the turning-radius relation uses the hinge to axle distances, can be given as:

$$R(\Phi) = \frac{a_h + b_h \sec\Phi}{\tan|\Phi|} \quad (4.10)$$

A small positive lower bound is imposed on the denominator to prevent the cause of near-zero divisions. The lateral acceleration is then computed as a_y and measured- a_y as shown in the Figure 16 using the basic equation:

$$a_y = \frac{V_{cg}^2}{R(\Phi)} \quad (4.11)$$

These definitions provide a_x and a_y with the correct signs for the subsequent load-transfer calculation.

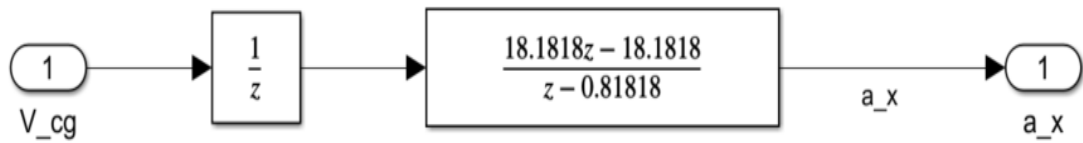


Figure 15: Computation of longitudinal acceleration a_x

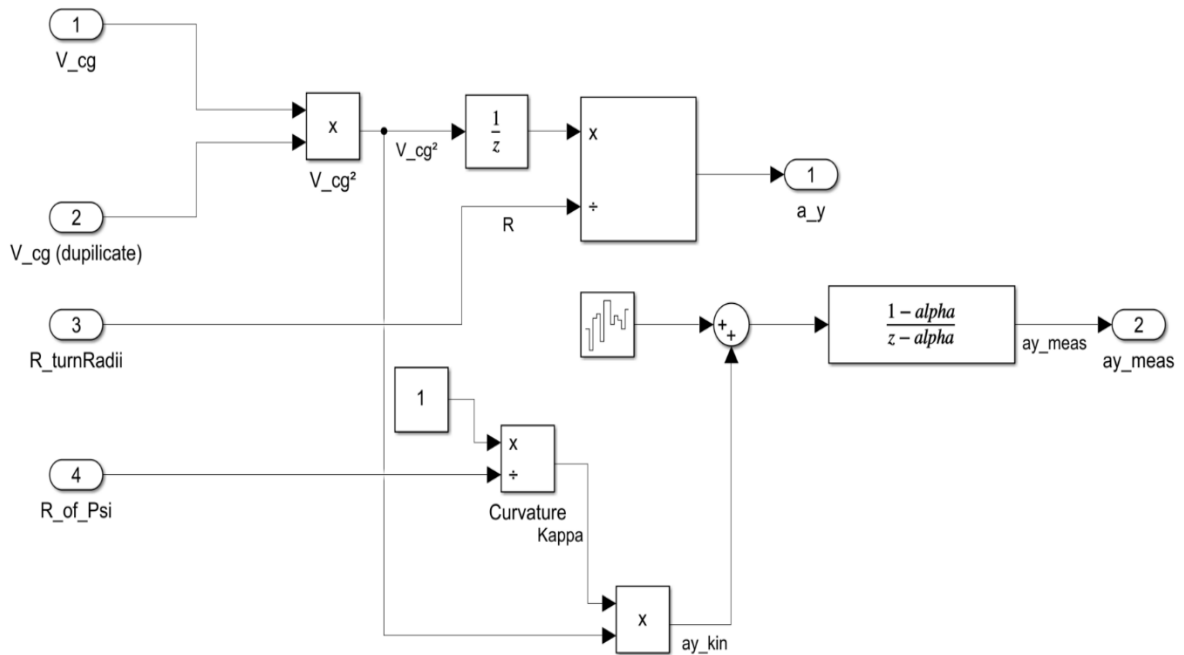


Figure 16: Computation of lateral acceleration a_y and measured a_y

- III. **Load transfer and per-wheel normal forces:** Each of the wheels normal forces combines a static component with a dynamic component from acceleration and roll. The wheelbase is given as the sum of $L = a_h + b_h$. Then the static per wheel loads can be given as:

$$F_{zf0} = \frac{1}{2}mg \frac{b}{L} \quad \& \quad F_{zr0} = \frac{1}{2}mg \frac{a}{L} \quad (4.12)$$

A positive a_x will shift the load towards the rear. With the rigid body approximation, the total front to rear load transfer is $mh_{CG}a_x/L$. This will be distributed equally to the two wheels on each of the axles, which yields the per-wheel increments given by:

$$\Delta F_f^x = -\frac{1}{2}mh_{CG}a_x/L, \quad \text{at the front} \quad (4.13)$$

$$\Delta F_r^x = +\frac{1}{2}mh_{CG}a_x/L, \quad \text{at the rear}$$

The lateral transfer is driven by the roll moment $M_y = mh_{CG}a_y$. If exact roll stiffness values are available, then the axle split becomes $\beta_f = K\phi_f/(K\phi_f + K\phi_r)$. This information is not available, hence it will be estimated as $\beta_f = b/L$ which preserves the static front and the rear proportion. The axle-level roll moments are $M_{yf} = \beta_f M_y$ and $M_{yr} = (1 - \beta_f)M_y$. Now dividing these by their corresponding track widths will give the total left-to-right transfer on each axle. The per-wheel roll-transfer terms adopts the usual convention that a positive a_y increases load on the right wheels:

$$\begin{aligned} \Delta F_{fL}^y &= -\frac{1}{2}\frac{M_{yf}}{t_f} \quad \& \quad \Delta F_{fR}^y = +\frac{1}{2}\frac{M_{yf}}{t_f}, \text{ for the front axle wheels} \\ \Delta F_{rL}^y &= -\frac{1}{2}\frac{M_{yr}}{t_f} \quad \& \quad \Delta F_{rR}^y = +\frac{1}{2}\frac{M_{yr}}{t_f}, \text{ for the rear axle wheels} \end{aligned} \quad (4.14)$$

The four normal forces follow by summation and clipping to non-negative values to represent the possible wheel lift as:

$$\begin{aligned} F_{zFL} &= \max(0, F_{zf0} + \Delta F_f^x + \Delta F_{fL}^y), \\ F_{zFR} &= \max(0, F_{zf0} + \Delta F_f^x + \Delta F_{fR}^y), \\ F_{zRL} &= \max(0, F_{zf0} + \Delta F_r^x + \Delta F_{rL}^y), \\ F_{zRR} &= \max(0, F_{zf0} + \Delta F_r^x + \Delta F_{rR}^y) \end{aligned} \quad (4.15)$$

In the Simulink plant model, this sequence of computations is defined within a MATLAB Function block that returns the outputs $\{F_{zFL}, F_{zFR}, F_{zRL}, F_{zRR}\}$. The corresponding list is defined as ‘Calc_LT’ in the layout and is provided in the appendix.

- IV. **Tire rolling radius under vertical load:** The deflection of tires is a common effect on these vehicles, and the rate of deflection depends on the load. These deflections reduce the effective rolling radius of the tires, a quasi-static estimate is retained for these diagnostics. The front axle load is $F_{z,front} = F_{zFL} + F_{zFR}$, with an analogous definition the rear axle. With a representative for per-tire vertical stiffness k_{tire} in newtons per meter, the axle level deflections are $F_{z,front}/2k_{tire}$ and $F_{z,rear}/2k_{tire}$. Subtracting these dynamically from the static radius $r_{static} = 1.013\text{m}$ gives the effective radii of the front and rear wheels, this is dynamically calculated based on the Simulink diagram in Figure 17. The parameter k_{tire} is exposed as a block mask and the default value is taken as $3 * 10^6 \text{N/m}$, calculated based on the sag of the tires from the real vehicle.

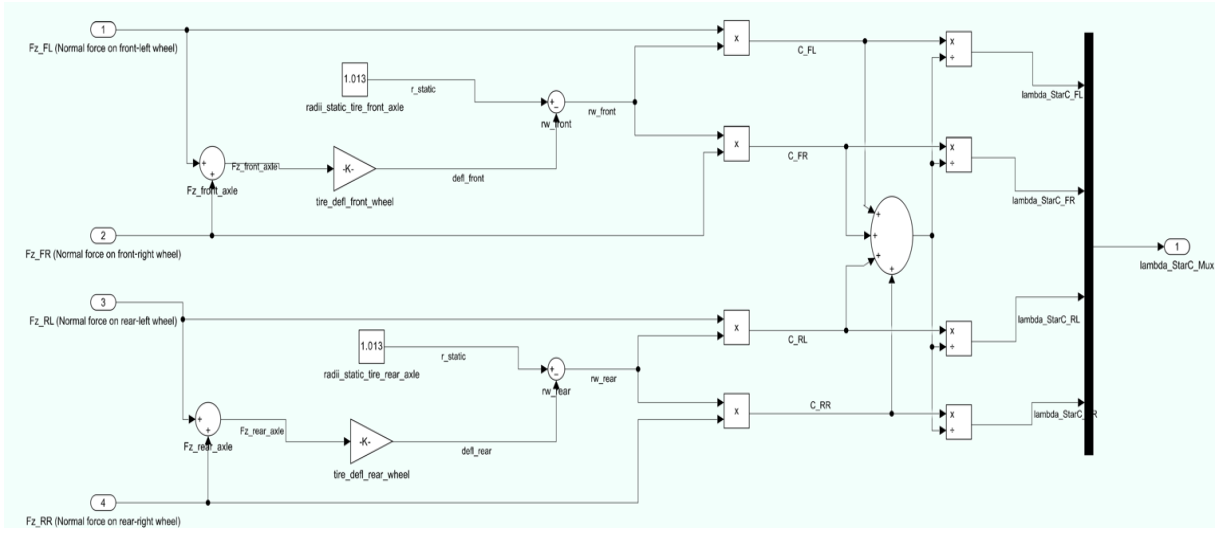


Figure 17: Effective rolling-radius estimation

- V. **Splitting total torque by normal-force share:** The load transfer information decides how the torque should be apportioned. The four normal forces are normalized by their sum, given as:

$$\lambda_i = \frac{F_{z,i}}{\sum_i F_{z,i}}, \text{ where } i = \{FL, FR, RL, RR\} \quad (4.16)$$

Which will yield fraction that sums to approximately one unless a wheel has been lifted and been clipped by zero. The split then blends the equal distribution with load-proportional distribution under a single tunable scalar $k_{Fbias} \in [0, 1]$:

$$\alpha_i = (1 - k_{Fbias}) * 0.25 + k_{Fbias} * \lambda_i \quad (4.17)$$

$$T_{i,pre} = \alpha_i * T_{PCU} \quad (4.18)$$

The setting $k_{Fbias} = 0$ will give an equal split of torque, when $k_{Fbias} = 1$ produces a purely load-based torque distribution among each wheels. This gives a control to decide how

responsive the load based torque distribution has to be. The implementation is provided in the appendix as torque split.

- VI. **Dynamic payload and center-of-gravity shift:** The operations with variable payload requires a run-time update of the geometry and the mass. An upstream subsystem ‘PayloadCG’, performs this update by writing new values of a , b and m into the machine data bus before it goes to the Torque Vectoring plant. This ensures that the ‘Calc_LT’ block will always read the current values for the quantities they consume. The ‘PayloadCG’ resides at the high-level inside the Machine Data block. It accepts the nominal ‘MachineData’ bus and an external payload input m_{load} in kilograms, defined within the loadDynamics block in Figure 18. It computes the payload dependent center of gravity locations and the total mass, then assigns these values to the bus files a , b and m .

An internal function named ‘loadDynamics’, maps the payload to the center of gravity migration and the total mass through a linear law. These parameters are the empty-vehicle CG distance from the front axle $a_0 = 1.72\text{ m}$, and the full-load CG distance $a_1 = 3.158\text{ m}$, then the wheelbase $L = 5.826\text{ m}$. The empty mass is taken as $M_{empty} = 43,031\text{ kg}$ and the maximum payload is $M_{payload,max} = 66,000\text{ kg}$. The payload inputs are clamped to this range. Then the dynamic distance from CG to axle can be given as:

$$a_{CG} = a_0 + \frac{a_1 - a_0}{M_{payload,max}} * m_{load} \quad \& \quad b_{CG} = L - a_{CG} \quad (4.19)$$

where m_{load} is the current payload added to the vehicle. Then the total mass $m = M_{empty} + m_{load}$. For monitoring the axle level mass shares $m_{f,axle} = mb_{CG}/L$ and $m_{r,axle} = ma_{CG}/L$ are computed, together with their per wheel halves. The Bus Assignment block writes the computed values of a_{CG} over a , and b_{CG} over b and the updated mass in every sampling rate.

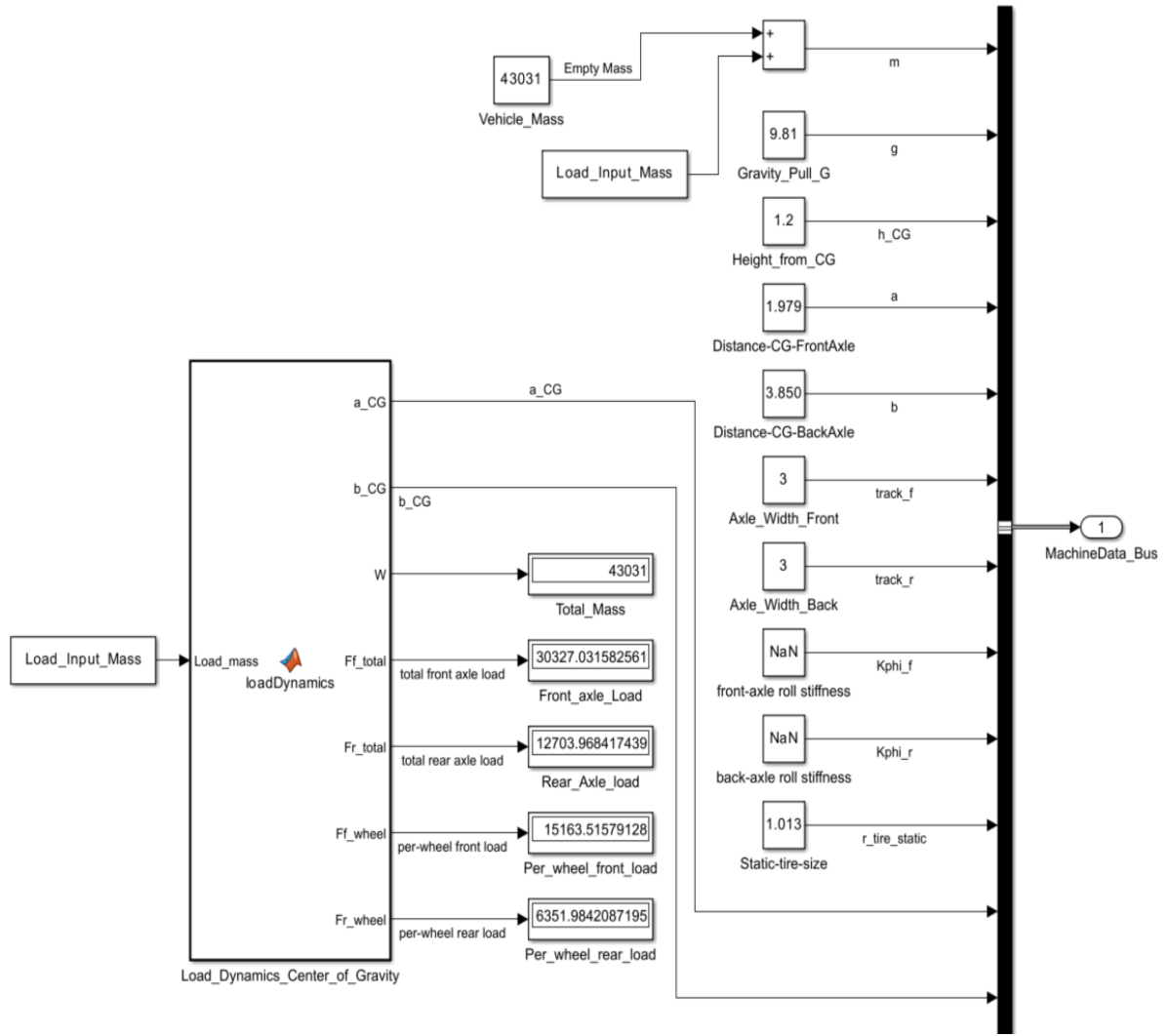


Figure 18: MachineData bus and the payload-geometry dynamics block

4.2.3 Allocation of wheel torques with yaw control and traction management

This section is the further extension of the normal-force based torque distribution into a complete allocator-based controller that tracks the commanded total drive torque and the achievable yaw moment, while enforcing the tire-friction limits, the actuator bounds, and the torque-rate constraints. The simulated block diagram of this third level of control is shown in Figure 19. The parameters, lookup tables and the bus definitions are all held within a single data directory as in Figure 20, to preserve everything in one source and to ensure that each symbol in the equations will map to a unique calibration in the model.

The data dictionary is saved as 'tv_params.sldd', which store all the parameters and its values like the vehicle geometry, masses, inertias, stiffnesses, bounds, and weights. There are two buses that are defined within directory as the 'TV_SensorsBus' and the 'TV_TorqueBus'. The 'TV_SensorsBus' carries the parameters velocity ' v_x ' in meters per second, the computed lateral acceleration ' a_y ' in meters per second squared, the yaw rate ' r ' in radians per second, the articulation ' ϕ ' in radians, the wheel speeds ' ω_i ' in radians per second, and the per-wheel

estimates of the vertical and the lateral forces $F_{z,i}$ and $F_{y,i}$ in newtons for $i \in \{FL, FR, RL, RR\}$. Whereas the 'TV_TorqueBus' carries the physical torques ' T_i ' in newton meters and their normalized forms in the unit interval.

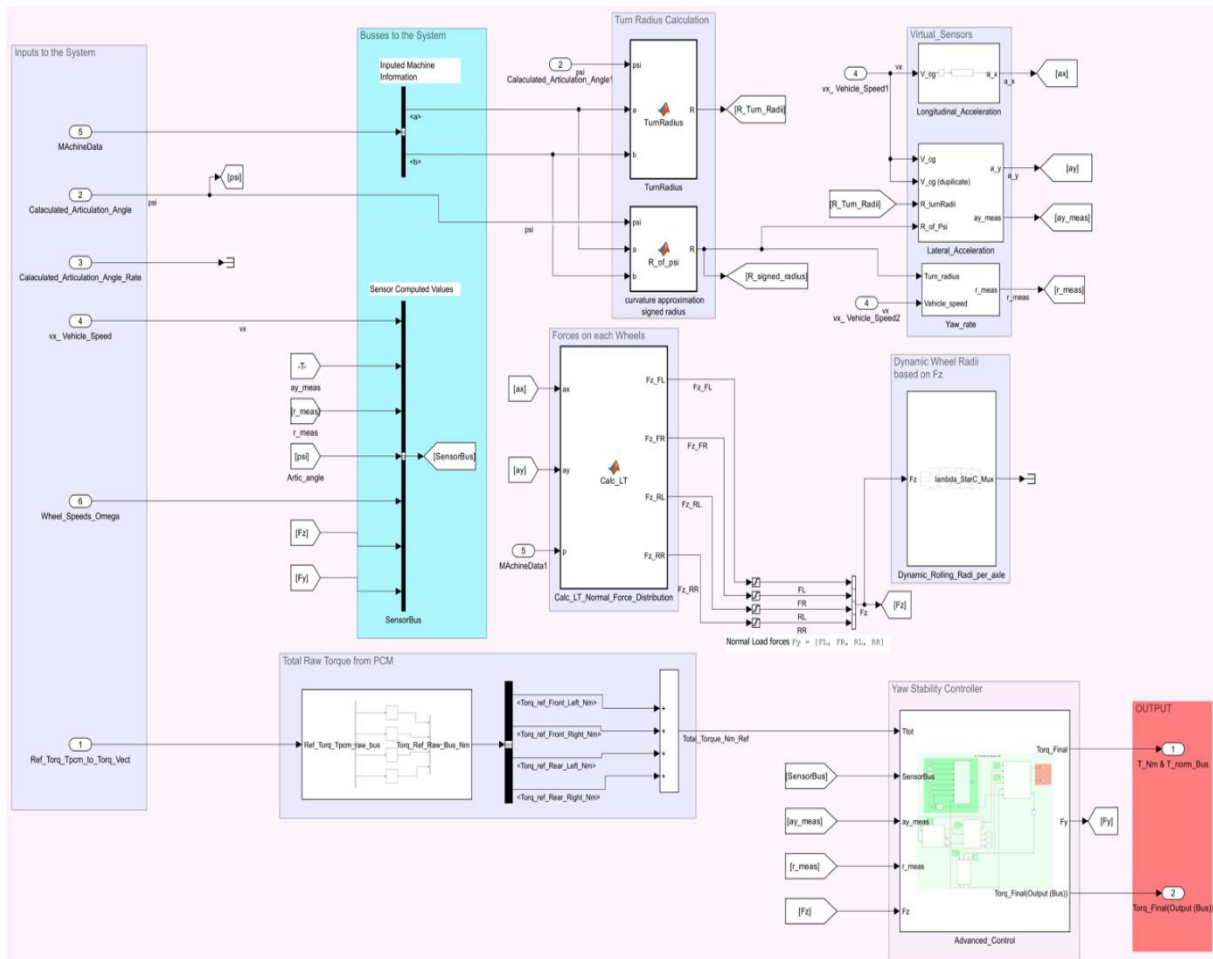


Figure 19: Integrated torque vectoring with yaw control - system layout

Name	Status	Value	DataType	Dimensions	Complexity	Min	Max	Unit	Argument	StorageClass	DataSource	tMod6	LastModified
Tr_EKF	0.01		double	[1 1]	real	[]	[]	s		Auto	tv_params.slidd	2025-0...	STORAGE
m	43000		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
m_min	43000		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
m_max	100000		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Iz	272189.193		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
a	1.979		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
b	3.85		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
tf	3		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Tr	3		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
kappa_max	0.1		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
sigTargets	[0.1 0.1 0.1 0.1]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
tmin	[0 0 0 0]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
dTmax	[500 500 500 500]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Wdiag	[1 1 1 1]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
rho	5		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
gamma	100		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
eta	50		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Q_EKF	[0.5 0.0 0.2]		double	[2 2]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
R_EKF	[0.5 0.0 0.02]		double	[2 2]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
epsi_vx	0.1		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
rate_trac_mps	0.1		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kx_bp_vx	[0 2 5 10]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kx_table	[0.6 0.4 0.22 0.12]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kb_bp_vx	[0 2 5 10]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kb_table	[0 0 0.25 0.4]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Mz_max_bp_vx	[0 2 5 10]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Mz_max_bp_ma	[0.3 0.7 1]		double	[1 3]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Mz_max_table	[100000 156000 260000 100000 145000 234000 85000 125000 200000 72000 100000 165000]		double	[4 3]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kdps_table	[150 400 700 900]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kx_LUT	[0.6 0.4 0.22 0.12]		auto	[1 4]		[]	[]			Auto	tv_params.slidd	2025-0...	abing
Kb_LUT	[0 0 0.25 0.4]		auto	[1 4]		[]	[]			Auto	tv_params.slidd	2025-0...	abing
Mz_max_LUT2D	[100000 156000 260000 100000 145000 234000 85000 125000 200000 72000 100000 165000]		auto	[4 3]		[]	[]			Auto	tv_params.slidd	2025-0...	abing
TV_SensorsBus											tv_params.slidd	2025-0...	abing
TV_TorqueBus											tv_params.slidd	2025-0...	abing
Cf	7.28E+6		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Cr	8.62E+6		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Rv	[0.965 0.965 0.965 0.965]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Tmax_base	[132500 132500 132500 132500]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Pmax	[440000 440000 440000 440000]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Pbus_max	700000		double	[1 1]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kps_bp_vx	[0 2 5 10]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing
Kps_table	[0 0 0 0]		double	[1 4]	real	[]	[]			Model default	tv_params.slidd	2025-0...	abing

Figure 20: Parameter datastore tv_params.slidd in Simulink Model Explorer

Other defined parameters includes the geometry and inertia, schedulers and limits, QP weights and damping. Inertia I_z is taken as $2.7219 \times 10^5 \text{ kg.m}^2$ based the mass, geometry and track width. Linear axle cornering stiffness for the estimator is taken as $C_f = 7.28 \text{ MN/rad}$ and $C_r = 8.62e6 \text{ MN/rad}$. For the Extended Kalman Filter, the process-noise covariance on lateral velocity v_y and yaw rate r can be given as, $Q_{EKF} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.2 \end{bmatrix}$. And the measurement-noise covariance of lateral acceleration a_y and yaw rate r_{meas} can be given as $R_{EKF} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.2 \end{bmatrix}$. Exact values can be decided based on the actual IMU log file information from the vehicle. Electrical and torque limits are defined with motor power caps $P_{max} = [420000, 420000, 420000, 420000] \text{ W}$ and total DC bus cap as $P_{bus,max} = 660 \text{ kW}$. Same way the bounds for torque minimum and maximum is defined.

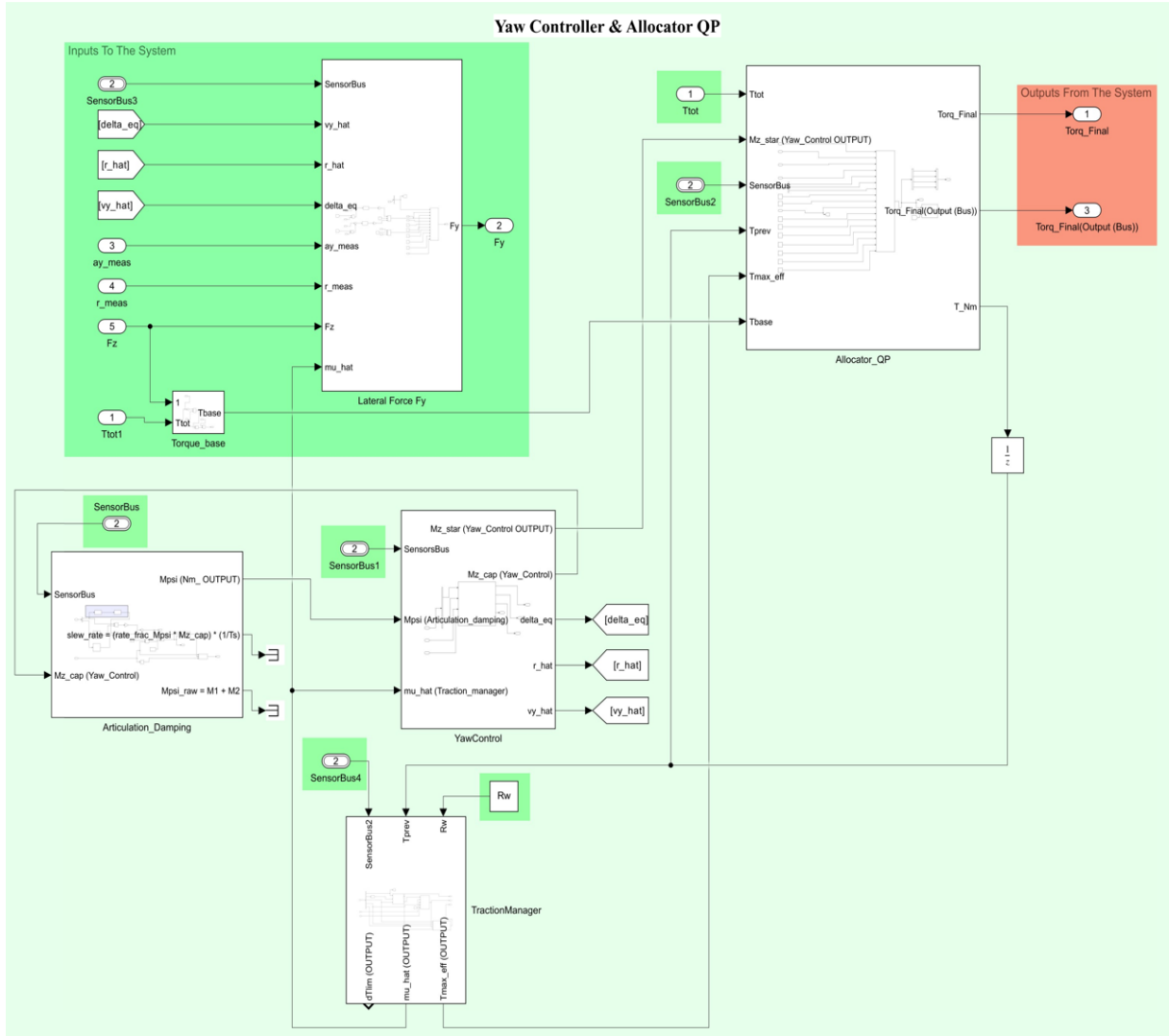


Figure 21: Yaw-Moment Controller block

I. Yaw control and desired yaw moment M_z^*

Reference yaw rate from articulation: In the preceding part equation (4.10) established the computing of turn radius $R(\Phi)$, with a small positive denominator floor for $|\Phi|$ near zero. The yaw-moment controller and torque allocator used in this section are summarized in Figure 21. The ideal yaw rate can be given as:

$$r_{ref} = \frac{v_x}{R(\Phi)} \quad (4.19)$$

Capping the curvature by $|k| \leq k_{max}$ with $k = 1/R(\Phi)$ enforces $|r_{ref}| \leq k_{max}v_x$. A discrete derivative $\dot{r}_{ref}$ provides the feedforward yaw moment $M_{z,ff} = I_z \dot{r}_{ref}$.

Estimation of the sideslip and filtered yaw rate is summarized in Figure 22. The Extended Kalman filter estimates $x = [v_y \quad r]^T$ using a continuous-time bicycle model with the

center of gravity geometry a , b and an equivalent steer $\delta_{eq} = \arctan((a+b)/R(\Phi))$. With $v_x = \begin{cases} \epsilon_{vx}, & \text{if } v_x < \epsilon_{vx} \\ v_x, & \text{otherwise} \end{cases}$, where ϵ_{vx} is just a speed floor of value 0.10 m/s.

$$F_y^f = C_f \left(\delta_{eq} - \frac{v_y + ar}{v_x} \right), \quad F_y^r = -C_r \left(\frac{v_y - br}{v_x} \right) \quad (4.20)$$

$$v_y = \frac{F_y^f + F_y^r}{m} - v_x r, \quad \dot{r} = \frac{aF_y^f + bF_y^r}{I_z} \quad (4.21)$$

And the measurements $y = [a_y \quad r]^T$ with $a_y = \dot{v}_y + v_x r$. The block runs at T_s , with Q_{EKF} and R_{EKF} from the directory. The sideslip estimate is given by $\hat{\beta} = \arctan(\hat{v}_y / \max(v_x, \epsilon_{vx}))$. The function listings for the transitions and the measurement model and the Jacobians for the EKF are provided in the appendix.

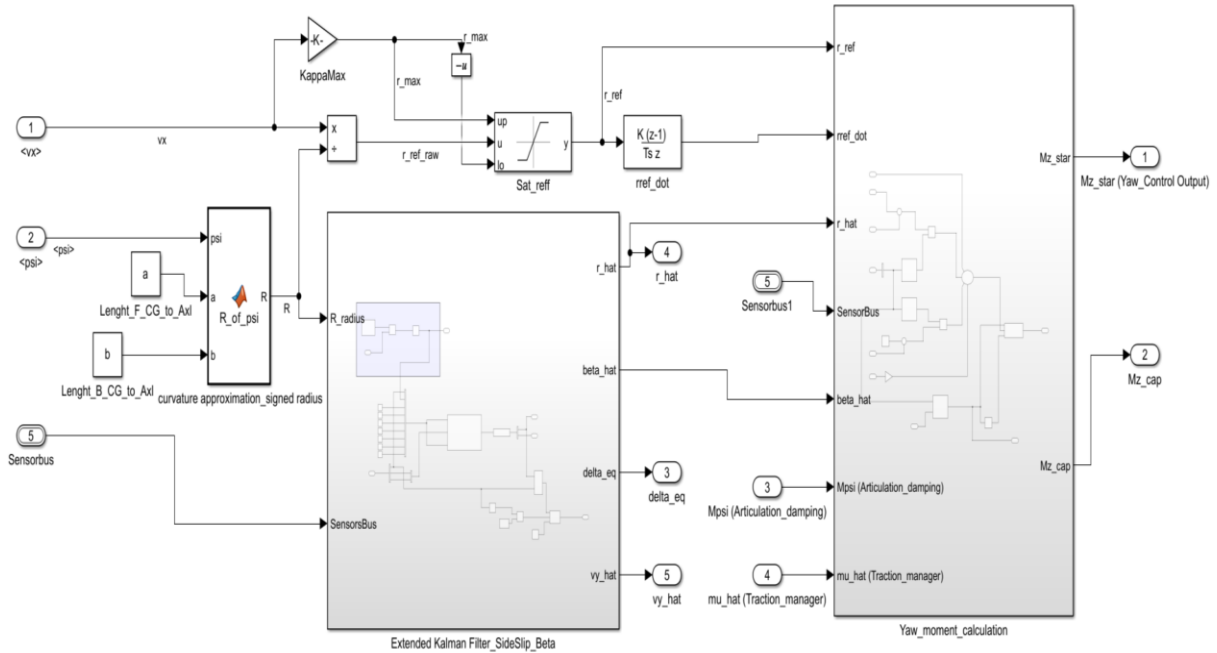


Figure 22: Yaw-state estimation with EKF and computation of yaw-moment command

Yaw-moment composition and feasibility capping: The yaw-rate and sideslip feedback combine with the feedforward, given by:

$$M_{z,raw} = K_r(v_x)(r_{ref} - \hat{r}) + K_\beta(v_x)(-\hat{\beta}) + I_z r_{ref} + M_\phi \quad (4.22)$$

Here, $K_r(v_x)$ and $K_\beta(v_x)$ are yaw-rate and sideslip gains. The desired yaw moment request is limited by a speed and friction dependent envelope, given by:

$$M_z^* = \text{sat}(M_{z,raw}; -M_{z,max}(v_x, \hat{\mu}), M_{z,max}(v_x, \hat{\mu})) \quad (4.23)$$

Which prevents any errors before allocation.

II. Articulation damping and stabilizing moment M_ϕ

The central articulation process is prone to lightly damped oscillations at moderate speeds. The implementation of this articulation-damping controller is shown in Figure 23. A modest stabilizing moment is applied here using the articulation angle Φ and filter rate. The law can given as:

$$M_\phi = -K_\phi(v_x)\Phi - K_{d\phi}(v_x)\dot{\Phi}_{filt} \quad (4.24)$$

Where $K_\phi(v_x)$, $K_{d\phi}(v_x)$ are damp schedules with brake points in m/s defined in the directory. The articulation rate $\dot{\Phi}$ can be generated by a discrete derivative at the controller with step $T_s = 0.01$ s. By using a first-order discrete low-pass filter with a cutoff near 2 Hz generates $\dot{\Phi}_{filt}$, which will remove the sensor noise without adding any noticeable phase lag at the frequencies of interest. To prevent any abrupt changes, a per-step rate limiter will scale the allowable change in M_ϕ as a fixed fraction of the current yaw-moment envelope. A final saturation at $\pm M_{z,max}$ will keep the damping action consistent with the feasible yaw authority that is already defined for the controller. The signal M_ϕ enters the yaw-moment sum before the last saturation stage so that all contributions will be capped together.

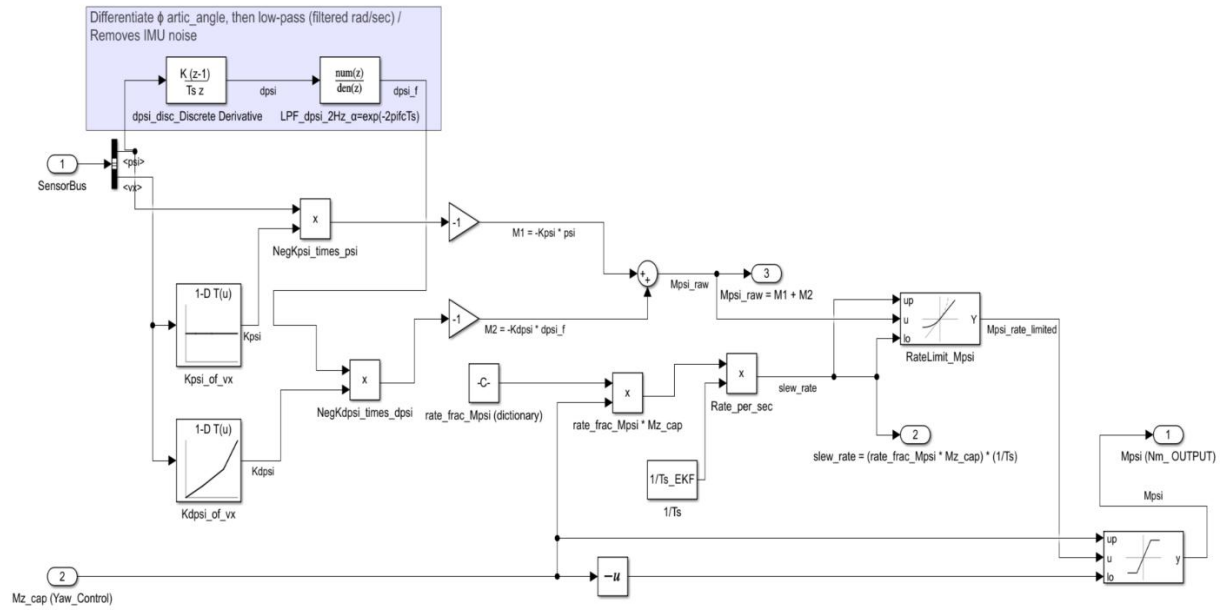


Figure 23: Articulation-damping block and hinge damping moment M_ϕ

III. Traction manager for slip regulation, friction estimation and torque ceilings

Per-wheel kinematics and slip control: The traction manager that implements the PI slip regulation and torque correction ΔT_i is shown in Figure 24. For estimating longitudinal ground speed on each wheel, we use standard yaw kinematics, that is:

$$v_{xi} = v_x - y_i r, \quad y_i \in \left\{ \frac{w_f}{2}, -\frac{w_f}{2}, \frac{w_r}{2}, -\frac{w_r}{2} \right\} \quad (4.25)$$

The guarded slip ratios can be given as:

$$k_i = \frac{R_{w,i} \omega_i - v_{x,i}}{\max(v_{x,i}, \epsilon_{vx})}, \quad (4.26)$$

With this mild clipping, it helps to avoid spurious extremes at very low speed. Each channel will run a discrete PI regulator on the error $e_{k,i} = k_i^* - k_i$. The output ΔT_i tightens the local torque ceiling rather than the commanding torque directly. Anti-windup will protect the integrators under limits, a per-step rate cap avoids any form of jerks, and hard bounds will keep the ΔT_i well regulated.

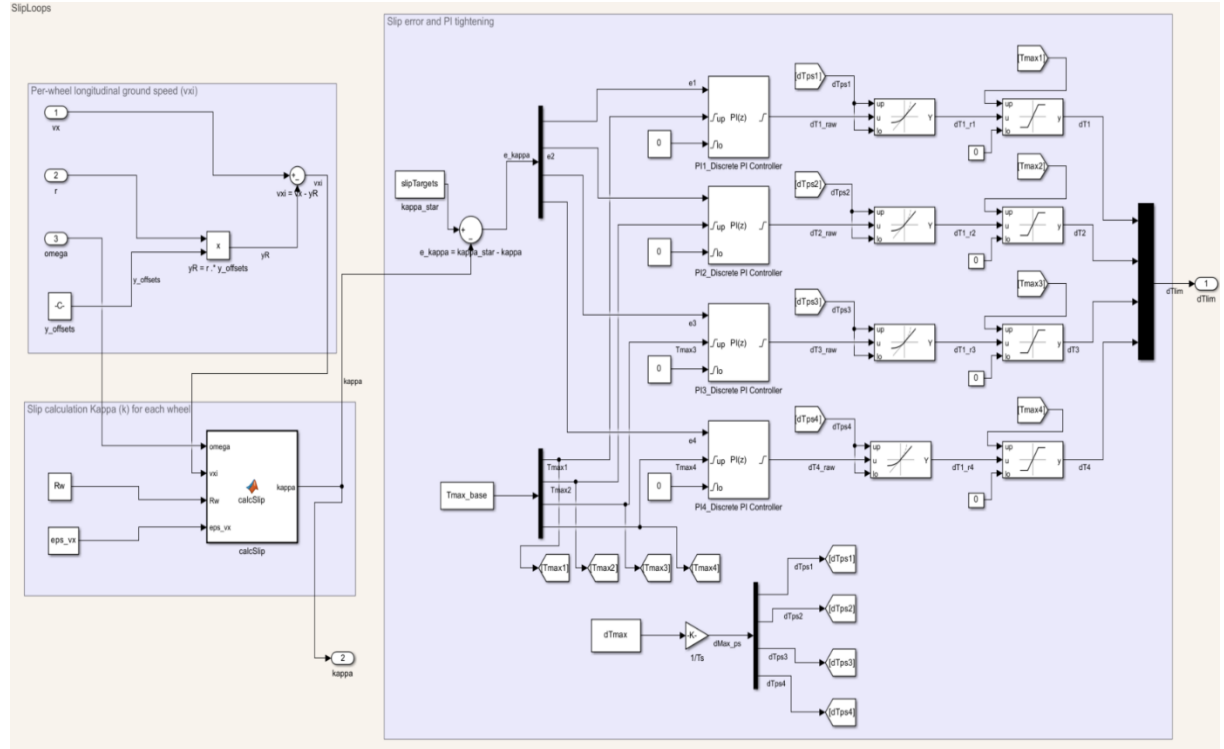


Figure 24: Traction-manager slip regulation and torque correction ΔT_i

Friction estimation: The longitudinal forces are estimated from the previously applied torque, that is, $F_{xi} \approx T_i / R_{w,i}$. The corresponding utilization can be given as:

$$\mu_{obs,i} = \frac{F_{x,i}}{\max(F_{z,i}, \epsilon)} \quad (4.27)$$

The online friction-estimation pipeline that produces $\hat{\mu}$ is shown in Figure 25. The samples with $|k_i|$ below a certain threshold and v_x above a small floor are only considered informative. These samples will update a decaying running maximum that tracks the achievable peak. A low-pass filter then yields $\hat{\mu}$, the results can adapt to surface changes over a few seconds and remain quiet enough to avoid the cause of chatter to the limits.

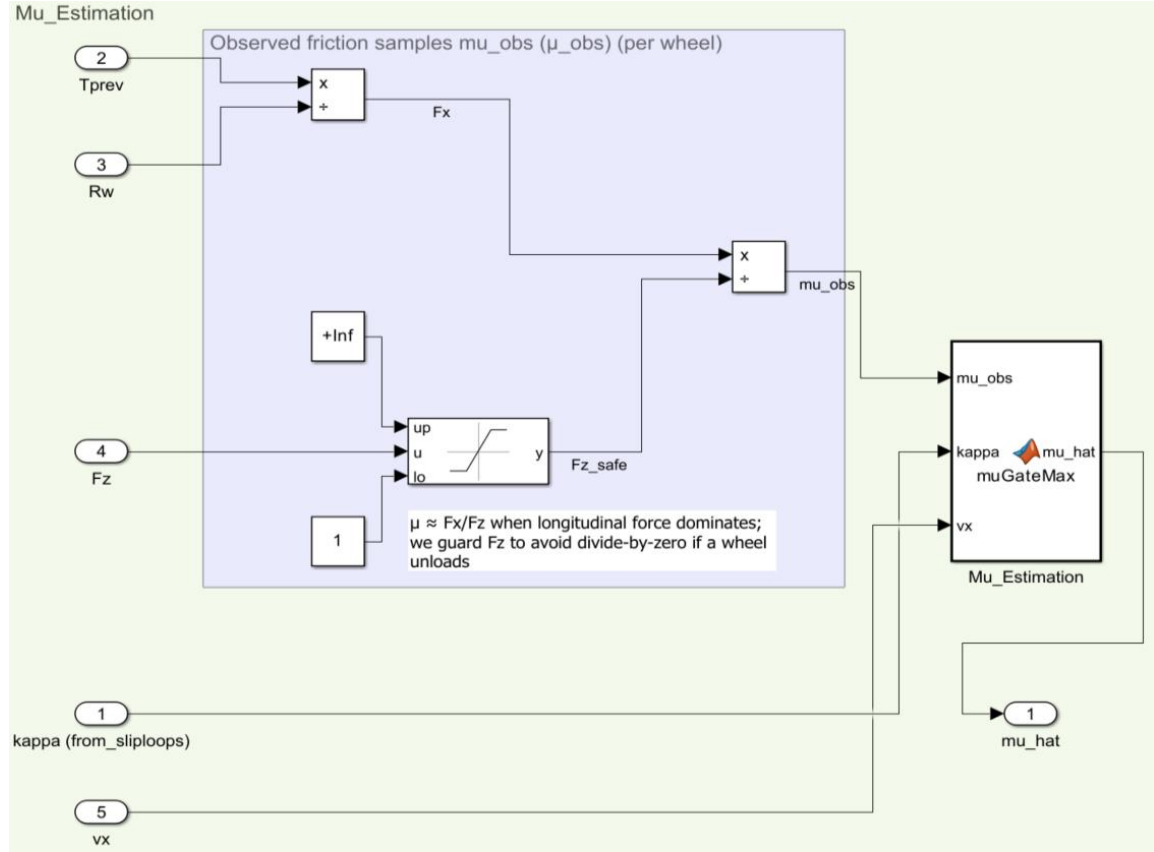


Figure 25: Online road-friction estimation $\hat{\mu}$ inside the traction manager

Effective torque ceilings: The per wheel torque ceilings combine the hardware limits, power limits, friction limits and the slip-tightening actions. Starting from the static torque cap $T_{max,i}^{base}$,

$$T_{P,i} = \frac{P_{max,i}}{\max(\omega_i, 1)} \quad (4.28)$$

enforces the power envelope. A friction-circle projection uses:

$$F_{x,i}^{max} = \sqrt{\max(0, (\hat{\mu} F_{z,i})^2 - F_{y,i}^2)}, \quad T_{\mu,i} = F_{x,i}^{max} R_{w,i} \quad (4.29)$$

Finally,

$$T_{max,i}^{eff} = \max(T_{min,i}, \min(T_{max,i}^{base}, T_{P,i}, T_{\mu,i}) - \Delta T_i) \quad (4.30)$$

The combined torque-ceiling computation from power, friction and rate bounds is summarized in Figure 26. This produces the live ceiling supplied to the allocator. The vector T_{max}^{eff} changes with the speed, load and the surface while preserving actuator safety.

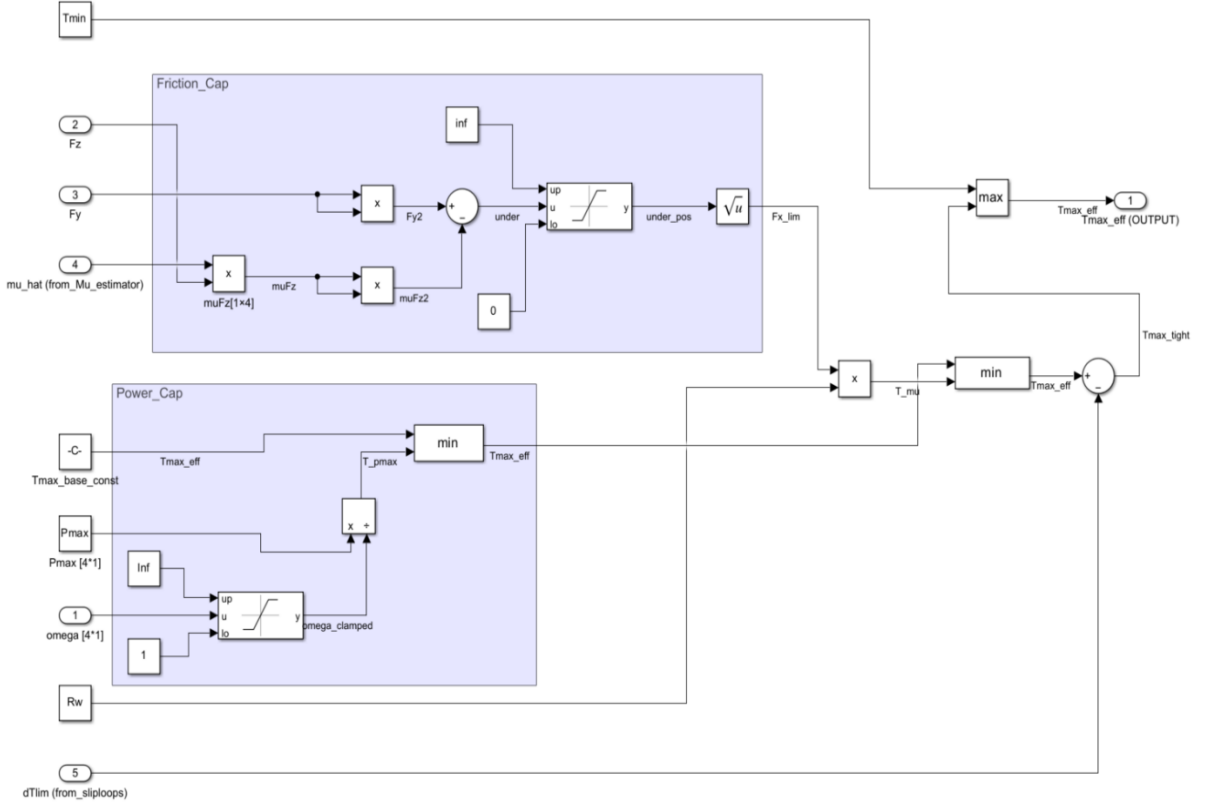


Figure 26: Effective torque ceiling from power, friction and rate bounds

IV. Allocator QP to per-wheel torque from T_{tot} and M_z^*

The quadratic-program torque allocator that maps $[T_{tot}, M_z^*]$ to per wheel torques is shown in Figure 27. The baseline torque distribution is from the computation based on the normal force from part 2. That is:

$$T_i^{base} = \frac{F_{z,i}}{\sum_j F_{z,j}} T_{tot} \quad (4.31)$$

The yaw-moment geometry follows the small-angle mapping:

$$a_{M_z} = \left[-\frac{w_f}{2R_{w,1}}, \frac{w_f}{2R_{w,2}}, \frac{w_r}{2R_{w,3}}, \frac{w_r}{2R_{w,4}} \right], \quad M_z = a_{M_z} T \quad (4.32)$$

Objective and constraints: Let T be the wheel torque vector (a 4-dimensional real vector that carries 4*1 column vector) and T_{prev} be the previous solution. The convex objective here is given as:

$$J(T) = (T - T^{base})^T * W(T - T^{base}) + \rho ||T - T_{prev}||_2^2 + \gamma (1^T T - T_{tot})^2 + \eta (a_{M_z} T - M_z^*)^2$$

where $W = \text{diag}([1, 1, 1, 1])$ is the baseline proximity, $\rho = 5$ is the smoothness, $\gamma = 100$ is the total torque tracking and $\eta = 50$ is the yaw-moment tracking, and these are the Quadratic

Program (QP) weights defined in the directory. These will penalize the deviation from the baseline, excessive step-to-step changes, and errors in total torque and yaw-moment. The bounds can be defined as:

$$T_{min} \leq T \leq T_{max}^{eff}$$

This will enforce the actuator limit, power limit and the tire limits. The implementation uses a small quadratic program for each sample. If the solver cannot return a solution within that step, the controller will fall back to the baseline split for continuity. The full configuration and listings are provided in appendix.

V. Normalization for inverter interfaces

The normalization of the torque reference command are formulated by:

$$T_{norm,i} = \frac{T_i - T_{min,i}}{\max(T_{max,i}^{eff} - T_{min,i}, 10^{-6})} \quad (4.33)$$

This is finally saturated to $[0, 1]$. The physical and normalized demands are both carried together on the ‘TV_TorqueBus’.

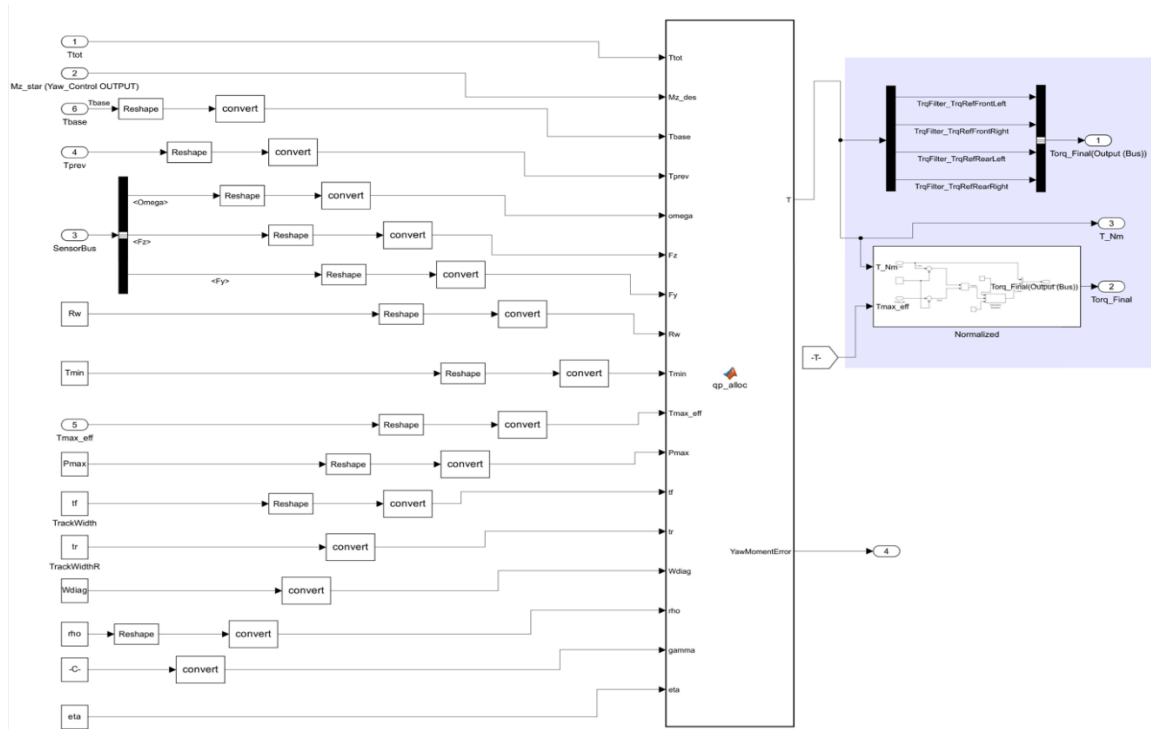


Figure 27: Quadratic-program allocator mapping $[T_{tot}, M_z]$ to vectored wheel torques

VI. Hardware-consistent saturation

Each motor is limited to $\pm 2500\text{Nm}$, thus the four raw torques are therefore saturated individually to obtain the final commands $\{T_{FL}, T_{FR}, T_{RL}, T_{RR}\}$. This will prevent the cause

of any unrealistic request to the inverter models and align the controller with known actuator limits.

4.2.4 Integration of the torque vectoring module within the top-level plant

The torque-vectoring function is inserted inside the Powertrain Controller Unit (PCU), within the propulsion code that contains the torque chain. The insertion point is downstream of pedal mapping, feasibility enforcement, brake blending, thermal derating, and the four per-wheel shaping filters. At this point the torque chain produces a four-element vector of filtered requests given as:

$$T_{req} = [T_{FL}^{ref}, T_{FR}^{ref}, T_{RL}^{ref}, T_{RR}^{ref}]^T$$

which is carried on the raw torque-reference bus shown in the Figure 28. This bus is connected to the torque-vectoring block as the input labeled T_{PCU} . Two measured states are supplied as auxiliary inputs: vehicle speed in meters per second and wheel speeds as a vector form. This signal is available on the same level of the model, as indicated in the diagram.

The torque-vectoring block transforms the four requested torques into a vectorized set, given as:

$$T_{vec} = V(T_{req}, v, \Phi),$$

The mapping $V(\cdot)$ is designed to be the sum preserving the nominal operation:

$$1^T T_{vec} = 1^T T_{req},$$

while respecting the per-wheel caps already imposed in the upstream plant for the thermal and voltage limits. The module will output a normalized torque-reference bus named as ‘Vectored_Torq_Norm_Bus’ as in the figure. A unit delay will align timing with the rest of the PCU and produces the ‘TorqueRefBus’ that leaves the torque chain.

The vectored torque bus output is then routed to the traction-safety layer, which holds the anti-spin logic and the virtual cross-axle coupling. These functions will construct the per-wheel speed windows and the coupling terms from the measured wheel speeds and articulation. They then gate or clip this vectored torques to ensure compatibility with available grip and with the modeled cross-axle constraint. In other words, the anti-spin and the virtual coupling sit after the vectoring block and leaves the final authority on the admissible torque before packaging it to the inverters.

After the traction gating, the ‘TorqueRefBus’ proceeds unmodified through the existing output packager in the PCU. The bus is serialized into network frames for the four motor controllers. Each of these controllers will dynamically run its own local checks for the electrical speed,

bus voltage, and device temperature, then only it tracks the commanded torque with its inner loop. This will preserve the original actuation pathway and diagnostics.

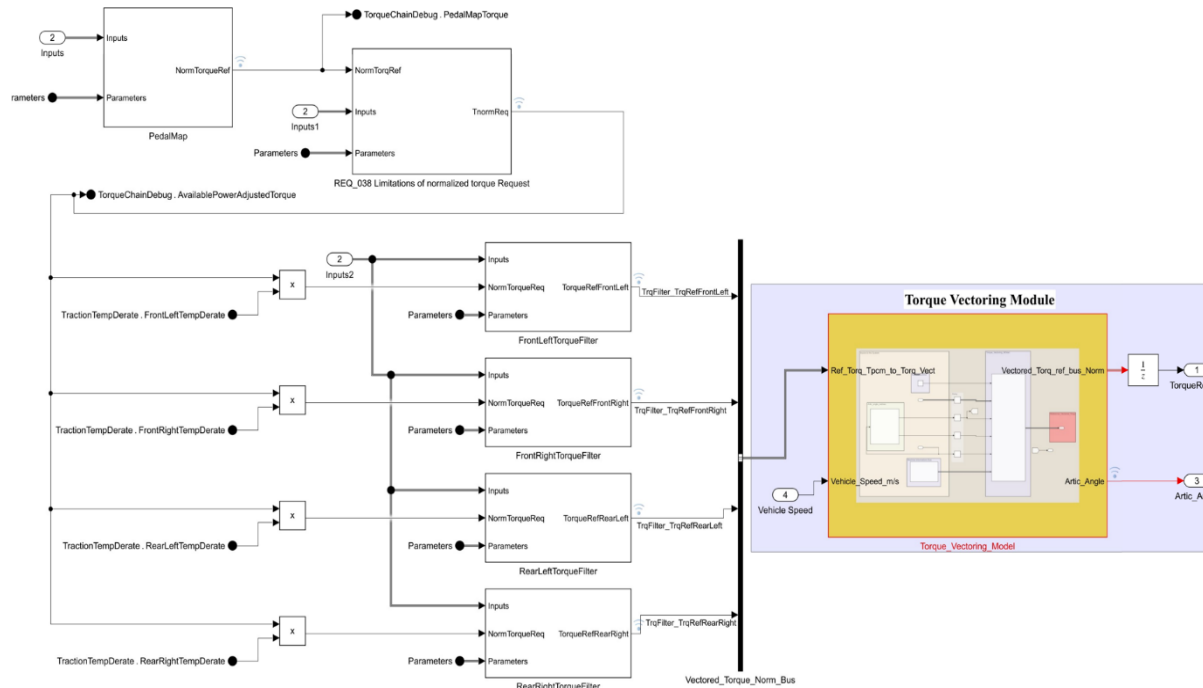


Figure 28: Integration of the torque-vectoring module within the MT66 eDrive model

4.2.5 Summary

The operator's input remain unchanged until the torque filtering block. The torque chain will still generate four filtered torque requests, but now it will pass through the torque-vectoring block, which redistributes them based on the normal forces, yaw moment and articulation angle, while preserving the upstream protections. The vectored bus will then enter the anti-spin and the virtual-coupling gate, after which the references are packaged and then sent to the inverters. The mechanical plant will finally convert the commanded torques to tractive effort.

Chapter 5

5. Results and Discussion

The results section will discuss the behavior of the torque-vectoring controller designed in the methodology across the drive cycle. The analysis links the observed signals to the design choices from part one (steering to articulation), part two (reference torques from normal forces) and part three (torque vectoring). This chapter reports the closed-loop simulation of MT66 eDrive with the full torque-vectoring stack over a single 0-300 s drive cycle. The route combines speed changes from 0 to 25 km/h, alternating left and right steering plateaus with articulation magnitude up to 0.785 rad, scheduled payload steps from 0 to $66 \cdot 10^3$ kg at different intervals and a 4° uphill segment from 248 to 268 s.

Evaluation follows the scenario timeline and uses plots of steering-to-articulation mapping and tracking and the PID effort. The kinematic consistency is checked through turning radius and curvature, together with measured longitudinal and lateral accelerations and yaw rate, as well as yaw-rate tracking with sideslip and the commanded yaw moment M_z^* and the damping term M_ϕ . The tire to road behavior is assessed via per-wheel lateral forces F_{yi} and Normal loads F_{zi} , their trends with payload and curvature, and the online friction estimate $\hat{\mu}$. Powertrain behavior is evaluated by comparing raw PCU torque requests with allocator outputs per wheel, verifying preservation of total drive torque, and marking intervals where traction, power, or rate limits bind, including DC-link power headroom. Together these signals show whether vectoring provides and preserves total demand, improves left–right distribution in turns, respects feasibility limits, and maintains torque to each wheel based on load forces acting over the route.

5.1. Drive Cycle, Steering Coordination and Articulation

The Figure 29 (a) presents the vehicle speed in km/h, (b) the steering command range in degrees, (c) the desired articulation ϕ in radians, and (d) the derivative $\dot{\phi}$ for articulation rate. The steering mapping in part one of the methodology converts the steering command from the operator input to its desired articulation angle ϕ and limits the curvature at higher speeds. The Figure 29 (c) shows that ϕ saturates near ± 0.78 rad on the longest plateau, matching the possible articulation angle about the hinge ± 45 degrees and staying within the chosen articulation bounds. The derivative signal remains within ± 0.6 rad/s, except having short spikes at reversal to neutral as seen in Figure 29 (d). These results confirm that the speed-dependent steering mapping discussed in the methodology and the rate limits are effective as intended. The measured speed of the vehicle varies between roughly 3 and 25 km/h, the slow phase changes in speed coincide with the periods of heavy payload and grade when we compare Figure 29 with Figure 31.

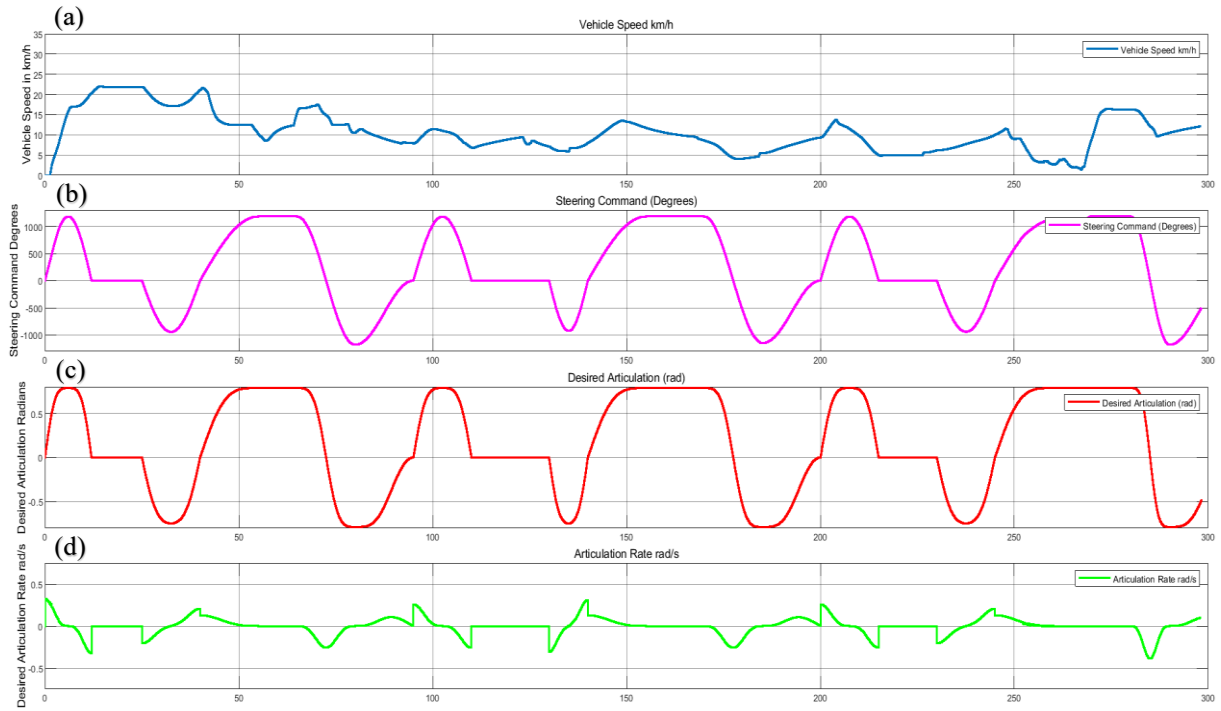


Figure 29: Drive cycle: speed profile and steering-to-articulation mapping. (a) Vehicle speed [km/h]. (b) Steering input [deg]. (c) Desired articulation [rad]. (d) Articulation rate [rad/s]

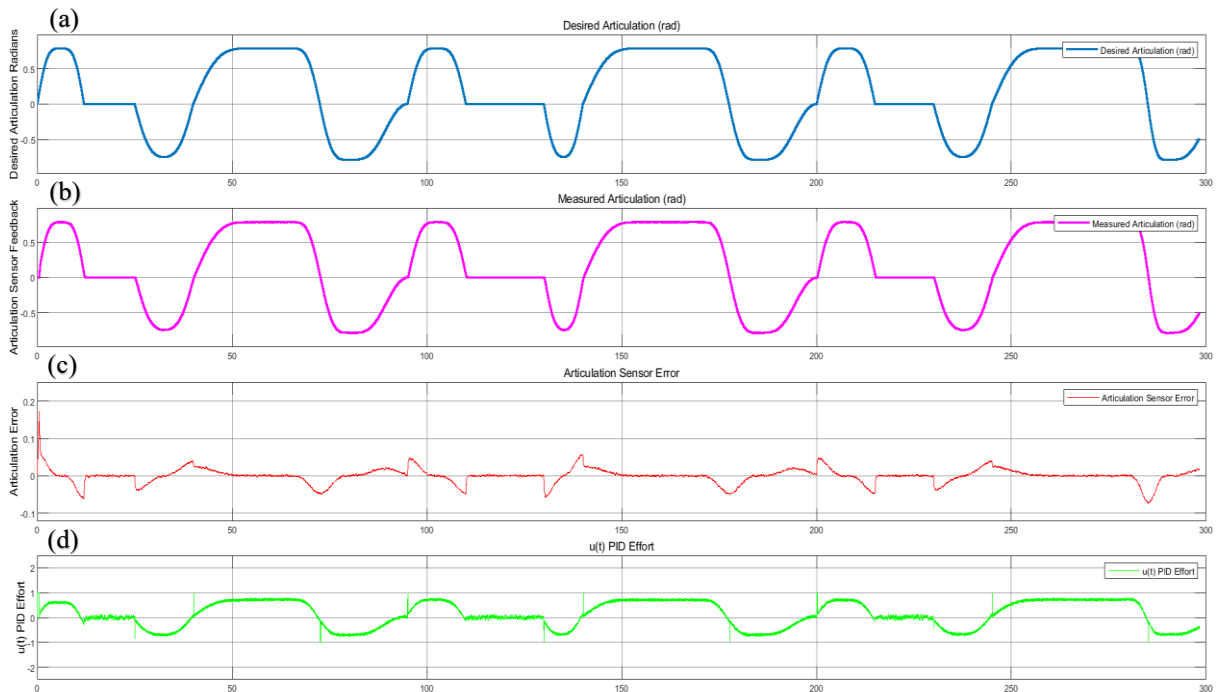


Figure 30: Articulation reference tracking, sensor feedback and PID effort. (a) Desired articulation [rad]. (b) Measured articulation [rad]. (c) Articulation error [rad]. (d) PID effort

In Figure 30 (a) the desired and (b) the measured ϕ from the virtual sensor are almost indistinguishable during the steady segment operation by only having a mean absolute error of nearly 0.02 radians as seen in Figure 30 (c). Error spikes occur only during the reversals where the hinge must change the sign and the low-pass filtered rate will impose a finite response during that period. The largest excursions are of the order 0.08 to 0.10 rad and it last for milliseconds. The bottom panel reports the PID effort $u(t)$. Short impulses appear at when steering reaches 0 radians, but there is no presence of any noticeable sustained oscillation as seen in Figure 30 (d), which indicates that the controller zero and the derivative filter are well matched to the hinge dynamics. This behavior matters because M_ϕ , the articulation damping moment added from part two responds in respect to ϕ and $\dot{\phi}$. So accurate tracking keeps M_ϕ predictable and avoids any cause of hidden loops inside the yaw controller.

5.2. Payload Schedule and Geometry Migration

The Figure 31 explains the payload program and its dependency with other geometrical parameters of the vehicle. The total mass of the vehicle steps from the empty payload value near 43,000 kg to 109,000 kg at the full load times listed in the scenario as in Figure 31 (a). The center-of-gravity abscissae a and b is migrating linearly with the load differences, as defined in the part 4.2.2. In the plot when the maximum payload is added at 42 s, the values of a increases from nearly 1.7 to 3.2 meters and b decreases from 4.1 to 2.65 meters as in Figure 31 (b) and (c). Which means the static load is shifted towards the rear axle. The right-hand side plots from the Figure 31 reports the axle and the single-wheel loads as equivalent kilograms. The rear axle load rises by roughly 50,000 kg during the full-load phases, whereas the front axle load increases only by a smaller amount, consistent with the CG movement toward the rear. Because at no load condition the power-frame of the vehicle sits mostly on the front axle. The variations in CG height with load is also reported in Figure 31 (d). This small rise amplifies roll moments at the same lateral acceleration, which is confirmed later in the lateral force and yaw-moment results.

The variations in the mass and geometric parameters are propagated through the MachineData bus, which means that the normal-force model from part two and the allocator in part three respond consistently among each other. All feedback paths are defined with unit delay to avoid the cause of discontinuities or algebraic loops in the results.

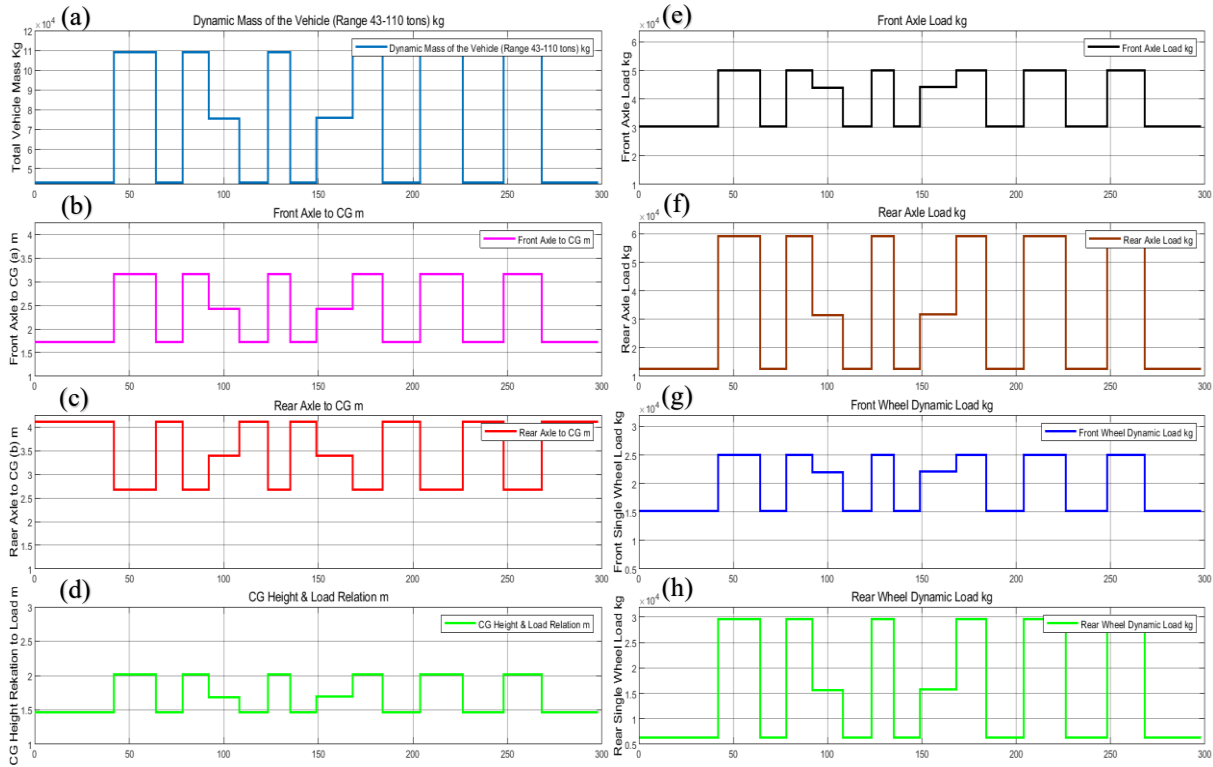


Figure 31: (a) Payload schedule [kg]. (b) Front axle CG migration [m]. (c) Rear axle CG migration [m]. (d) CG height [m]. (e) Front axle load [kg]. (f) Rear axle load [kg]. (g) Front wheel load [kg]. (h) Rear wheel load [kg]

5.3. Kinematic consistency with Turning Radius and Curvature

The Figure 32 compares the (a) articulation angle, (b) vehicle speed in m/s, (c) turn-radius about the hinge and (d) the path curvature approximation. The turning-radius plot shows larger spikes only when articulation angle ϕ crosses zero. This is expected because $R(\phi) = (a_h + b_h \sec \phi) / \tan |\phi|$ grows unbounded near zero articulation, it is not important to have limited $R(\phi)$ some saturation value, because a small steering input beyond the dead band at a slow rate also means the vehicle should start articulating at that slow rate. The curvature plot follows $\kappa = \tan \phi / (a_h + b_h)$ and it is smooth across that same interval. The signs of curvature will always remain in relation with the ϕ as seen in Figure 32 (a) and (d). The vehicle holds nearly constant curvature during the plateaus, which validates the rate limiter and the curvature cap in the steering map. This capping helps to prevent the cause of excessive curvature at higher speed, visible at 150–160 s where speed increases and $|\kappa|$ remains bounded.

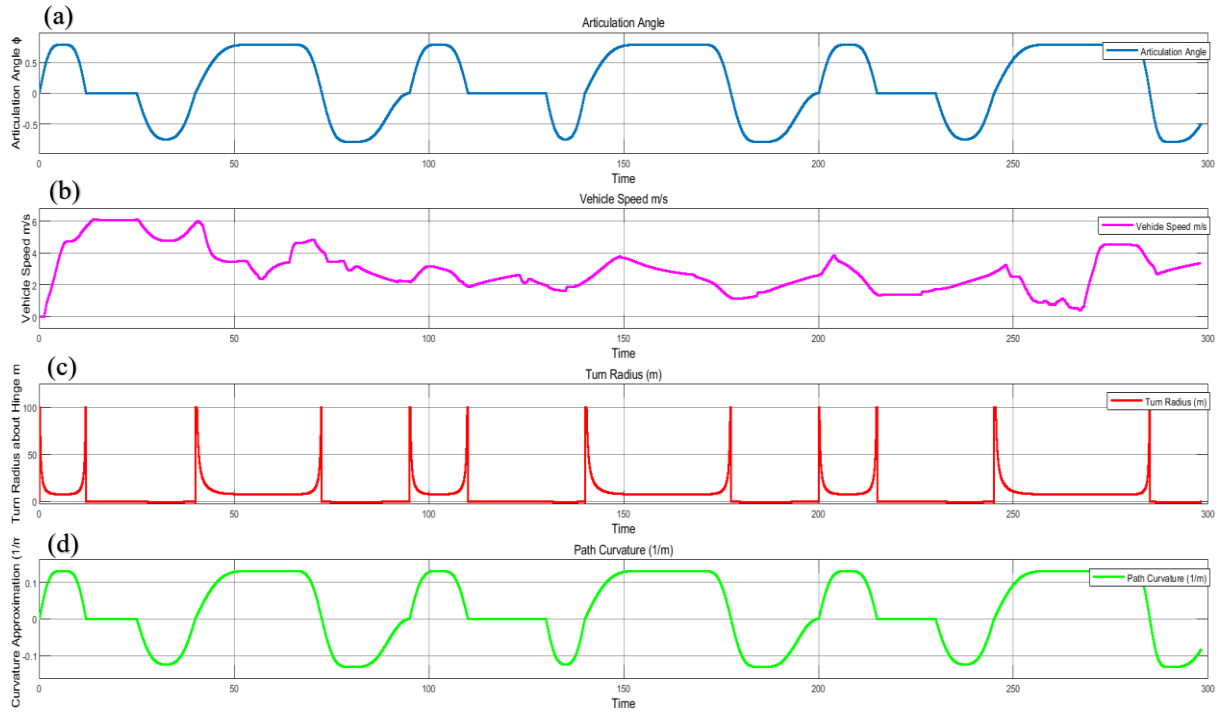


Figure 32: Turning radius about the hinge and path curvature. (a) Articulation angle [rad]. (b) Vehicle speed [m/s]. (c) Turn radius of vehicle [m]. (d) Path curvature [1/m]

5.4. Accelerations and Yaw Rate

The results from Figure 33 presents (a) the longitudinal acceleration a_x , (b) the lateral acceleration a_y , (c) the high-pass and low-pass blend used for diagnostics a_{meas} , and (d) the measured yaw rate r_{meas} . Having a positive ϕ correlates with having positive values of r_{meas} and positive a_y . The largest lateral accelerations peaks can be seen occurring when the articulation angle holds its extreme values, particularly during heavy payload windows, though the peak a_y values remain moderate due to the curvature cap. The longitudinal acceleration is small on average, with brief positive transients that coincide with the throttle steps or when applying the sudden brakes and the onset of the 4° grade at 248–268 s shows sudden transience as seen in Figure 33 (a). No drift or bias is visible, which indicates that the discrete differentiation and the complementary filters are tuned appropriately.

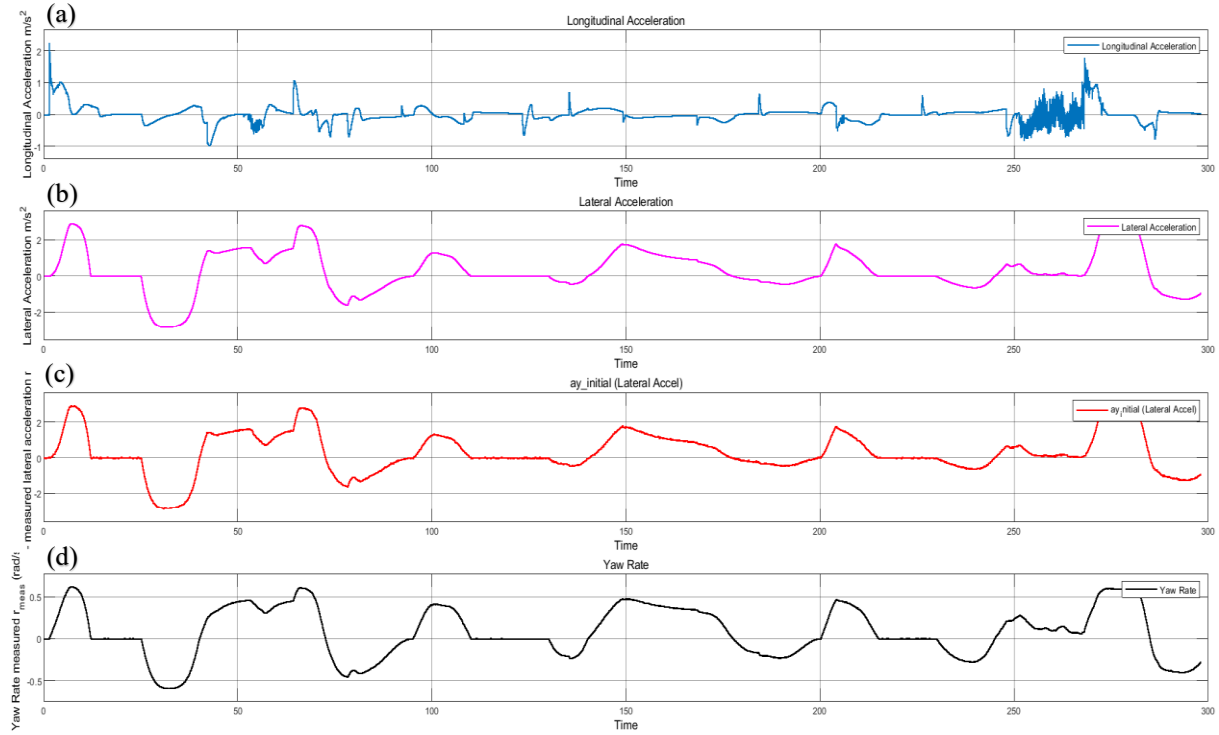


Figure 33: (a) Measured longitudinal acceleration [m/s²]. (b) Lateral acceleration [m/s²]. (c) Measured lateral acceleration [m/s²]. (d) Measured yaw rate [rad/s].

5.5. Lateral Tire forces and Lateral Load transfer

Figure 34 shows the lateral force $F_{y,i}$ for all the four wheels on the vehicle. The sign convention matches the model, that is, when the value of ϕ is positive, the right wheels are experiencing positive F_y , consistent with an outward shift of the vertical loads to the right under the positive lateral acceleration. As payload increases, the magnitudes of both the left and right F_y rises because the available friction $\hat{\mu}F_z$ increases with F_z the normal forces. This is most visible in the plot during the time periods from 168–184 s, 204–226 s, and 248–268 s, which the payload changes. At each transition to full load it can be seen that the right-hand peaks increase by roughly 20–30 percent relative to empty, which aligns with the increase in F_z shown later in Figure 37. A brief reduction in F_y accompany steering reversals and match the finite rate of change of ϕ . No evidence of saturation-induced clipping could be seen on the outer wheels, which confirms that the allocator enforces the friction circle without requesting unrealizable moments.

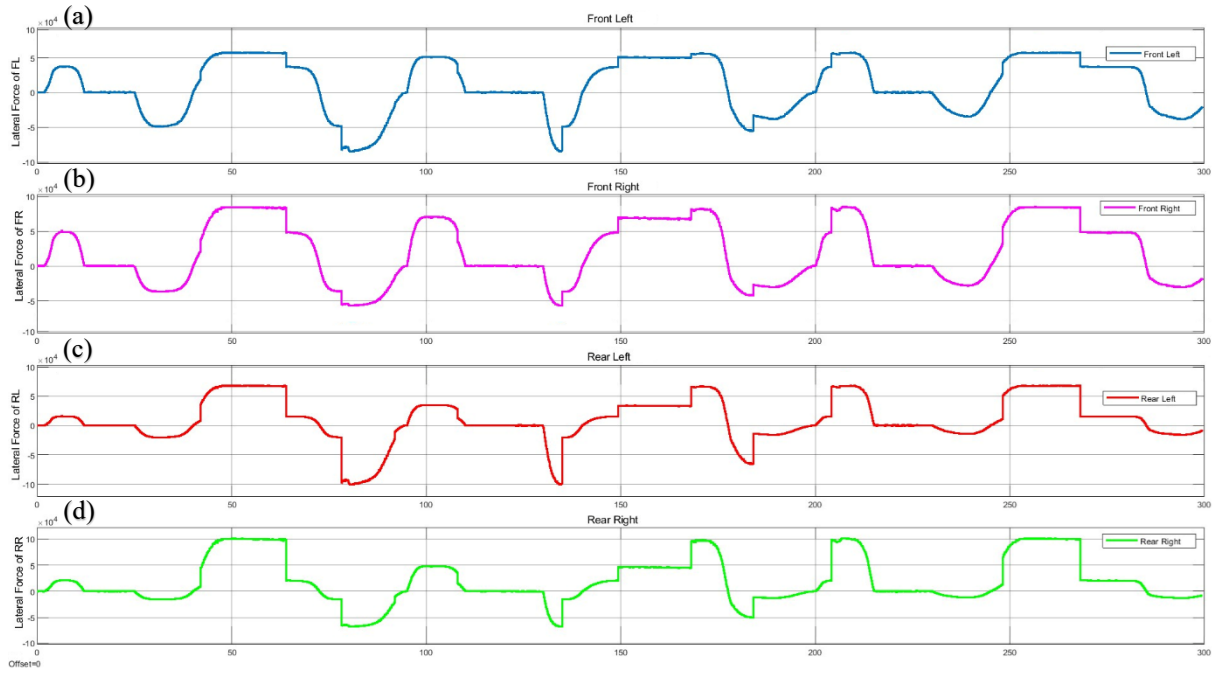


Figure 34: Per-wheel lateral forces [N]. (a) Front left wheel. (b) Front right wheel. (c) Rear left wheel. (d) Rear right wheel

5.6. Yaw-Moment command, Yaw tracking and Sideslip

The Figure 35 links to (a) the estimated absolute yaw moment M_z^* , (b) the yaw rate r , (c) the sideslip $\hat{\beta}$, and (d) the yaw moment error $e_r = r_{ref} - \hat{r}$. The yaw-moment command follows the expected square-wave pattern which is set by the steering program. Its magnitudes grows in the heavy payload phases, in order for the controller to overcome the larger lateral inertia. During the time period between 248–268 s, when the vehicle is fully loaded to 66,000 kg and with a climbing of 4° grade, it is clearly visible that the M_z^* briefly increases as seen Figure 35 (a), same time the allocator also respects the power limits. Sideslip shows larger plateaus near +0.04 rad appears during high ϕ and moderate speed, which matches the linear bicycle model's prediction. The sideslip estimate remains within ± 0.04 rad for most part of the cycle as seen in Figure 35 (c). The yaw-moment errors are very small values of nearly $\pm 4 \cdot 10^4$ which is almost negligible and biased slightly negative at high curvature, indicating that the cap on $M_{z,max}(v_x, \hat{\mu})$ is occasionally active. At instance when the vehicle suddenly starts to articulate between the right turn and left turn at a faster rate, then the error spikes seem to increase. This happens because the system has to quickly change states and then settles in few milliseconds. This behavior shows a match to the design choice of soft equality in the QP objective and the bounded articulation damping M_ϕ .

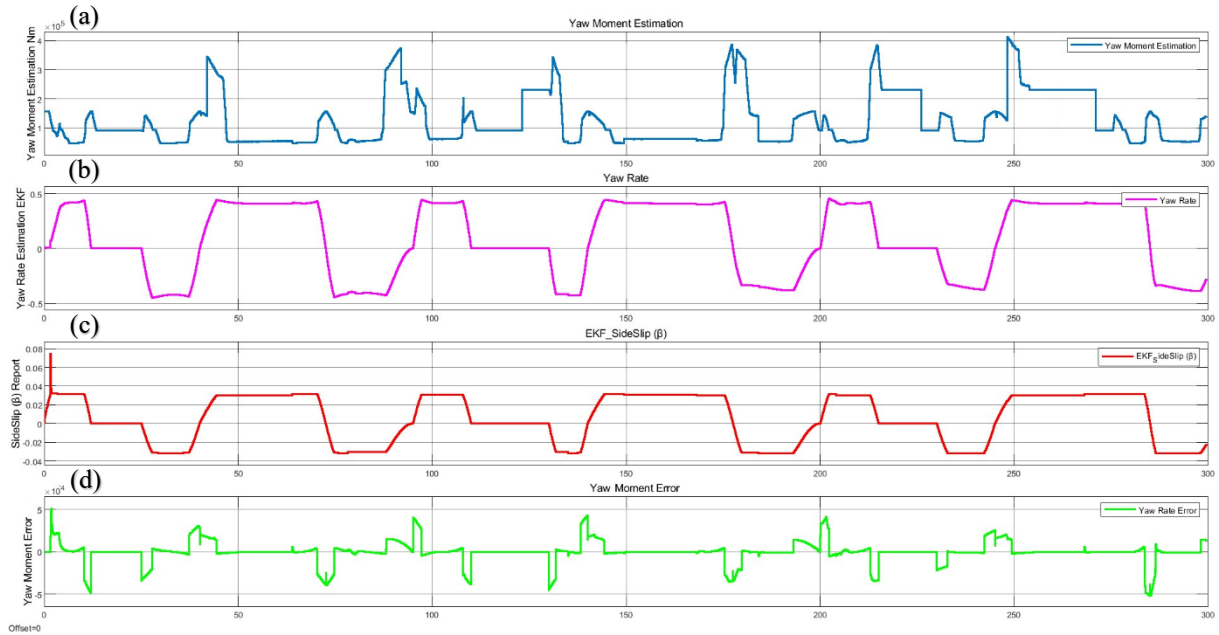


Figure 35: (a) Demanded yaw moment [Nm]. (b) Yaw rate tracking [rad/s]. (c) Sideslip estimate. (d) Yaw moment error [Nm]

5.7. Articulation Damping, Lateral Velocity and Friction estimate

In Figure 36, it provides four traces that clarifies the role of the damping term. The articulation angle repeats the earlier program. The lateral velocity $\hat{v}_y$ recorded from the EKF follows the sign of the articulation angle ϕ and it peaks around 0.6 m/s on heavy load turns at moderate speeds as shown in Figure 36 (b). The friction estimates $\hat{\mu}$ lays between a value of 0.35 and 0.65, with having shorter increases towards the upper end when there is sustained slip being recorded and the speed permits the reliable observation. During the low speed operations, the allocator estimates and holds friction factor near 0.35 due to the possible gate on usable samples, see Figure 36 (c). The articulated damping M_ϕ shows large impulses at sign changes of ϕ and reduced steady values during plateaus. The per-step rate limit scales with the $M_{z,max}$, so that the impulses are always bounded and there are no ringing appearing in the yaw rate. During 248–268 s the damping pulses grow because $\hat{\mu}$ is lower under grade with heavy load, which reduces $M_{z,max}$ and forces the controller to use more of the available envelope.

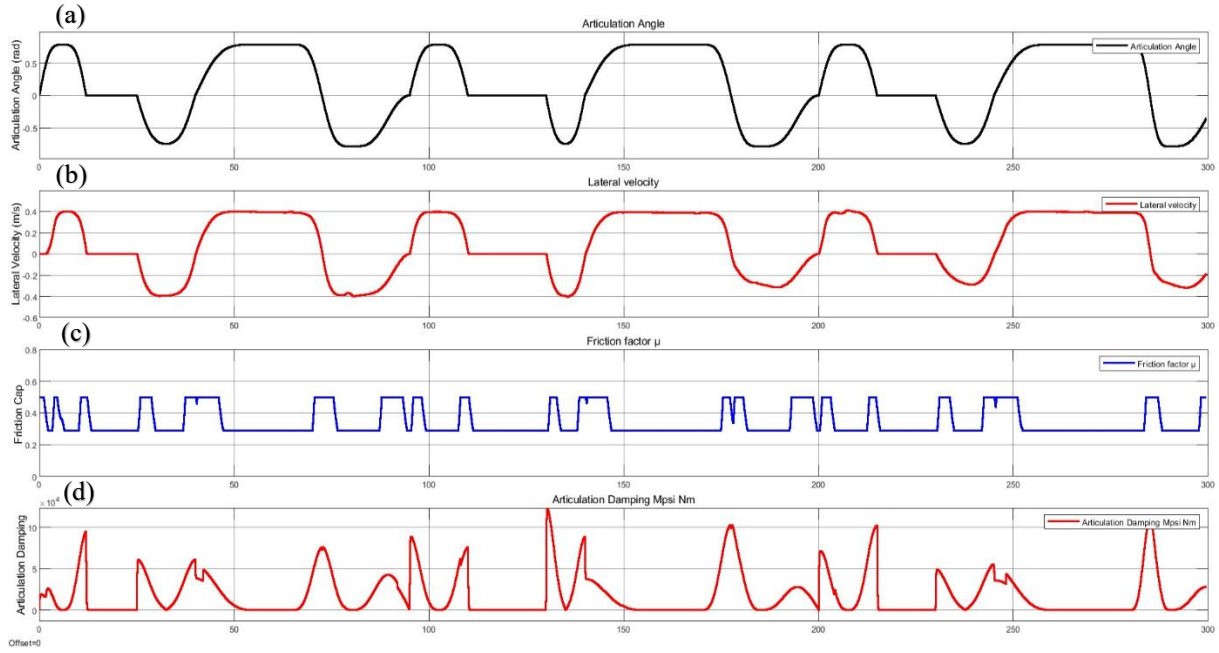


Figure 36: (a) Articulation angle [rad]. (b) Lateral velocity [m/s]. (c) Friction cap. (d) Articulation damping [Nm]

5.8. Normal Forces and Load Transfer consistency

The Figure 37 reports the normal forces on each four-wheel corner. From the plots, the right wheels carry more load during a positive ϕ , as seen in Figure 37 (b) and (d). Conversely the left wheels during a negative ϕ , as seen in Figure 37 (a) and (c). Across the payload steps, all the four baselines rise in proportion to the mass and to the CG migration. The rear right corner exhibits the largest absolute values and the largest dynamic range during the excitation time. This follows from the rearward CG shift and the roll moment distribution $\beta_f = b/(a + b)$ used due to roll stiffness data are absent. The step increases in F_z at 168 s, 204 s, and 248 s align exactly with the scheduled payload additions, see Figure 31 (a). The magnitudes agree with the axle-load plots in Figure 31 (e) and (f). These forces are taken in as the major torque-split baseline estimator and the friction caps inside Traction Manager, so that the allocator responds immediately to any changes in payload.

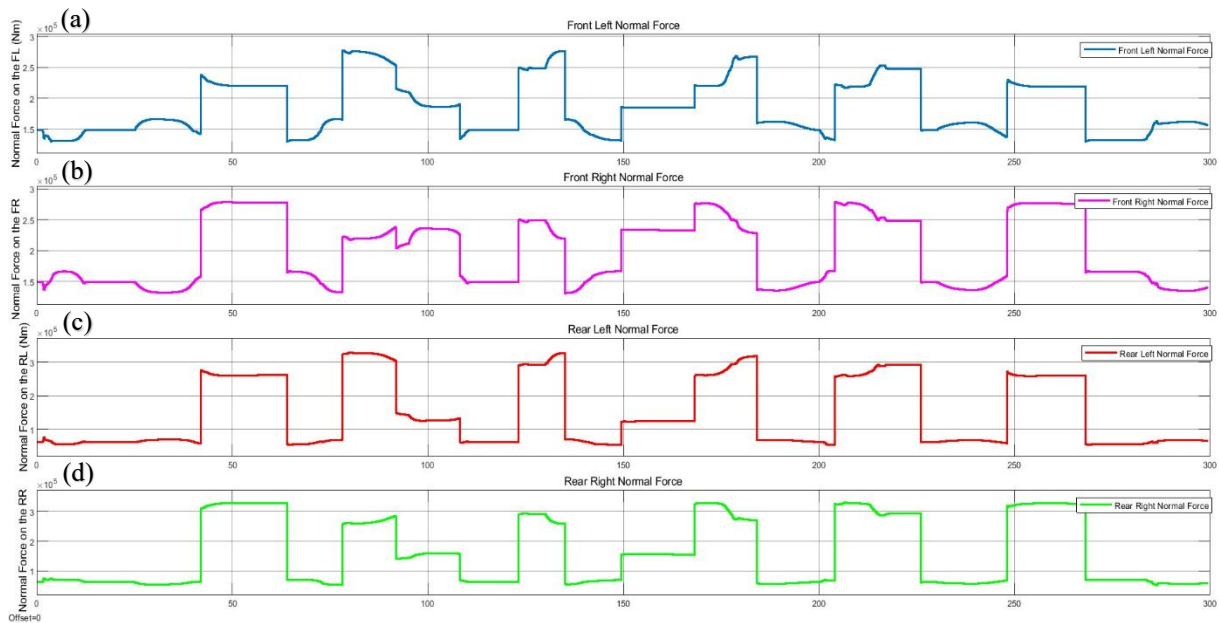


Figure 37: Per-wheel normal forces under payload and curvature [Nm]. (a) Front left wheel. (b) Front right wheel. (c) Rear left wheel. (d) Rear right wheel

5.9. Vectoring behavior versus the Raw Powertrain commands

The combined panel in the Figure 38 compares (a) the available power limit, (b) articulation, (c) speed, (d) mass, against (e) the per-wheel torques from the PCU command and the Torque vectoring unit. The black dashed lines represent the raw PCU torque demand. The solid red lines show the vectored torques delivered by the allocator. During straight running, the vectoring solution gives outputs in close orientation to the raw demand see Figure 38 (e), adjusted purely based on the normal forces acting on each axle, power capping and rate limits. During turns, the allocator redistributes the torque toward the outer wheels, by respecting the normal force due to axle load. The sum of the four torques remains almost close to the total demand, but at no load condition the power used by the rear wheel motors would be lesser, except at instances where power limits or friction caps bind. Those instances coincide with dips in the “Power Available Limit” trace and with the steps in $\hat{u}$. The effect is most visible between 255 and 265 s, where the vehicle climbs a 4° grade at full load. The allocator keeps the longitudinal demand by drawing the additional torque from the rear pair, while still producing the required yaw moment by a left–right bias. This confirms the value of soft equality penalties in the QP allocator, which avoid infeasibility while keeping the outputs smooth.

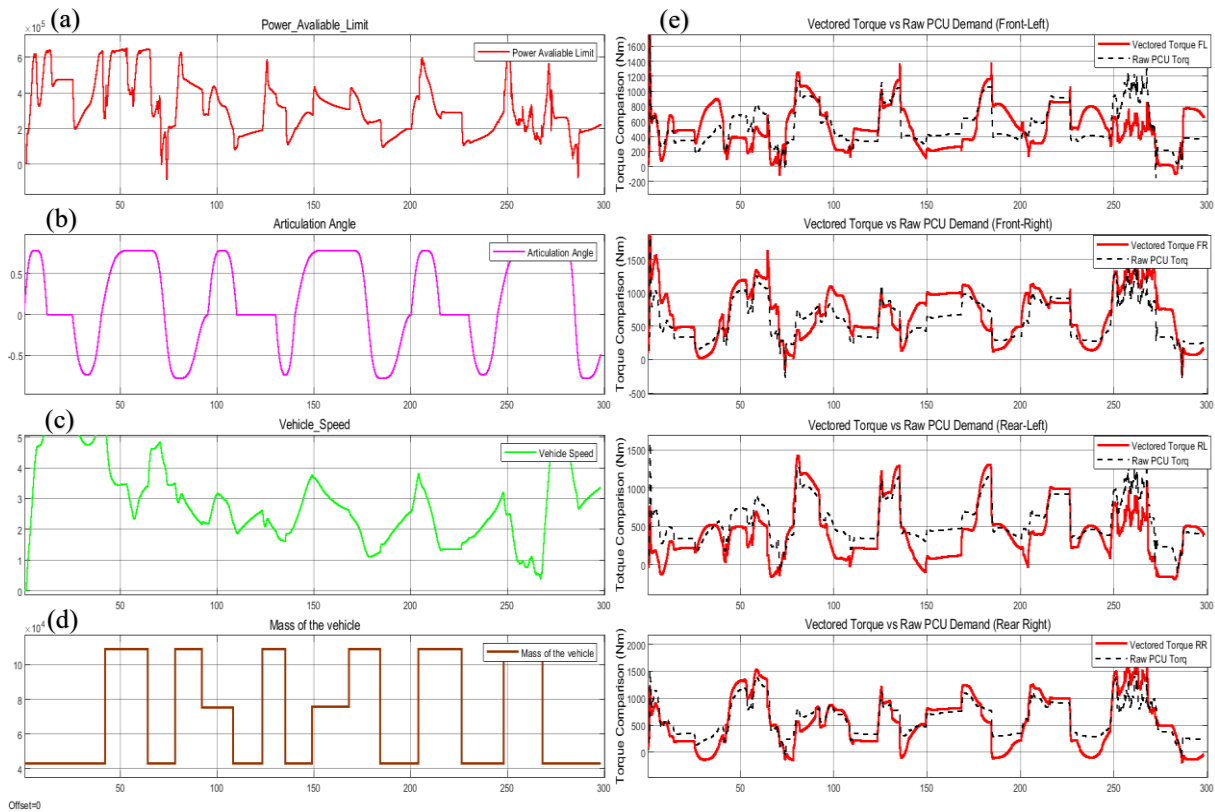


Figure 38: (a) Available electrical power limit [W]. (b) Articulation angle [rad]. (c) Vehicle speed about CoG [m/s]. (d) Mass of vehicle [kg]. (e) PCU vs vectored torques per wheel (FL, FR, RL, RR)

Torque distribution and sum consistency:

- Moderate left turn, medium power headroom (Figure 38):

Time period from 42 - 50 s. Power-Available trace sits near $6.1 \cdot 10^5$ W, so the power is not binding. Total load is 66000 kg, and speed is 3.5 m/s.

Raw PCU demand is [940, 1000, 800, 1100] Nm.

Vectored Torques is [400, 1250, 500, 1400] Nm.

The outer wheel bias is clear. FR and RR are about +250 Nm above the PCU demand, while the FL and the RL are about -300Nm below.

PCU sum ≈ 3860 Nm, vectored sum ≈ 3550 Nm ($\approx 7\text{-}8\%$ low), consistent with a soft sum penalty rather than a hard equality.
- Sharp left turn, high power headroom (Figure 38):

Time period from 155 - 160 s. Power-Available trace sits near $4 \cdot 10^5$ W, so the power is not binding. Total load is 33000 kg, and speed is 3.8 m/s.

Raw PCU demand is [400, 400, 460, 650] Nm

Vectored Torques is [200, 1000, 240, 850] Nm

The outer wheel bias is clear. FR and RR are about +600 Nm and + 200 Nm above the PCU demand, while the FL and the RL about -200 Nm and -220 Nm below.

PCU sum ≈ 2000 Nm, vectored sum ≈ 2200 Nm ($\approx 9\%$ high), consistent with a soft sum penalty rather than a hard equality.

- Zero articulation with full load (Figure 38):
Time period from 220 - 235 s. Power-Available trace sits near $3 \cdot 10^5$ W, so the power much less the binding limit. Total load is 66000 kg, and speed is 2.6 m/s.
Raw PCU demand is [900, 900, 900, 930] Nm
Vectored Torques is [800, 810, 1000, 1000] Nm
FL and FR are about -100 Nm below the PCU demand, while RL and RR are about +100 Nm above the PCM demand.
PCU sum ≈ 3600 Nm, vectored sum ≈ 3600 Nm (Equality)
During full load condition the rear axle loads are much higher than the front axle, hence pure vectoring based on normal loads.
- Moderate left turns with full load and 4° grade climb (Figure 38):
Time period from 250 - 270 s. Power-Available trace sits near $4 \cdot 10^5$ W. Total load is 66000 kg and speed is 1 m/s.
Raw PCU demand is [1200, 1500, 1200, 1500] Nm
Vectored Torques is [750, 1500, 1000, 1600] Nm
The outer wheel bias is clear. FL and RL are about -400 Nm and -200 below the PCU demand, while the FR and the RR about almost equal.
PCU sum ≈ 5400 Nm, vectored sum ≈ 4850 Nm (7% less)
During full load condition with a 4° grade climb the rear axle loads are very high than the front axle, hence vectoring based on normal loads and articulation.
- Extreme right turn at very high rate with full load (Figure 38):
Time period from 135 - 140 s. Power-Available trace sits near $2 \cdot 10^5$ W. Total load is 66000 kg and speed is 1.9 m/s.
Raw PCU demand is [800, 500, 860, 500] Nm
Vectored Torques is [1400, 50, 1350, -100] Nm
The outer wheel bias is very clear in this situation. FL and RL are about +600 Nm above the PCU demand, while FR and RR about -500.
PCU sum ≈ 2600 Nm, vectored sum ≈ 2700 Nm (with a negative braking torque of -100Nm).
During full load condition with a 4° grade climb the rear axle loads are very high than the front axle, hence vectoring based on normal loads and articulation.

5.10. Torque demand at Motors

The Figure 39 plot reports the commanded motor torque at each wheel. The changes in torque respect the 10 milliseconds discrete step, and the residual high-frequency ripples is considerably small with no evidence of limit cycles. During high-load operations the rear axle carries the larger share, which follows the rise in the rear normal force from the payload and the shift of center-of-gravity. At lower load and during higher speeds the front axle contributes more. Short dips during steering reversals shows the deliberate swap in left–right moment needed to change yaw sign. The ranking of magnitudes across the wheels FL, FR, RL, and RR matches the

normal-force trends from Figure 37 and the lateral-force from Figure 34. Which confirms the internal consistency of the vectoring solution on vehicles for smoother maneuverability, especially for such mine trucks that operate in the confined environment.

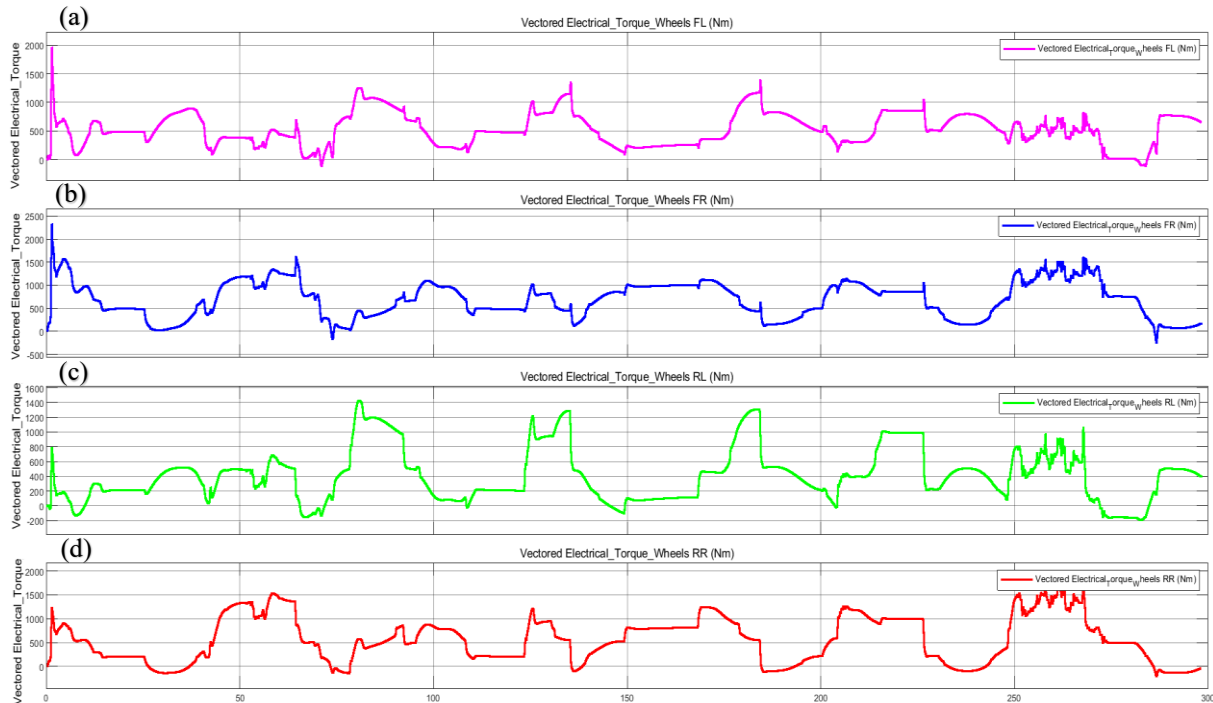


Figure 39: Commanded electrical motor torque demand per wheel [Nm]. (a) Torque demand to front left. (b) Torque demand to front right. (c) Torque demand to rear left. (d) Torque demand to rear right

5.11. Summary

The dataset shows a coherent control story from request to actuation. The steering program produces bounded articulation and curvature. The payload schedule alters a , b , m , and height in the expected directions. Normal forces and lateral forces change accordingly and feed the allocator. The EKF yields smooth yaw-rate and sideslip estimates. The yaw-moment command and the articulation damping achieve good tracking of the yaw rate with small transient errors at reversals and during brief capacity limits. The allocator distributes torque across the four wheels to meet both the total demand and the desired yaw moment while respecting friction and power bounds. The response scales logically with load and with grade. The controller therefore provides a stable and practical solution for torque vectoring on a heavy, centrally articulated platform and is ready for hardware-in-the-loop evaluation with the same parameter set.

Chapter 6

6. Conclusion

This thesis was aimed at designing and simulating a supervisory torque-vectoring module and a yaw-moment controller architecture for a centrally articulated series-hybrid mine truck with four independent in-wheel motors. The problem was framed around a realistic industrial constraint. Only the torque requests to the traction inverters are available, while the low-level motor and inverter parameters are confidential. This work has addressed two research questions. First, how to integrate the torque vectoring control with the existing MT66 eDrive Simulink plant model. Second, how to manage the articulation and the variable load so that the traction, stability, and maneuverability improves under the typical underground duty.

The project contributions can be summarized as threefold. First a steering-to-articulation module converts the operator's demand into a bounded articulation reference angle Φ and rate, these are scheduled by speed and shaped by the actuator limits. A yaw-moment controller forms a feasible direct yaw-moment request M_z from a steady yaw-rate target and feedback from the estimated yaw rate r and the sideslip β . A quadratic programming allocator maps the pair $[T_{tot}, M_z]$ to per-wheel torques, while enforcing the parameter constraints like the machine torque, available power, max and min rates, and the tire-friction bounds. Normal forces are estimated from the geometry of the vehicle, payload, grade, and the measured accelerations, which closes the loop between the lateral dynamics and the traction limits.

The simulation results shows consistent behavior across 0 to 300 s drive cycle with multiple payload steps, plus a 4° uphill segment from 248 to 268 s. The steering program produces the intended articulation plateaus and the closed-loop articulation, tracking errors remain nearly less than 6.3% across steady steering operation and no drift is observed during throttle inputs. The yaw-rate tracking is accurate in the linear tire region, with small bias when the curvature rises and the tire or the power limits starts to tighten. The allocator converts the M_z into its corresponding left and right torque bias which lifts the outer-wheel torque and decreases inner-wheel torque during cornering, while keeping the total near the requested T_{tot} from the powertrain controller unit. For example, during 155–160 s at $v = 3.8$ m/s and light load, the PCU total is 2000 Nm whereas the vectored total is 2200 Nm, about 9% higher, which reflects the soft-sum penalty. Under full load at 220–235 s, both totals are 3600 Nm, indicating equality.

When the payload increases or if the grade rises, the allocator protects feasibility by shifting more drive towards the rear axle. A moderate left turn at full load on a 4° grade (Figure 38) yields a PCU total torque request of about 5400 Nm, while the vectored total delivered is about 4850 Nm (a 7% reduction). Within the vectored total, the rear axle receives about 3100 Nm and the front 1750 Nm, corresponding to roughly 64% versus 36%. Thus the rear torque is approximately 77% higher than the front, indicating a clear rearward dominance under grade.

In the most demanding regions, namely full load with a modest uphill, the allocator accepts a controlled yaw-moment shortfall to preserve longitudinal propulsion. This coincides with a reduction in available DC-link power to nearly 400 kW and recovers as the available DC-link power increases to almost 610 kW and the operating limits relax. The sideslip stays within a narrow band, which is consistent with the small-slip design region. The ordering of the electrical torques, the mechanical torques at the wheels, lateral forces, and the normal forces is mutually consistent, which indicates that the modeling chain is nearly coherent.

These findings answer the research questions. The entire architecture integrates cleanly with the existing top-level plant model of the vehicle and uses only the torque interfaces as the major variable, along with other constraints like the vehicle speed, available power, wheel speeds, operational environment from the Simscape Multibody of the plant. This manages the articulation and the variable load by combining reference shaping, state estimation, and constrained allocation that respects the realistic limits. The novelty lies in adapting a two-effort allocation $[T_{tot}, M_z]$ to a hinge-steered, heavy articulated platform, while aligning yaw authority with the hinge geometry and the load transfer. The practical values are a modular controller that can be implemented with the existing signals, schedules its authority as margins change, and remains solvable when the constraints bind.

There are clear implications for both theoretical study and in practice. On the theory side, the study shows that a compact linear model $\dot{x} = Dx + Fu$ with small-slip tires can support reference generation and feedback, provided that the allocator exposes feasibility through limits that reflects the normal-force and the friction estimates. On the practical side, the results show how an industrial plant model can host the layered controller that improves the path following without touching the low-level electrical inverter codes. This approach helps to de-risk the field deployment by revealing when and why the moment shortfalls occurs, and by demonstrating the smooth torque redistribution under saturation.

The work had multiple limitations towards accessing real world parameters. Validation is purely simulation-based study. Regenerative braking was not modeled for negative longitudinal effort as the real plant model does employ the use of regeneration. The tire forces were treated in the linear region and were only scheduled by the load. The hydraulic actuation was represented by the reference shaping without a closed hydraulic loop. The thermal and the DC-link dynamics were approximated by the power caps and were not closed with the temperature or voltage states. The parameter values for inertias were estimated from public data and may differ from the actual production values. The estimator uses an extended Kalman filter and simple kinematics. Its performance near the adhesion limit was not stress-tested with measurement noise representative of underground sensors.

In summary, the thesis demonstrates that a layered torque-vectoring and yaw-moment controller, built around feasible effort requests and a constrained allocator, can improve stability and maneuverability of an articulated electric haul truck while respecting strict actuator limits. The architecture is implementable, uses the available signals, and yields internally consistent responses across the payload and grade changes. It provides a credible base for hardware-in-the-loop testing in the future and later field trials.

6.1. Future Work and Recommendations

To have further extensions is both natural and valuable. Having a more advanced predictive yaw control layer based on model predictive control could enforce the short horizon constraints on sideslip, friction, and actuators. This addition could reduce the transient yaw moment deficits when the physical limits change quickly. A nonlinear tire model with load and slip dependent stiffness can widen the valid operating region. If the regenerative braking and blended brake coordination are not limited on the real vehicle, the allocator can also be modeled to handle negative total torque. Negative T_{tot} corresponds to braking. The thermal and DC link states are already represented in the top plant model and should be coupled to the power caps during the allocation process. State estimation can be strengthened with an adaptive filter or an interacting multiple model filter that switches across friction and payload regimes. Learning based friction observers that fuse wheel speed and acceleration features should also be explored. A snaking mode damping channel that biases the rear torques at higher speed could be integrated with the yaw loop. Hardware in the loop experiments, parameter identification on a test vehicle, and controlled underground trials are required to confirm robustness and refine limits. These studies should calibrate the authority schedule between articulation and torque vectoring. Further development requires stronger coordination from the industry and should share parameters, values, and limits so researchers can model and validate the system accurately.

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Appendix

```

function [Fz_FL, Fz_FR, Fz_RL, Fz_RR] = Calc_LT(ax, ay, p) %#codegen
% Calc_LT - Compute the normal (vertical) load on each wheel of a rigid, centrally-articulated 4x4.
%
% Given body accelerations ax (longitudinal) and ay (lateral), and a struct/bus p with
% vehicle geometry and mass properties, it splits the total weight across the four wheels.
% It starts from the static distribution set by the CG location, then adds:
%   • pitch transfer from ax (forward/back weight shift), and
%   • roll transfer from ay (left/right weight shift).
%
% Call signature
% [Fz_FL, Fz_FR, Fz_RL, Fz_RR] = Calc_LT(ax, ay, p)
%
% Inputs
% ax : Longitudinal body acceleration [m/s^2].
%     ax > 0 means accelerating forward (weight tends to move rearward).
% ay : Lateral body acceleration [m/s^2].
%     ay > 0 means accelerating to the left (e.g., left turn → weight shifts to the right).
% p : MachineData struct/bus (SI units) with:
%     - m : total mass [kg]
%     - g : gravity [m/s^2]
%     - h_CG : CG height above ground [m]
%     - a_CG : distance CG → front axle [m]
%     - b_CG : distance CG → rear axle [m]
%     - track_f : front track width [m]
%     - track_r : rear track width [m]
%     - Kphi_f : front roll stiffness [N·m/rad] (optional; NaN → fallback)
%     - Kphi_r : rear roll stiffness [N·m/rad] (optional; NaN → fallback)
%     - r_tire_static : static tire radius [m] (not used here but common in the bus)
%
% Outputs (nonnegative, in newtons)
% Fz_FL, Fz_FR, Fz_RL, Fz_RR : normal loads at front-left, front-right, rear-left, rear-right.
%
% Inline helper: get(f,def) → p.(f) if present, otherwise def.
get = @(f, def) (isfield(p, f) * p.(f) + (~isfield(p, f)) * def);

% Basic constants and geometry (defaults chosen to be reasonable if fields are missing)
g = get('g', 9.81); % gravity [m/s^2]
m = get('m', 1); % mass [kg]
h = get('h_CG', 1); % CG height [m]

% CG-to-axle distances (front/back) and wheelbase
a = get('a_CG', 1.796); % CG → front axle [m]
b = get('b_CG', 3.85); % CG → rear axle [m]
L = a + b; % wheelbase [m]

% Track widths
tf = get('track_f', 1.5); % front track [m]
tr = get('track_r', 1.5); % rear track [m]

% Roll stiffness (optional). If NaN, we'll fall back to axle-weight share.
Kpf = get('Kphi_f', nan); % front roll stiffness [N·m/rad]
Kpr = get('Kphi_r', nan); % rear roll stiffness [N·m/rad]

% -----
% 1) Static per-wheel loads (front and rear)

% Static axle loads: front axle = m*g*(b/L), rear axle = m*g*(a/L).
% Per-wheel means divide each axle load by 2.
Fzf0 = 0.5 * m * g * b / L; % front per-wheel static load [N]
Fzr0 = 0.5 * m * g * a / L; % rear per-wheel static load [N]

% -----
% 2) Longitudinal load transfer (pitch)

```

```

% Pitch moment m*h*ax/L shifts load rearward when ax>0.
dFtot = m * h * ax / L;      % total transfer magnitude [N]
dFf    = -0.5 * dFtot;       % per front wheel change (front loses under accel)
dFr    = 0.5 * dFtot;       % per rear wheel change (rear gains under accel)

% -----
% 3) Lateral load transfer (roll)

% Total roll moment from lateral acceleration.
My = m * h * ay;            % [N·m]

% Split roll moment front/rear:
% - If both Kphi_f and Kphi_r are provided, use stiffness share.
% - Otherwise, use static axle weight share (b/L to front, a/L to rear).
if ~isnan(Kpf) && ~isnan(Kpr)
    beta_f = Kpf / (Kpf + Kpr);
else
    beta_f = b / L;
end

Myf = beta_f * My;          % front share of roll moment [N·m]
Myr = My - Myf;            % rear share of roll moment [N·m]

% Convert each axle's roll moment to a left/right pair transfer (divide by track).
dFy_f = Myf / tf;          % front pair transfer [N]
dFy_r = Myr / tr;          % rear pair transfer [N]

% Split each axle's pair transfer to individual wheels.
% For ay > 0 (left turn), right side gains, left side loses.
dFFL = -0.5 * dFy_f;      % front-left change [N]
dFFR = 0.5 * dFy_f;       % front-right change [N]
dFrL = -0.5 * dFy_r;      % rear-left change [N]
dFrR = 0.5 * dFy_r;       % rear-right change [N]

% -----
% 4) Final per-wheel loads (clipped to be nonnegative)

Fz_FL = Fzf0 + dFf + dFFL; % front-left
Fz_FL = max(0, Fz_FL);

Fz_FR = Fzf0 + dFf + dFFR; % front-right
Fz_FR = max(0, Fz_FR);

Fz_RL = Fzr0 + dFr + dFrL; % rear-left
Fz_RL = max(0, Fz_RL);

Fz_RR = Fzr0 + dFr + dFrR; % rear-right
Fz_RR = max(0, Fz_RR);
end

```

Listing 1: Normal force computation MATLAB Function

```

function Fy = fy_from_ekf(vx, vy_hat, r_hat, delta_eq, Fz, a, b, Cf, Cr, eps_vx, mu_hat)
%#codegen
% Inputs: scalars vx, vy_hat, r_hat, delta_eq, a, b, Cf, Cr, eps_vx, mu_hat
%         Fz is 1x4 [FL FR RL RR]
% Output: Fy is 1x4 [FL FR RL RR]

vx_eff = max(vx, eps_vx);

% Axle forces (linear bicycle)
Fyf = Cf * (delta_eq - (vy_hat + a*r_hat)/vx_eff);
Fyr = -Cr * ((vy_hat - b*r_hat)/vx_eff);

% Load shares
FzF = max(Fz(1)+Fz(2), 1); % avoid div by zero
FzR = max(Fz(3)+Fz(4), 1);
sFL = Fz(1)/FzF; sFR = Fz(2)/FzF;
sRL = Fz(3)/FzR; sRR = Fz(4)/FzR;

% Split to wheels
Fy = [ sFL*Fyf, sFR*Fyf, sRL*Fyr, sRR*Fyr ];

% Optional friction cap
if ~isempty(mu_hat)
    muFz = mu_hat .* Fz;
    Fy = max(-muFz, min(Fy, muFz));
end
end

```

Listing 2: Lateral force computation MATLAB Function

```

function T = qp_alloc(Ttot,Mz_des,Tbase,Tprev,omega,Fz,Fy,Rw,Tmin,Tmax_eff,Pmax,tf,tr,Wdiag,rho,gamma,eta)
%#codegen
persistent x0; if isempty(x0), x0 = Tbase; end

% Quadratic cost on (T - Tbase) and (T - Tprev)
W = diag(Wdiag);
H = 2*(W + rho*eye(4));
f = -2*(W*Tbase' + rho*Tprev'); % make Tbase,Tprev columns for math

% Add two slack variables (s_T and s_M) to softly enforce sum and yaw
H = blkdiag(H, 2*gamma, 2*eta);
f = [f; -2*gamma*Ttot; -2*eta*Mz_des];

% Yaw mapping row
aMz = [-tf/(2*Rw(1)), tf/(2*Rw(2)), -tr/(2*Rw(3)), tr/(2*Rw(4))];

% Equalities tying slacks to actual sum(T) and aMz*T
% [1 1 1 1 -1 0] * [T; sT; sM] = 0 => sT = sum(T)
% [aMz 0 -1] * [T; sT; sM] = 0 => sM = aMz*T
Aeq = [1 1 1 1 -1 0;
       aMz 0 -1];
beq = [0; 0];

% Safety pass to re-tighten Tmax_eff
for i=1:4
    Tmax_eff(i) = min(Tmax_eff(i), Pmax(i)/max(omega(i),1)); % power
    Fx_lim = sqrt(max(0, (1.0*Fz(i))^2 - Fy(i)^2)); % ellipse (mu already used upstream)
    Tmax_eff(i) = min(Tmax_eff(i), Fx_lim*Rw(i)); % torque from Fx_lim
end

% Bounds include T (4) + sT + sM (2)
lb = [Tmin(:); -inf; -inf];
ub = [Tmax_eff(:); inf; inf];

% Solve
opts = optimoptions('quadprog','Display','off', ...
    'Algorithm','interior-point-convex','MaxIterations',50,'OptimalityTolerance',1e-6);

z0 = [x0(:); 0; 0]; % warm start (previous solution)
z = quadprog(H,f,[],[],Aeq,beq,lb,ub,z0,opts);

if isempty(z)
    T = Tbase; % safe fallback if solver fails
else
    T = z(1:4).'; % return row vector
    x0 = T(:); % store warm start
end

```

Listing 3: Allocation computation

```

function omega_a = fcn(theta_sw, omega_sw, theta_sw_max)
% Calculate cosine term (convert theta_sw from degrees to get a normalized value)
cos_term = cos((pi/2) * (theta_sw / rad2deg(theta_sw_max)));

% Compute articulation angular velocity:
% Multiply the scaling constant (pi^2/(8*theta_sw_max)), the cosine term,
% and convert omega_sw from deg/s to rad/s using deg2rad.
omega_a = (pi^2 / (8 * theta_sw_max)) * cos_term * deg2rad(omega_sw);

```

Listing 3: Articulation rate mapping

```

function theta_a_desired = fcn(theta_sw_deg, theta_sw_max_rad)
% Compute Desired Articulation Angle (Static Mapping)
%
% This function implements Equation (14) of the dynamic articulation-control model:
%  $\theta_{a,desired}(t) = A_{max} \cdot \sin(\pi/2 \cdot x(t))$ 
% where  $x(t) = \theta_{sw}(t) / \theta_{sw\_max}$  (both in degrees),
% and  $A_{max}$  is your physical articulation limit in radians.
%
% Inputs:
%   theta_sw_deg      - Steering-wheel angle (degrees)
%   theta_sw_max_rad  - Speed-dependent half-lock steering limit (radians)
%
% Output:
%   theta_a_desired   - Desired articulation angle (radians)

%% 1) Define your true max articulation limit (in degrees → radians)
A_max_deg = 45; % e.g. your vehicle's ±45° articulation range
A_max     = deg2rad(A_max_deg); % convert to radians (≈0.7854 radians)

%% 2) Normalize steering-wheel input to [-1, +1]
%   theta_sw_max_rad is half-lock in radians, so convert to degrees first:
theta_sw_max_deg = rad2deg(theta_sw_max_rad);

%   x =  $\theta_{sw} / \theta_{sw\_max}$  (unitless ratio)
x = theta_sw_deg ./ theta_sw_max_deg;

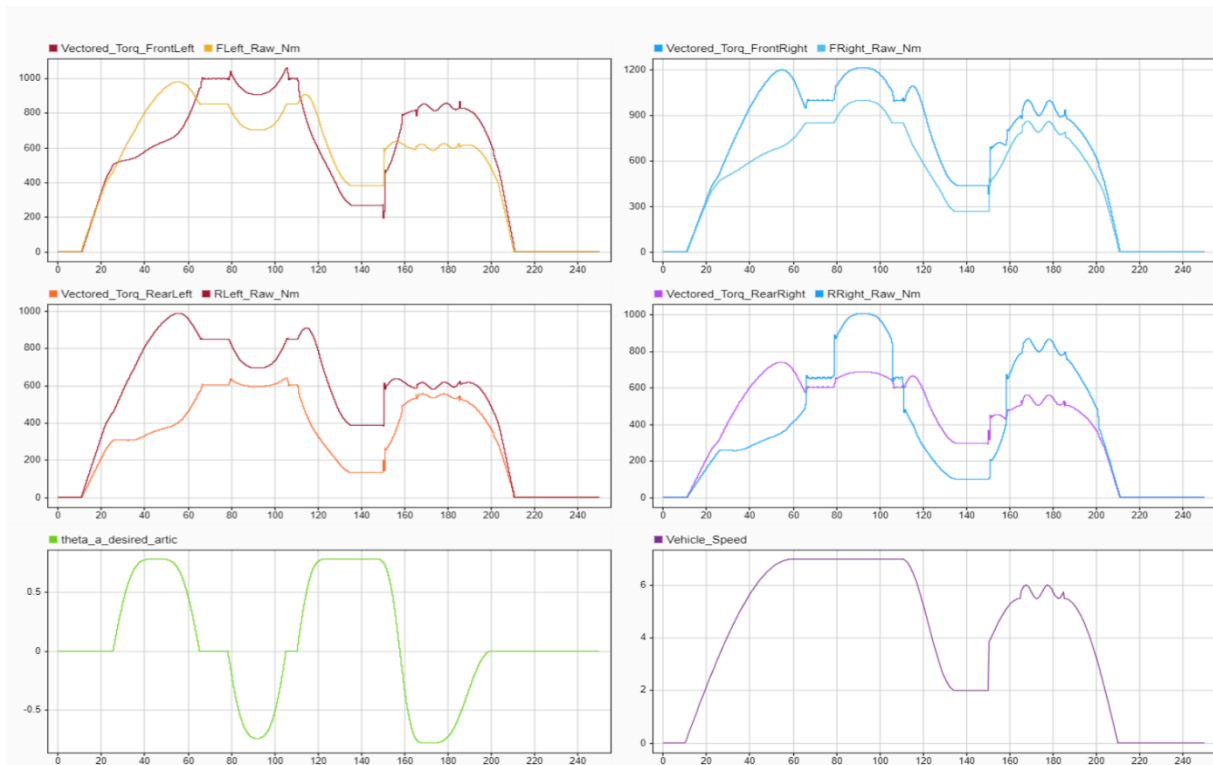
%% 3) Static sine-mapping into articulation range
%    $\sin(\pi/2 \cdot x) \rightarrow$  ranges smoothly between -1 and +1 as x goes from -1 to +1
theta_a_desired = A_max .* sin( (pi/2) .* x );

%% 4) (Optional) Saturate x to [-1, +1] to guard against overshoots
%   x_clamped = max(min(x,1),-1);
%   theta_a_desired = A_max * sin((pi/2)*x_clamped);

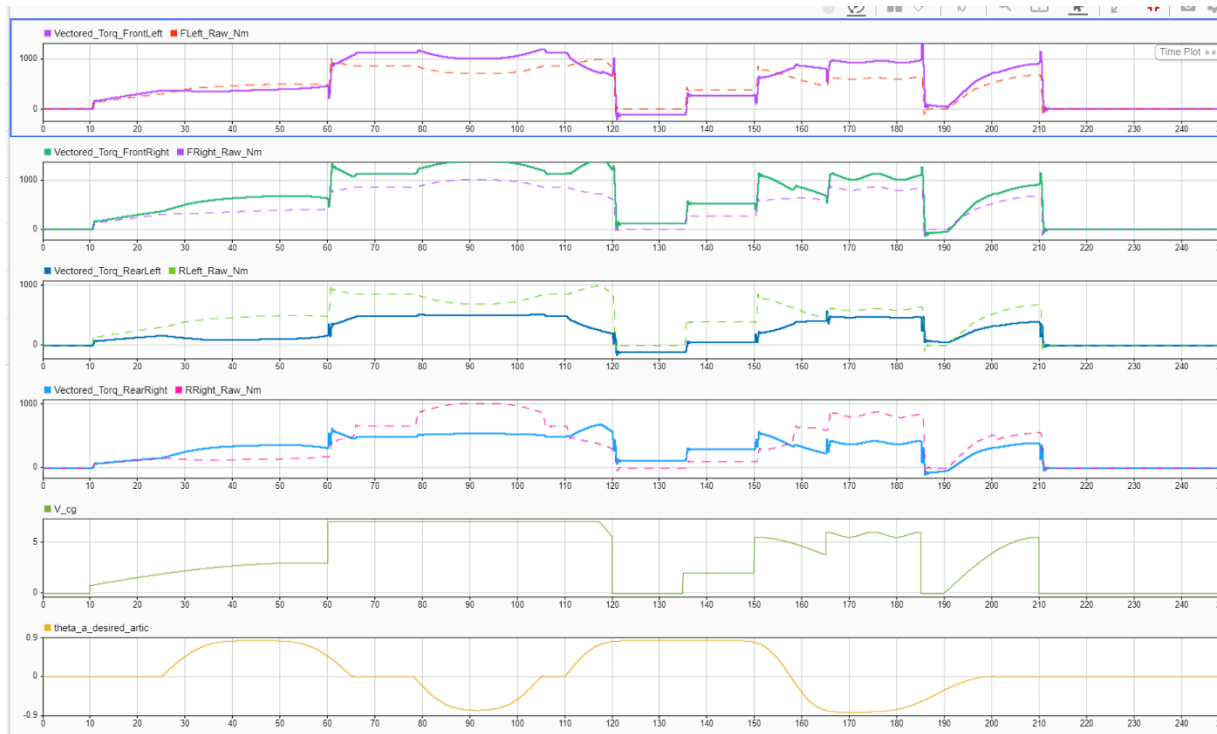
%% 5) Units:
%   - theta_a_desired is in **radians**,
%   - with maximum magnitude A_max (your ±45° physical limit).
end

```

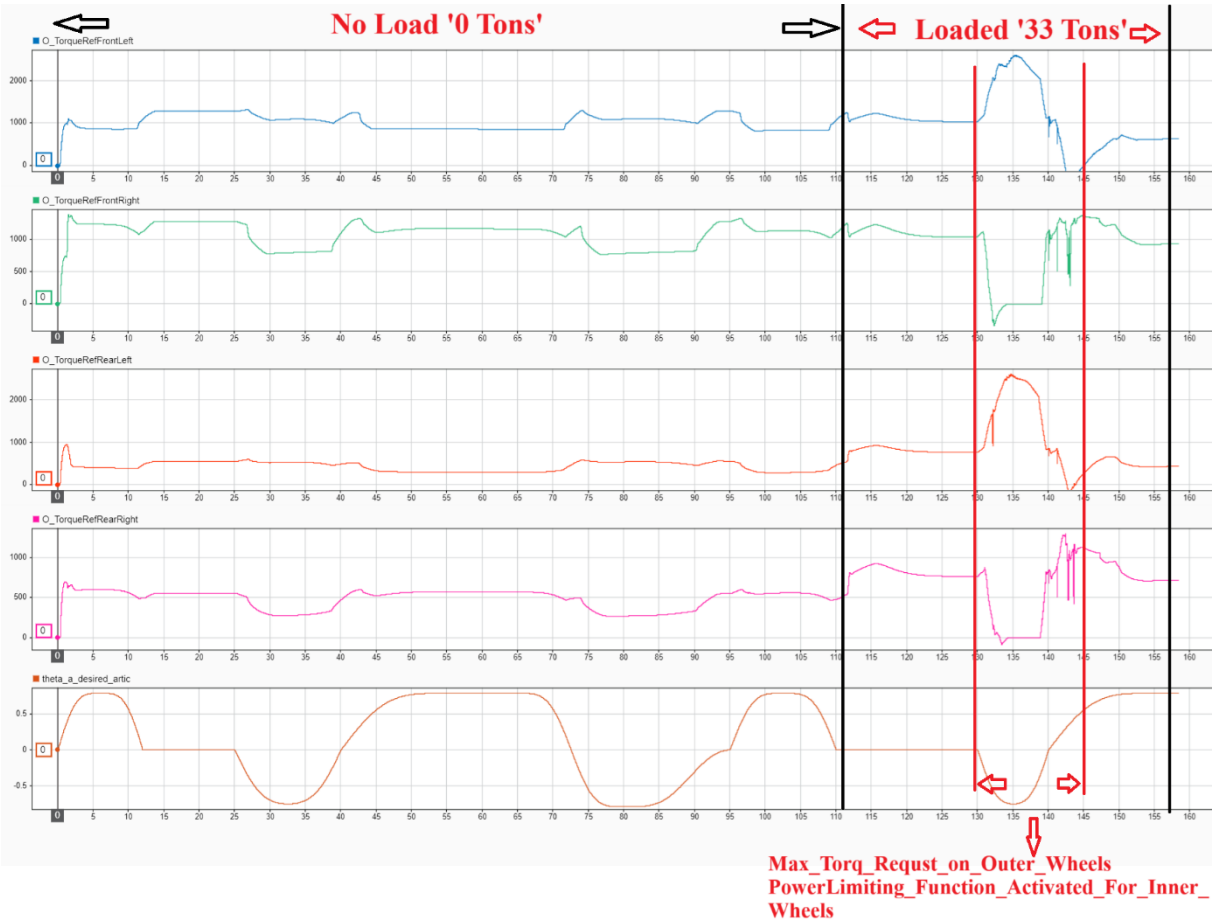
Listing 4: Desired articulation mapping



Listing 5: Torque comparison between inner and outer wheels during articulation compared to the PCU commanded torques



Listing 6: Torque comparison between inner and outer wheels during articulation compared to the PCU commanded torques



Listing 7: Torque comparison between the front and rear axles based on loads